

Program overview

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Year 2022/2023
Organization Applied Sciences
Education Master Chemical Engineering

Code	Omschrijving	ECTS	p1	p2	p3	p4	p5
Master Chemical Engineering 2022							
Chemical Engineering Core Programme 2022							
Core Courses 2022							
CH3044A	Process, Dynamics and Control (PD&C)	4					
CH3133	Computational Practicum	6					
CH3153	Molecular Transport Phenomena	4					
Advanced ChE Courses (choose 3 courses from 6) 2022							
Advanced Chemical Engineering courses (12 EC, students choose 3 compulsory courses out of 6) 2022							
CH3013	Interfaces and Particles	4					
CH3051	Applied Transport Phenomena (ATP)	4					
CH3143	Advanced Thermodynamics	4					
CH3175	Solid State Materials	4					
CH3373	Soft Materials Engineering	4					
CH3682A	Reactor & Kinetics (R&K)	4					
Chemical Engineering Electives 2022							
Chemical Engineering Electives 2022 (12 EC, students choose 12 EC from the courses listed below)							
AP3171	Advanced Physical Transport Phenomena	6					
AP3252	Electron Microscopy Characterization of the Nanoscale	3					
AP3352	Introduction to Nuclear Science and Engineering	6					
CH3061	Multiphase Reactor Engineering	4					
CH3073	Separation Processes, Design and Operation (SPDO)	3					
CH3082	Chemical Technology	3					
CH3101	Heterogeneous Catalysis for Chemical Engineers	3					
CH3222	Energy Storage in Batteries	4					
CH3382	Molecular Engineering of Soft Materials in Health Care	4					
CH3421	Computational Transport Phenomena	6					
CH3513	Electrochemistry for Renewable Energy	4					
CH3531	Functional Ceramics	3					
CH3542	Inorganic Materials	3					
CH3564	Particle Technology for Health and Energy	3					
CH3622	Process Intensification (PI)	3					
CH3632	Chemistry and Physics of Solar Cells	6					
CH3672	Computational Materials Science	3					
CH3763	Nuclear Medicine	3					
CH3765	Advanced Materials Analysis by Radiation Instrumentation	3					
CH3771	Nuclear Chemistry	6					
CH3783	Materials Chemistry for The Nuclear Fuel Cycle	3					
CH3921	Sustainable Polymer Materials	3					
LM3311	Green Chemistry and Sustainable Technology	3					
LM3731	Advanced Biocatalysis	6					
ME45203	Electrochemical Energy Storage 2: Analytical Modelling	4					
SEN1121	Complex Systems Engineering	5					
SET3070	Thermochemistry of Biomass Conversion	4					
SET3085	Hydrogen Technology	4					
Thesis Project 2022							
Thesis Project 2022 (students choose a MEP of 30EC or 42 EC. PSO and MEP together should be a minimum 60EC)							
Design Module 2022							
CH3803	Product & Process Design	6					
CH3843	Design Project	12					
TPM330A	Engineering Ethics and Risk	4					
Professional & Social Orientation 2022							
Research and Development 2022							
CH3702	Industrial Internship	18					
Study Abroad 2022							
Study Abroad 2022 (only in combination with a 30EC MEP, CH3930)							
Education 2022							
Education 2022 (only in combination with a 30EC MEP, CH3930)							
Education Basis 2022							
Education Basis 2022 (only in combination with a 30EC MEP, CH3930)							
SL3462	Educational Sciences	6					

SL4201	Pedagogy of STEM education	4	
SL4202	Professional Learning Community	1	
SL4230	Chemistry Teaching Methodology	4	
SL4235	Chemistry Foundation Teaching Placement	15	
Education Verdieping 2022			
Education Verdieping 2022 (only in combination with a 30EC MEP, CH3930)			
SL3012	Personal Professional Development	3	
SL4300	Design and Research in Education	10	
SL4330	Advanced Chemistry Teaching Methodology	5	
SL4335	Advanced Chemistry Teaching Placement	12	
Management of Technology 2022			
Management of Technology 2022 (only in combination with a 30EC MEP, CH3930)			
1st Semester MoT Modules 2022			
1st Semester MoT Modules 2022 (only in combination with a 30EC MEP, CH3930)			
MOT1412	Technology Dynamics	5	
MOT1421	Economic Foundations	5	
MOT1442	Social and Scientific Values	5	
MOT1461	Financial Management	5	
MOT1524	Leadership and Technology Management	5	
MOT2313	Research Methods	5	
2nd Semester MoT Modules 2022			
2nd Semester MoT Modules 2022 (only in combination with a 30EC MEP, CH3930)			
MOT1003	Integration Moment	5	
MOT1435	Technology, Strategy and Entrepreneurship	5	
MOT1452	Inter- and intra-organisational decision making	5	
MOT1531	Digital Business Process Management	5	
MOT1534	High-tech Marketing	5	
MOT2421	Emerging and Breakthrough Technologies	5	
Science & Engineering 2022			
TUD4040	Joint Interdisciplinary Project	15	
Free electives			
AS3111	ATHENS	2	
AS3121	Scientific Writing and Argumentation	3	
AS3131	Art, Empathy & Ethics	4	
TPM301B	Spoken English for Academic Purposes Intermediate	2	
TPM302B	Spoken English for Academic Purposes Advanced	2	
TPM303A	Intermediate Writing in English for the University	2	
TPM304A	Advanced Writing in English for the University	2	
TPM305A	Writing a Masters Thesis in English	2	
WM0203TU	Oral Presentations	2	
WM1115TU	Dutch Elementary 1	3	
WM1116TU	Dutch Elementary 2	3	
WM1117TU	Dutch Intermediate 1	3	
WM1135TU	Advanced English for the University	3	

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Master Chemical Engineering 2022

Responsible Program Employee	Prof.dr. J.H. van Esch
Program Coordinator	J. Quak
Contact for students	Academic Counselor: Drs. Carina Vonk (C.C.Vonk@tudelft.nl)
Program Title	Chemical Engineering
In association with the Faculty of	TNW
EC Program	The Chemical Engineering programme is a two-year master programme and comprises 120 EC. The study load of modules is expressed in European Credits (EC). This is a result of the European Credit Transfer System (ECTS), which encourages acknowledgement of study results between higher education institutions within the European Union. ECs give an indication of the weight of a certain part of the programme. One EC involves approximately 28 hours of study. It includes all time spent on the course: lectures, self-study, assignments, examinations, etc.
Introduction 1	Note: use the menu at the left for information on each subset of the programme.
Program Structure 1	Year 1: The first year of the Chemical Engineering programme comprises of a mixture of core courses, electives and a design module. The Core Programme consist of the following modules: - Obligatory Core Courses (14EC) - Advanced Chemical Engineering Courses (12EC) - Chemical Engineering Electives (12EC) - Design modules (22EC) Year 2 The Core Programme (60EC) is combined with one of the following combinations of Professional and Societal Orientation completes the program: - a 30EC Professional and Societal Orientation and a 30EC Master Thesis Project - an 18EC Professional and Societal Orientation and a 42EC Master Thesis Project . Students have the opportunity to personalise a significant part of their programme by selecting one of the following for their Professional & Societal Orientation (PSO). - Research & Development - Science & Technology - Management of Technology - Education - Study Abroad Please be advised that students will not receive a separate title or addendum when finishing a science & engineering orientation. It will not be reflected on the degree. PROGRAMME ADDITIONS - Honours track: This is an individual programme of at least 20 EC on top of the full Chemical Engineering programme, which contains specially developed modules for all TU Delft honours track students. The full Chemical Engineering programme including the additional honours track (140 EC in total) should be completed within 2 years. REGULATIONS See https://www.tudelft.nl/en/student/faculties/as-student-portal/education/regulations-degree-programmes-as/ (or Brightspace) for the official regulations of the programme (note that the regulations for next academic year may not be available yet).
Fail or Pass Regulation	The student meets the requirements for the degree audit when all the required modules of the programme have been completed with: a PASS (V), an exemption (VR) or a grade of 6 or higher. Exemptions can only be granted by the Board of Examiners (BoardofExaminers-CE-AS@tudelft.nl). IMPORTANT: Due to restrictions imposed as a result of the corona crisis, the method of assessment may deviate from the one mentioned per course. Students are informed timely about changes.
With Honours Regulation	The designation with distinction (cum laude) for the Masters degree audits can be granted by the Board of Examiners if the student meets the following criteria: - The weighted average of the results of the courses (excluding the MSc final project) is at least 8.0; passes (V) and exemptions (VR) will not be taken into consideration; - The number of credits for the courses for which a pass (V) an exemption (VR) has been granted does not exceed 40 credits in total; - The result for the MSc final project is at least 9.0; - None of the marks is lower than 7.0; - The students degree programme has not taken longer than two years between the first and last grade earned. Any study delay that falls under the Graduation Support Scheme (RAS) is taken into account. In special cases the Board of Examiners may decide to grant the designation with distinction to a student who does not meet all the requirements if the student in question has shown exceptional skills in the degree programme.

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Chemical Engineering Core Programme 2022

Program Structure 1	The Core Programme comprises 60 credits (60EC) and is the same for all students. The Core Programme consist of the following modules: - Obligatory Core Courses (14EC) - Advanced Chemical Engineering Courses (12EC) - Chemical Engineering Electives (12EC) - Design modules (22EC) Across the electives and core courses students are free to select from any combination of courses (subject to scheduling limitations), however the department does make some recommendations of different combinations depending on the interest of the student. These combinations are selected to align within 4 different Science & Engineering Orientations: - Circularity - Energy - Health - Nuclear
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Core Courses 2022

CH3044A	Process, Dynamics and Control (PD&C)	4
Responsible Instructor	Dr. B. Bera	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 3	
Course Language	English	
Expected prior knowledge	Basic knowledge of industrial process (equipments) for mass and heat transfer, separation and chemical reaction. Knowledge of calculus, linear algebra, complex numbers Knowledge of numerical techniques like integration and their implementation in Python.	
Course Contents	Every single process at a (chemical) engineering plant needs to be controlled. And control is not possible until the dynamics of the process is thoroughly understood, modelled and analysed. In this course, the students will be given a taste of understanding, modelling, analysing and controlling processes that they are already familiar with from their bachelor courses. The course makes this goal (of giving a complete overview of the task of a control engineer) very clear in the introductory lectures. Subsequently, the classic modelling cycle is addressed where each and every step of the modelling cycle; such as: making assumptions, formulating the mathematical equations, applying the correct boundary conditions, solving these mathematical equations (numerically), interpreting the degrees of freedom of the system etc. Once the student is equipped with this knowledge, the added tool of 'Analysis' is provided to the students, where non-linearity in dynamic processes, linearization, point of linearization (and its relation with point of operation) are discussed in detail. Stability of a system is also given some emphasis in this part, and connections are drawn with other M. Chem. Engineering courses such as diffusion-reaction equations etc. Subsequently, Laplace Transforms, Transfer functions, and analysis in the laplace domain are discussed in detail although Analysis in frequency domain is not covered seeing that frequency response is not the primary thing that a chemical engineer is interested in. Model Approximation, parametrization and interaction are subsequently discussed. Interaction among the various parts in a process lays the path to the control part of the course where Single Input Single Output (SISO) is first discussed. Proportional feedback control, feedback and feedforward control are introduced, then root locus diagrams, PID control and IMC control are also discussed. During the interaction part, it is clearly shown that in a real process, one part not only influences another part but also vice-versa. This is picked up further in the Multiple input multiple output (MIMO) part. Plantwide control and batch control are also touched upon along with optimization as a tool for control as well as preliminaries of engineering instrumentation.	
Study Goals	Demonstrate understanding of a typical process or operation from the chemical industry. Apply various steps of a modelling cycle to a process system. Analyse the dynamics of such (above-mentioned) process. Design and simulate a control system for a unit operation. Extend the understanding and modeling for a single process to plant level.	
Education Method	Lectures and Tutorials	
Computer Use		
Literature and Study Materials		
Assessment	50% of the final grade will be from a group assignment, where students will work together in groups of three. The other 50% of the final grade will be from a written final exam.	
Permitted Materials during Tests		

CH3133	Computational Practicum	6
Responsible Instructor	Dr. B. Bera	
Instructor	Dr. B. Bera	
Instructor	A.M. Schweidtmann	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor level linear algebra: vector, matrix, linear mapping, basis vectors, solving linear systems, computing eigenvalues and eigenvectors, application of linear algebra; Bachelor level calculus: differentiation and integration of a function in one and more variables, complex numbers, analytical methods for ordinary and partial differential equations, Fourier transformation, applications of calculus; Bachelor level computer programming: elementary instructions, conditional statements and loops, functions and main program;	
Course Contents	Three half days for intro in programming techniques (space to improve via autograding/gamification) Linear Systems; Non-Linear Systems; Ordinary Differential Equations Partial Differential Equations; Data Handling Optimization.	
Study Goals	Demonstrate clear understanding of why numerical modeling is required in various aspects of Chemical Engineering e.g., reaction engineering, control systems, convective flow etc. statement on mathematical modeling. Evaluate the connection among problems from various fields of chemical engineering Solving mathematical problems using numerical methods i.e., working out the 'input-process in-between-output' cycle Implement a given numerical method using Python (or Julia) Evaluate the properties of a given numerical method (computational complexity, convergence, stability properties); assess their scope, shortcoming, advantage and disadvantage Relate and apply concepts from numerical solving of mathematical models to processes in different kinds of transport and process systems (relevant for a chemical engineer)	
Education Method	In Q1, more classical lecture then tutorial format. In Q2, mostly tutorials (since this is a practical), to work on the assignments, with help from the instructors and the TAs.	
Computer Use	Python is the programming language for this course	
Literature and Study Materials		
Assessment	There are 6 assignments, to be made individually, during the course. Each quarter will have 3 assignments. The assignments are connected to the other courses being given in that quarter. In Q1, there is a general introductory assignment, followed by two assignments connected to the other two courses (MTP and PDC) in Q1. In Q2, there will be total 6 available assignments, out of which a student has to choose three (based on which three courses they are following in Q2). Each assignment is 1 ECT point's worth. Excluding the lectures, and tutorials and self-study, a student is expected to complete the assignment in 10-12 hours. There is no final written/oral exam for this course, neither in Q1, nor in Q2.	
Permitted Materials during Tests		

CH3153	Molecular Transport Phenomena	4
Responsible Instructor	Dr. V. Garbin	
Instructor	P. Boukany	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 3	
Course Language	English	
Expected prior knowledge	- Physical Transport Phenomena (4052FYSTRY) - Physical Chemistry (4052FYSCKY) - Thermodynamics (4051CHTHEY and 4052STATHY) - Calculus I and II (4051CALC1Y and 4051CALC2Y) - Statistical methods (4052STAMEY) - Differential Equations (4052LADIFY).	
Course Contents	1. Introduction and motivation for the course. 2. Review of course prerequisites: Review of BSc transport phenomena (e.g. Ficks law, Fouriers law) and approaches to problem solving (e.g. shell balances). Continuum scale and link to molecular scale. 3. Kinetic Theory: Basic assumptions of kinetic gas theory. Mean free path. Maxwell velocity distribution. Calculating pressure from microscopic picture. Effect of intermolecular forces (van der Waals). 4. Binary diffusion: Einstein-Smoluchowski theory - Diffusion as a random walk. Langevin theory. Link to macroscopic description of diffusion. 5. Multicomponent diffusion: Limitations of Ficks first law of diffusion. Diffusion driven by a gradient in chemical potential. Generalized Maxwell-Stefan diffusion model. Multicomponent mass transport. 6. Ion transport and electrohydrodynamics: Transport of an isolated ion. Transport of non-isolated ion. Nernst-Planck transport equations. The Debye-Hückel double layer. Electrophoresis for non-isolated particles. Examples of applications.	
Study Goals	1. Reproduce the basic laws of transport phenomena; solve problems involving the transport of mass, species, momentum, and energy, including setting up the problem using a shell balance, deriving the corresponding differential equation(s), formulating of the initial and boundary conditions, and solving basic ordinary differential equations. 2. Use microscopic models (kinetic gas theory, random walk, Langevins theory) to derive expressions for macroscopic quantities such as pressure and diffusion coefficients; use these expressions to estimate the order of magnitude of these macroscopic quantities. 3. Explain the microscopic framework of multi-component diffusion and solve simple quasi-static one-dimensional problems using the Maxwell-Stefan equations, without and with additional forces (such as wall friction and electric fields). 4. Describe the motion of isolated and non-isolated charged particles using a molecular model; solve basic problems that involve electrophoretic flows.	
Education Method	Lectures: theory and applications. Working classes: solving problems, applying concepts to new examples.	
Computer Use		
Literature and Study Materials	Handouts including theory and exercises with (partial) solutions.	
Assessment	The assessment includes a mid-term assignment and a final exam. The assignment will enable students to apply the concepts learned in the lectures and working classes to an unfamiliar problem. The assignment can count towards 20% of the course grade, and the final exam in principle counts for 80% of the course grade. In case the course grade based on both the assignment and the exam is lower than the grade of the exam itself, then the exam counts for 100% of the final grade. Students who do not secure a passing mark for the course based on the assignments and the final exam, can do a resit. In this case, the resit alone counts for 100% of the final course grade.	
Permitted Materials during Tests	The exam is closed-book. Only a non-programmable calculator is allowed.	

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Advanced ChE Courses (choose 3 courses from 6) 2022

CH3013	Interfaces and Particles	4
Responsible Instructor	L. Rossi	
Instructor	Dr. V. Garbin	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Topic 1: Introduction Introduction to course, study goals Colloidal state Colloidal stability (intermolecular forces) Topic 2: Surface tension and surfactants Surface tension: Definition and molecular origin Surface tension as a force, energy, or pressure Measurement methods Surfactants: Definition Gibbs adsorption equation Micelles/CMC Wetting phenomena: Contact angle (equilibrium, advancing, receding) Methods to control wetting and contact angle Topic 3: Emulsions Intro and emulsion formation Emulsion stabilization (surfactant and solid particles) Ostwald ripening Topic 4: Particle preparation methods Bottom-up: Nucleation and growth Polymerization Top-down: Emulsion-based Lithography Milling Topic 5: Magnetic colloids Magnetism and magnetic interactions Magnetic materials Magnetic colloids Topic 6: Granular matter Intro to granular matter (length scales) Packing and flow Segregation, mixing, fluidization Topic 7: Industrial applications 2 guest lectures from industry explaining direct application (used for case study in groups)	
Study Goals	At the conclusion of the course, the student will be able to: 1. Understand and explain basic concepts of particulate systems and interfacial phenomena 2. Analyze and solve problems related to nanostructured systems and interfacial phenomena 3. Formulate plausible solutions to real industrial problems	
Education Method	Lectures, tutorials, case study based on industrial applications (in groups).	
Books	Relevant chapters of textbooks and topic-specific scientific articles and material will be provided. All books will be available as E-books available from the TUDelft library will be provided by the teachers during the course.	
Assessment	The course will be assessed base on a case study in groups (40%) and a final exam (60%). To pass the course the student must obtain a sufficient mark (6) in the exam. Students who do not secure a passing mark in the exam can take the resit. In this case the mark for the case study will remain valid to calculate the final mark.	

CH3051	Applied Transport Phenomena (ATP)	4
Responsible Instructor	Dr. T.E. Burdyny	
Instructor	Dr. A.Y. Rwei	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Transport Phenomena (on BSc level) Molecular Transport Phenomena Differential Equations	
Course Contents	Overview: In many processes in (bio)chemical industrial, as well as in health and energy related applications, fluid flow, heat transfer, mass transfer, and chemical reactions interact in a complex way. These equations can relatively easily be described by differential equations, but their solution requires substantial computational requirements without providing insights into the system at hand. To reduce complexity, generic rules are taught in ATP with an emphasis on simplification and deriving valuable interpretations about the underlying operation of the system. Concepts such characteristic times, scales and regimes are dealt with to simplify problems enough to make them tangible or solvable analytically. Several techniques are also introduced for finding approximate solutions to partial differential equations of transport phenomena systems. Content: The course content is split into three sections: 1. A recap of key course material but with an Applied perspective. 2. A Methods section that teaches simplifying concepts including Scaling, Reduction in Dimensionality and Timescales, Perturbation Theory and Self-Similarity 3. An applications section that covers a set of common physical systems. This section also combines conservation laws to apply the concept of Scaling across equations.	
Study Goals	At the end of this course, the student can: - Determine approximate solutions to chemical engineering transport phenomena problems - Analyse the behavior of a transport phenomena system without an analytical or numerical solution - Identify small secondary phenomena and determine whether their inclusion or exclusion greatly impacts the system. - Reduce complex problems to simpler ones using the method of order-one scaling applied to both length and time scales.	
Education Method	Standard 2-hour lectures are followed or preceded by self-study sessions. Study Guides and teaching assistants support these self-study sessions. The lectures provide learning activities to introduce the conceptual material, and for students to test this out in example. These relate to the above learning objectives. The tutorials/self-study sessions will provide additional learning activities supporting the material learnt in class the same week. In 2022-23 the original step-by-step study guides traditionally used will be replaced with a conceptually summary section, and then more concrete transport phenomena material representative of examination questions. We use Brightspace forums as an outside of class hours medium for discussion and questions.	
Assessment	100% Final Exam Here the learning objectives associated to the methods and transport phenomena applications will be tested in a 3-hour exam format. Some methods will be directly tested conceptually in shorter problems, while other will be tested in an application-centered problem. All the above learning objectives can then be tested within a 3-hour examination.	

CH3143	Advanced Thermodynamics	4
Responsible Instructor	A.L. Smith	
Instructor	Dr.ir. M.A. van der Veen	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	- o Physical chemistry - o Classical thermodynamics (fundamental equation, Gibbs Duhem, relationships between thermodynamic quantities, activity and fugacity coefficients, phase equilibria, equilibrium constants) for students holding a Molecular Science and Technology bachelor degree from TU Delft this means that they need to have passed the course Scheidingstechnologie (4052SCHTEY). - o Basic notions of quantum mechanics - o Calculus and numerical analysis at the bachelor level of chemical engineering	
Course Contents	This is a 4 ECTS course composed of lecture hours, self-study, two to three assignments, and a final examination. Chemical thermodynamics dictates the behaviour and performance of materials and processes for a wide range of applications, and is central to many fields of relevance for chemical engineers. This course will address advanced macroscopic thermodynamics and will introduce statistical thermodynamics that makes the link between the atomic (molecular) and macroscopic scales. It will cover both the thermodynamics of materials (Part I) and thermodynamics of process engineering (Part II). Part I: Thermodynamics of materials The first part of the course will introduce key experimental techniques used for the determination of thermodynamic properties and phase diagram equilibria, including calorimetry, mass spectrometry and laser heating techniques. While classical thermodynamics and its experimental methods deal mostly with the macroscopic scale and are well adapted for condensed matter (solids and liquids), they are not always well suited for the determination of thermodynamic properties of gases, which require knowledge of the internal energy at the atomic (molecular) scale and the use of statistical methods. This course will thus also introduce the basis of statistical thermodynamics and statistical-mechanical evaluation of thermodynamic properties (e.g. heat capacity, entropy) of gases. Furthermore, more general thermodynamic modelling methods based on Gibbs energy minimization will be introduced, as well the development of multi-component thermodynamic databases, that integrate the experimental and statistical information as input for their development and optimization. The understanding and proper use of such databases is key to predict the performance of modern materials. Part II: Thermodynamics of process engineering The second part of the course bridges the gap between the theory of classical thermodynamics and the practice of chemical and process engineering. Reliable values of properties of fluids are necessary to design an industrial process, such as separation units and reactors. Often experimental data are limited, and estimations and predictions are necessary. This part of the course will focus on estimation methods, and, this is key, expects the student to achieve the ability to critically assess the reliability of their estimated values. The focus is on PVT and thermodynamic properties of gasses and liquid, as pure component and mixtures, and their phase equilibria. Many of these methods rely heavily on empirical formulations, but the most reliable ones often have a theoretical basis. As the physical properties of every substance are based on their molecular structure, often such theory is molecular based. Where relevant, such theory will be covered. The course will first focus on pure components properties, low pressure vapour-liquid equilibria, high pressure vapour-liquid equilibria and reaction equilibria. The course will cover estimations of vapour pressure, heat of vaporisation, activity and fugacity coefficient models, equations of state, enthalpy and entropy of formation at relevant temperature and pressures. The course will finally stress the importance and use of theoretical and experimental data, as well as thermodynamic databases in various fields of relevance for chemical engineers through practical assessments and case studies.	
Study Goals	1. Students have basic knowledge of key experimental techniques to measure a given thermochemical or thermophysical property of materials. 2. Students can explain the basic concepts of statistical thermodynamics. 3. Students can relate bulk macroscopic properties to microscopic properties such as intermolecular interactions. 4. Students can perform basic calculations/derivations necessary to use statistical thermodynamics as a tool. 5. Students have basic knowledge and skills on the application of statistical thermodynamics in various fields of relevance for chemical engineers. 6. Students have basic knowledge of the methods used to build thermodynamic databases of materials and are able to use them while using proper conditions/hypotheses for the calculations. 7. Students can apply and critically assess estimations method for PVT and thermodynamic properties relevant for the chemical and process engineer. They understand to which extent these methods are based on theory, and in which conditions they are proper to use. 8. Students have a sound knowledge of the underlying thermodynamical framework, and can apply the theory correctly to different contexts of chemical processes. 9. Students have knowledge of and can correctly use thermodynamic databases relevant for the chemical and process engineer.	
Education Method	- o Lectures, tutorials, supported self-study - o Group and/or individual assignments	
Computer Use	Use of personal computer/laptop. Part II can in principle be done with Excel.	
Literature and Study Materials	A teaching assistant will provide relevant output from Aspen Plus needed for the assignments of part II of the course. - o H. Czichos, T. Saito, L. R. Smith, Springer handbook of materials measurement methods (2006), Springer (eBook available from the TU library) - o B. Widom, Statistical mechanics: a concise introduction for chemists, (2002), Cambridge University Press (eBook available from the TU library) - o A. Kroupa, Modelling of phase diagrams and thermodynamic properties using Calphad method Development of thermodynamic databases, Computational Materials Science 66 (2013) 3-13 - o B.E. Poling, J.M. Prausnitz, J.P. O'Connell, The properties of Gases and Liquids, Fifth Edition (2001), McGraw-Hill. - o D.W. Green, M.Z. Southard, Perrys Chemical Engineers Handbook (2018), McGraw-Hill. Only chapter 4: thermodynamics.	
Assessment	The final grade is composed of the weighted average of the group and/or individual assignments (30%) and final exam (70%).	
Permitted Materials during Tests	Calculator	

CH3175	Solid State Materials	4
Responsible Instructor	Prof.dr. L.D.A. Siebbeles	
Instructor	Dr. A.J. Houtepen	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Quantum Chemistry & Physics (e.g. 1st year Leiden/Delft MST course 'Quantum Chemie en Fysica' (4051QCHFYY)).	
Course Contents	Topics: Material structure and defects Chemical bonding and electronic structure of molecules and solids Optical & electrical properties Materials for electrochemical cells or batteries Zeolites and metal organic frameworks for catalysis	
Study Goals	After successful completion of this course you will be able to describe and understand: - the relation between composition and crystal structure of materials - the origin and effects of different kinds of defects due to atomic dislocations and impurities - atomic bonding and molecular electronic energy (quantum) levels in a molecule on the basis of atomic orbitals - the electronic band structure in solids on the basis of atomic orbitals - the relation between the electronic band structure and the electrical and optical properties of metals, semiconductors and insulators - the absorption of light by electronic excitation in molecules and solid state materials, followed by motion of the electronic excited state and final decay by light emission or non-radiative processes - the effects of the electronic band structure and scattering of electrons in a material on the electrical conductivity - the basics of electrochemical cells, batteries and fuel cells - ionic transport and intercalation and their relevance for batteries - porous materials such as zeolites and metal-organic frameworks, and their application in heterogeneous catalysis This course is useful for follow-up electives on: - Computational Materials Science - Chemistry and Physics of Solar Cells - Electrochemistry for Renewable Energy - Heterogeneous Catalysis	
Education Method	Lectures and classroom exercises	
Computer Use		
Literature and Study Materials	Book: Solid State Materials Chemistry, by P.M. Woodward et al., Cambridge University Press (2021). https://www.amazon.nl/Solid-State-Materials-Chemistry-Evans/dp/0521873258/ref=asc_df_0521873258/?tag=nlshogostdde-21&linkCode=df0&hvadid=494649812806&hvpos=&hvnetw=g&hvrnd=12293455246517546115&hvpone=&hvptwo=&hvmqmt=&hvdev=c&hvdvcmdl=&hvlocint=&hvlocphy=1010704&hvtargid=pla-978713652170&psc=1	
Assessment	Written exam	
Permitted Materials during Tests		

CH3373	Soft Materials Engineering	4
Responsible Instructor	Dr. E. Mendes	
Instructor	Prof.dr. S.J. Picken	
Instructor	P. Boukany	
Contact Hours / Week x/x/x/x	x/6/x/x	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basic knowledge on physical-chemistry	
Course Contents	This course considers the relation between soft material composition and structure and how they can be used or modified to take part in various applications. The course provides core competence for students aiming at either a R&D-industry career or an academic one. In this course you will learn the foundations underlying the behavior of soft matter, (i.e. polymers, self-assembly structures). Fundamental knowledge is then related to advanced applications in various fields ranging from sensing, coatings, to bio-medical and health care materials as well as materials for energy such as fuel cell membranes. An introduction on the theories describing the structure and dynamics of soft materials will be given as well as how those properties can be determined experimentally. The topics covered in the course provide an introduction to structures, energies and timescales typical of soft condensed matter. Supra-molecular structures and material properties arising from those interactions will then be considered. Those relations are in turn, illustrated with high-end applications and you will be thought how the control of various kinds of molecular interactions can lead to desired advanced material properties and structures. Some specific topics: Molecular forces and binding energies in Soft Materials Mechanical properties of soft materials (viscoelasticity) Polymer dynamics Gels and rubbers Rheology techniques for studying soft materials Glass transition Polymer mixtures: phase diagram and phase separation Self-assembling in solution Self-assembled nano materials High Tech Topics on Soft Materials Engineering (considered in guest lectures or in group assignments - examples: gels for tissue engineering, responsive coatings, ...)	
Study Goals	After completing this course, the student can: Knowledge: 1) explain with your own words basic concepts of soft materials; 2) explain how different experimental techniques can be applied to determine the structure, dynamical response and properties of a soft material; Understanding: 3) identify various kinds of intermolecular interactions which determine the structure and dynamics of soft materials at various length scales; 4) explain how structure and dynamics of soft materials (e.g. polymers, self-assembled systems) can be used for a given product or application; Analysis (critical thinking): 5) interpret quantitatively (characterization) experimental data on soft materials; 6) calculate basic quantities related to structure and dynamics of soft materials; Design: 7) identify how specific properties of a given soft material determine or influence the process method chosen for a given material application 8) describe various ways of tuning material properties for desired performance in different applications.	
Education Method	Lectures Exercises Group assignment per duo of students (either seminar or report)	
Literature and Study Materials	Book: Soft Condensed Matter Oxford Master Series in Condensed Matter Physics, Vol. 6 Richard A. L. Jones Course slides Readers to support specific points Papers and / or guest lectures (topics: High Tech materials (i.e. composites), Soft Materials in Health, Soft Materials in Energy)	
Assessment	(80%) Written exam (20%) Assignment (report or presentation per group of 2-3 students)	
Permitted Materials during Tests		

CH3682A	Reactor & Kinetics (R&K)	4
Responsible Instructor	Dr.ir. J.R. van Ommen	
Instructor	Prof.dr. A. Urakawa	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	BSc level Chemical Reaction Engineering; Numerical Methods, Python. Note: it is absolutely required to have a good working understanding of BSc (undergrad) level chemical reaction engineering, thermodynamics and transport phenomena.	
Course Contents	Kinetics - Homogeneous reactions: Equilibrium and kinetics - Heterogeneous reactions: Adsorption, surface reactions, kinetics - Catalysis: Transport limitations, effectiveness, heat effects, catalyst deactivation Reactors - Brief review of conversion and selectivity in ideal reactor types - Energy balance and isothermal/non-isothermal reactors - Non-ideal reactors and residence time distribution - Fixed-bed reactors: Transport limitations - Advanced reactors such as fluidised-bed reactors	
Study Goals	This course deals with experimental and theoretical aspects of kinetics and reactor theory. After the course, the student will be able to: - formulate kinetic models for mechanisms of complex reactions - interpret kinetic studies and use reactor measurements to validate such models - describe and model complex reactors with diffusion, dispersion and heat effects - propose reactor designs based on kinetic insights	
Education Method	Lectures, homework (theoretical and computer based, often use of Python).	
Computer Use		
Literature and Study Materials	Chemical Reactor Analysis and Design Fundamentals, James B. Rawlings & John G. Ekerdt, among others + Course Handouts	
Assessment	The final grade is based on: * the final written exam (70%) that is based on theory (no coding/programming); * results of multiple exercises/homeworks that involve analytical problems and programming (30%).	
Permitted Materials during Tests		

Year	2022/2023
Organization	Applied Sciences
Education	Master Chemical Engineering

Chemical Engineering Electives 2022

AP3171	Advanced Physical Transport Phenomena	6
Responsible Instructor	Dr.ing. S. Kenjeres	
Instructor	Dr. B. Bera	
Contact Hours / Week x/x/x/x	0/0/6/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Fundamental Physical Transport Phenomena	
Course Contents	Analytical/Numerical/Modelling Aspects of Advanced Physical Transport Phenomena (Fluid Flow, Heat Transfer and Turbulence); 1. Basic Equations of Transport Phenomena - Field Description; 2. Mathematical Methods for Solving Transport Equations (PDE, separation of variables, eigenfunctions and eigenvalues, Bessel functions, Laplace transformation, Error-Gamma functions, integral methods); 3. Transport in Stagnant Media (diffusion, moving front problems, diffusion with source terms); 4. Momentum Transport (potential flows, creeping flows, boundary layers); 5. Transport in Flowing Media (stationary transport in flows with uniform velocity, heat transfer in laminar pipe flow, natural convection); 6. Numerical Heat and Fluid Flow (discretization methods for heat conduction, convection and diffusion; differencing schemes, numerical diffusion; steady and time-dependent convection and diffusion; calculation of flow field/velocity-pressure coupling, SIMPLE algorithm); 7. Turbulence: Some Features and Rationale for Modelling (some generic types of turbulent flows and convective processes, wall-bounded turbulent flows: velocity and temperature distributions/wall functions, Reynolds decomposition, RANS); 8. Turbulence Modelling (closure problem, eddy viscosity/diffusivity models, k-e model, other two-equation eddy-viscosity models).	
Study Goals	1. to be able to identify and to mathematically define particular physical mechanisms of the complex transport phenomena; 2. to be able to specify and to analytically solve characteristic PDEs describing simplified transport phenomena; 3. to be able to discretise the system of governing transport equations by using a finite volume method by performing term-by-term analysis (time-dependent, diffusive, convective, source/sink terms) for one-, two- and three-dimensional generic cases; 4. to be able to understand basic mechanisms of turbulence and to derive the transport equations for fluctuating field variables (Reynolds-decomposition); 5. to learn characteristic classes of the turbulence-models (zero-, one-, two-equations, full-stress models); 6. to be able to computationally perform some basic generic cases of flow, heat and mass transfer (CFD) (channel flow, back-step flow, differentially heated enclosure, in both laminar- and turbulent-regimes).	
Education Method	Combination of Lectures (4 Lectures per week, covering theoretical aspects) and practical exercises (both analytical and computational/computer exercises, 2 hours per week).	
Computer Use	Computational exercises covering step-by-step solving some of basic generic cases of flow, heat and mass transfer on Linux computers (mesh-definition, specification of the boundary and initial conditions, solving, postprocessing, analysis and critical assessment of results).	
Literature and Study Materials	1. Book: "Analysis and Modelling of Physical Transport Phenomena", Hanjalic K., Kenjeres S., Tummers M.J., Jonker H.J.J., VSSD Book, ISBN-13 978-90-6526-165-8, Second Edition, December 2009. 2. Book: "Transport Phenomena", Bird R.B., Stewart W.E., Lighfoot E.N., 2nd edition, Wiley (2002) 3. Book: "Fysische Transportverschijnselen II", Hoogendoorn C.J. and van der Meer, Th.H., Delftse Uitgevers Maatschappij (1991) 3. Handouts: For computational/computer exercises a reference manual and quick start manuals will be provided.	
Assessment	Written Exam (two times per year). The exam contains three distinct parts: (1) analytical; (2) numerical discretization; (3) turbulence modeling. At least 50% of each part needs to be completed correctly to successfully pass this course.	
Permitted Materials during Tests		
Elective	Yes	
Tags	Fluid Mechanics Physics Transport phenomena	

AP3252	Electron Microscopy Characterization of the Nanoscale	3
Responsible Instructor	S.C. Conesa Boj	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge		
Course Contents	The current revolution in nanoscience lies in the possibility to fabricate smaller and smaller structures (nanoscale level), together with the ability to tailor their specific set of properties. One of the factors responsible for this progress has been the improvement of old and the introduction of new instrumentation for characterizing nanostructures. In this sense, Electron Microscopy has revolutionized our understanding of materials by complementing the processing-structure-properties links down to atomic level thanks to the recent breakthroughs. The aim of this course is to present the principles behind the operation of electron microscopes and to delineate their different capabilities within the various imaging modes. Based on that, by the end of this course the students should be able to answer two fundamental questions: 1. Why should we use a particular electron microscopy technique? 2. How do we put this specific technique into practice? To answer the first question, we will establish first of all a solid theoretical foundation. And then we will use this knowledge to answer the second question by explaining operational details in a generic sense.	
Study Goals	* To describe the different TEM imaging modes, their advantages and limitations. * To extract and interpret quantitative structural information from TEM images. * To interpret the crystalline structure of the specimen from the TEM diffraction patterns in terms of Bragg reflections. * To associate TEM measurements with complementary data analysis techniques such as atomic modeling and TEM image simulation tools.	
Education Method	Lectures, expert lectures, examples of TEM problem-solving, practical sessions in the laboratory, and student presentations.	
Literature and Study Materials	Transmission Electron Microscopy. M.D.B. Williams, C.B. Carter. Plenum Press, New York, (1996). ISBN: 0-306-45247-2. In addition, Lecture handouts will be added to the Course documents folder after the lectures.	
Assessment	The assessment is performed through a combination of quizzes (10%), problem-solving activities (30%), laboratory reports (30%), and final presentation (30%).	

AP3352	Introduction to Nuclear Science and Engineering	6
Responsible Instructor	Dr.ir. M. Rohde	
Instructor	Dr. L. van Eijck	
Instructor	Dr.ir. M.C. Goorden	
Instructor	M. Schouwenburg	
Instructor	Dr. H. Schut	
Instructor	Dr.ir. A.G. Denkova	
Instructor	A.L. Smith	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge		
Summary	This course is designed as an introduction for Applied Physics, Biomedical Engineering, Chemical Engineering and Sustainable Energy Technology students to the broad range of topics that comprise nuclear science. These include; radioactive decay, radiation dosimetry, neutron and positron beams, nuclear reactor physics and designs, nuclear waste disposal, radioactivity in health sciences, and more. This Introductory course highlights the annotation Nuclear Science and Engineering; available as a separate certificate that accompanies the university diploma. The course centers on teaching the fundamental concepts that are necessary to move forward with a more in-depth exploration of these topics. As such, this course draws on faculty and staff from all the sections within the Department of Radiation Science and Technology (RST). Students should complete this course with a greater understanding and appreciation for the relevance of nuclear science and technology in today's global society.	
Course Contents	Subjects include (tentative): - The history of radioactivity - Modes of decay / de-excitation - Interactions of radiation with matter - Radiation dosimetry - Applications of research reactors - Neutron beams - Neutron scattering techniques - Positron beams - Nuclear reactors - Nuclear waste disposal - Health application of nuclear science - Medical imaging - Radiotracers	
Study Goals	Sub-goal (total % for sub-goal) : Weekly exercises(10%)/Exam(75%)/Presentation(25%)* Has insight into the nature of radiation and radioactivity and its interaction with matter (37%) : 5%/37%/0% Understands, in a broad sense, how nuclear science is/will be applied in fields of energy, health, medicine industry and others (38%) : 5%/38%/0% Is able to read, understand and present a scientific paper on a nuclear science subject (20%) : 0%/0%/20% Is able to collaborate with another student, while preparing the presentation (5%) : 0%/0%/5% * The sum is larger than 100% to indicate that the weekly exercises are not compulsory.	
Education Method	Oral lectures, guided tours (if possible) and a presentations given by the students on a subject related to nuclear science.	
Computer Use	Some homework exercises may require basic knowledge on Matlab, Python or Fortran (or any other programming code)	
Literature and Study Materials	J. Kenneth Shultis, Richard E. Faw: Fundamentals of Nuclear Science and Engineering Third Edition (available as e-book. Link will be provided on Brightspacd); Lecture slides.	
Assessment	10 (bi)weekly homework exercises, a written exam + a presentation of a scientific paper. Each (bi)weekly homework exercise deals with one to three lectures (2 to 6 hours) on a specific subject. The homework exercises are meant to train you for the written exam. A bonus can be obtained (B) if the student has met certain requirements (i.e. pass for a certain number of homework assessments). All students give a presentation in couples about a paper that is related to an NSE subject. The presentation should contain sufficient physics/chemistry (hence, no general/managerial talk). Grade is P. The exam is a written exam. Questions are similar (but not the same!) to the (bi)weekly tests. Grade is E. The final grade is calculated as $G = (P + 3E)/4 + B$ The grade for the presentation can be re-used for the resit in the same academic year. If you fail for this course (either with the exam or resit), you need to redo everything in the next academic year.	
Permitted Materials during Tests	During the tests and exam: calculator + pens	
Elective	Yes	

CH3061	Multiphase Reactor Engineering	4
Responsible Instructor	Dr.ir. J.R. van Ommen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The student is expected to: 1. have followed a course on mass and energy balances (such as Molecular Transport Phenomena) 2. have followed a basic course in reactor engineering (such as Reactors and Kinetics) 3. is able to work with Python (or similar software; e.g., has followed a course on numerical techniques)	
Course Contents	Multiphase reactor types: Fixed beds, trickle beds, fluidized beds, bubble columns, slurry reactors, microreactors, structured (monolith) reactors, electrocatalytic reactors. Engineering aspects: Flow regimes, mass transfer, conversion models. Hatta number, reaction enhancement. Residence time distribution. Catalytic reactors structured in time and space. Unifying concepts. Design aspects: reactor selection, decoupling of kinetics and transport phenomena, combination of reaction and separation, reaction coupling, separating catalytic steps, controlled energy input, circularity of materials.	
Study Goals	1. The student is able to describe for each of the archetypes of multiphase reactors, the various embodiments with practical relevance. 2. The student is able to use the scientific literature to find relevant information for the various reactor types. 3. The student is able to analyse a multiphase reactor and to describe the physical and chemical process steps in words. 4. The student is able to explain the basic measurement techniques used in studying multiphase reactors. 5. The student is able to write down the mass, energy, and momentum balances for the overall system and the steps identified under objective 3. 6. The student is able to implement customary models in the balances mentioned under 5, such that a system of equations describing the reactor is obtained. 7. The student is able to simplify the system of equations based on a quantitative analysis and identification the rate-determining steps. 8. The student is able to use current software packages to solve the systems of equations numerically for the basic types of reactors. 9. The student is able to use the developed models to analyse defined cases. 10. The student is able to make a quantitative design of a multiphase reactor for a given application. 11. The student is able to present, both orally and in writing, the outcomes of modeling, calculations, and evaluation.	
Education Method	Each lecture will give a short overview of a specific multiphase reactor type (only 45 minutes). The slides available on Brightspace. You will get references to papers and/or chapters from textbooks that give more information about the reactor type. These will be made available via Brightspace. After the lecture, an assignment (about 8 hours work) will be available via Brightspace for the specific reactor type treated. These assignments are made in teams of four students; this number may be changed depending on the number of participants. You are free to choose which software package you want to use for making the assignments, but Python is recommended. After two weeks, each team hands in a detailed solution of the problem on paper the day before the discussion before the announced deadline. Moreover, each team prepares a summary of the solution for presentation in PowerPoint. Please bring this presentation on a memory stick. Each week, a few teams present their solution; the instructors choose the student that is presenting. The other teams challenge this solution. At the end, each team will make a larger final project.	
Computer Use	You are free to choose which software package you want to use for making the assignments, but Python is recommended.	
Literature and Study Materials	All material is provided via Brightspace.	
Assessment	The final grade consists of three parts: * the solution to the minor assignments * an individual part; the nature of this part will depend on the number of participants * the final assignment Note: the exact format of assessment may change due to Coronavirus (COVID-19) regulations.	

CH3073	Separation Processes, Design and Operation (SPDO)	3
Responsible Instructor	Dr. H.B. Eral	
Contact Hours / Week x/x/x/x	0/0/4/0 (together with the mass transfer lectures of course ME45165)	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Summary	Separation processes are very important in all sectors of modern process industry. A reaction section in a chemical plant is typically surrounded by 3-6 separation units, which perform a variety of functions (e.g. feed concentration, product purification, solvent-recycling, off-gas treatment and water-recovery). Separation processes consume approximately 40% of the energy consumption and 75% of the investment cost in the process industry. Consequently, you are likely to come across the design and/or operation of separation units in your future career. Basic principles and methods for the process and equipment design of the main thermal separation processes in the chemical industry are discussed. Underlying theory on thermodynamics of mixtures and mass transfer by convection and diffusion across interfaces is shortly repeated Characteristics and design methods for vapour liquid, liquid-liquid, solid liquid separation processes or in processes in which mass separation agents and physical barriers are applied are given. Equipment characteristics and design for the mentioned separation processes is shortly discussed.	
Course Contents	Thermodynamics of separation processes, convective mass transfer, mass transfer by diffusion, film theory. Single equilibrium stages and cascades. Examples of separation processes. Basic principles and design methods for equipment used in equilibrium stage separation processes, such as distillation, absorption, stripping (desorption), extraction, crystallization and mechanical separation processes, such as sedimentation, filtration and membrane separations.	
Study Goals	The student is able to make a design thermal separation processes, including the dimensioning of the equipment. More specifically, the student: 1. Can describe the characteristics and principles of thermal separation processes, 2. Can describe and formulate thermodynamics equations and models for the selection and design of thermal separation processes, 3. Can quantify the mass transfer rate over an interface by diffusion and convection, 4. Knows the role of mass transfer in the dimensioning of equipment for separation processes 5. Can determine the separation efficiency for single and multi-stage separations and can quantify the influence of the process conditions and the contacting modes on the separation. 6. Can describe principles and design methods for processes and equipment used in equilibrium stage separation processes, such as distillation, absorption, stripping (desorption), extraction, crystallization and membrane separations 7. Can use computer based process simulation tools to analyse and design separation equipment 8. Can design and analyse thermal separation process including dimensioning of the equipment.	
Education Method	Lectures, assignments	
Course Relations	This course is not recommended for students that have passed the MST Bachelor course Separation Technology (4052SCHTEY).	
Books	1) Industrial Separation Technologies: Fundamentals Second Edition, de Guyter by A.de Haan, B.Schuur and H.B. Eral, 2020 2) "Separation Process Principles", J.D.Seader & E.J.Henley, John Wiley & Sons, 2nd Ed., 2006 or 3rd Ed., 2011	
Assessment	The developed knowledge and acquired skills are tested by means of homework assignments, which can be worked out in groups of two students. Upon passing all assignments, an written examination on the obtained separation technology knowledge finalizes the course. Note: the exact format of assessment may change due to Coronavirus (COVID-19) regulations.	

CH3082	Chemical Technology	3
Responsible Instructor	Dr. A.B. Bansode	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	The essentials of process technology are the integration of various disciplines, ranging from chemical engineering, chemistry, catalysis to biotechnology. The course builds an essential bridge between the chemical sciences and the chemical industry. The course is aimed at developing fundamental understanding of actual industrial processes by studying the chemical and technological concepts and unit operations to build the basis for conceptual process design. Topics. Oil refinery- physical, thermal, and catalytic processes in oil refinery Production of lower alkenes- Industrial steam cracking, product processing, novel developments in lower alkenes production Syngas production steam and autothermal reforming of natural gas, coal gasification, cleaning and conditioning of syngas Bulk chemicals & synthetic fuels- ammonia production, Fischer-Tropsch process, methanol to gasoline, fuel additives production Industrial homogeneous and heterogeneous catalysis- reactor types and characteristics, hydroformylation, shape selectivity Biomass and biotechnology- production of biofuels and bio-based chemicals, fermentation technology, cell biomass, enzyme technology Fine chemicals- atom economy, role of catalysis, production plants and production campaigns, batch reactor selection	
Study Goals	The student should be able to use the gained knowledge in the analysis of existing processes and in the design of new conceptual conversion systems. At the end of the course students should be able to: - Identify major chemical conversion processes & steps involved in raw materials conversion. - Describe the structure of chemical processes and important unit operations. - Explain the different conversion routes for raw material conversion. - Apply the technological insights to design/adapt the conceptual conversion systems.	
Education Method	Lectures, home-work questions, assignments	
Computer Use	No	
Literature and Study Materials	Chemical Process Technology second edition by J.A. Moulijn, M. Makkee, and A.E van Diepen, Publisher John Wiley @ Sons Ltd. ISBN 978 1 444 32025 1 Lecture slides	
Books	Chemical Process Technology second edition by J.A. Moulijn, M. Makkee, and A.E van Diepen, Publisher John Wiley @ Sons Ltd. ISBN 978 1 444 32025 1	
Assessment	Final Written exam(80%)+ Assignment (20%)	

CH3101	Heterogeneous Catalysis for Chemical Engineers	3
Responsible Instructor	Prof.dr. E.A. Pidko	
Instructor	Prof.dr. A. Urakawa	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge		
Course Contents	History and relevance of Catalysis; Catalysis and sustainability; Catalysis toolbox; Catalyst Preparation; Catalyst Performance Testing; Mechanism of Catalysis; Reaction Kinetics; Property - Performance relations; Texture and Morphology; Active Site Determination; Characterization (Infrared Spectroscopy, Temperature Programmed Techniques, X-Ray Diffraction, Electron Microscopy, and X-Ray Photoelectron Spectroscopy).	
Study Goals	Bring to life the black box that the catalyst often is to Chemical Engineers.This means gaining knowledge of: a. How catalysts work; b. How they are prepared and what happens to them in the reactor; c. What the key characteristics of catalysts are; d. How they are prepared, tested and evaluated; e. Which techniques are used to characterize catalysts.	
Education Method	Lectures, assignment, poster presentations (during plant visit or at TU Delft)	
Literature and Study Materials	Catalysis and Catalysts by J.A. Moulijn, F. Kapteijn, and G. Mul, Handouts Lectures	
Prerequisites	Bachelor Molecular Science and Technology; major Technology.	
Assessment	Written examination (80%) + assignment (20%)	
	Note: the exact format of assessment may change due to Coronavirus (COVID-19) regulations.	

CH3222	Energy Storage in Batteries	4
Responsible Instructor	Dr.ir. M. Wagemaker	
Instructor	Dr. E.M. Kelder	
Instructor	Prof.dr. F.M. Mulder	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Efficient energy storage in batteries is of major importance for electronic equipment, electrification of cars and buses, grid stability and for the introduction of fluctuating renewable energy sources including wind and solar. In this course we will learn about the thermodynamics and charge kinetics in batteries including the electrochemical driving force, thermochemistry, phase transitions and charge transport kinetics. The principles of state of the art and future rechargeable batteries, also touching super capacitors, will be discussed. The aim is to develop understanding of the battery performance parameters and to relate these with the with underlying atomic scale material properties. Along these topics current battery developments and research directions will be discussed. After seven lectures on the fundamental properties of batteries six dedicated working lectures will focus on important battery technologies including: Redox Flow Na-ion next generation Li-ion batteries Solid State batteries Supercapacitors	
Study Goals	At the end of the battery lectures students should be able to: - List the major types of Li and other batteries and their characteristics. - Reproduce the major challenges in batteries. - Explain how a battery works and the functioning of each component. - List and explain the different reaction types in batteries. - Explain the different charge transfer processes in batteries. - List and explain the different mechanisms that limit the cycle life of batteries. - Apply Nernst Law describing redox reactions. - Calculate conductivities and overpotentials. - Apply the Tafel equation describing the charge transfer between electrode and electrolyte. - Calculate energy densities and power densities from observed properties. - Derive from a phase diagram the different phases that will occur upon lithiation. - Apply the Gibbs Phase rule to predict constant of variable output voltage. - Derive the voltage curve from the shape of the Gibbs Free energy. - Predict the basic redox reaction of two electrodes given the formation energy or the standard potentials. - Compare different battery systems and differentiate. - Discuss scientific results observed in battery performance and to relate this to material properties.	
Education Method	The course contains fourteen 2-hour lectures. The first seven 2-hour lectures focusses on building the underlying thermodynamics, material science and electrochemistry chemistry and the principles of energy storage in batteries. This knowledge will be used to develop understanding of the material properties that determine performance of current and next generation battery. In the second seven lectures will be working lectures with focus on six important battery technologies. The first half hour will be a lecture on the specific battery technology followed by half an hour assignment. In the final half hour of the working lecture the assignment will be discussed. In the final lecture example exams will be discussed.	
Course Relations	This course is a core course in the Sustainable Energy Energy Storage profile.	
Literature and Study Materials	R.A. Huggins: Advanced Batteries. Available via the library as pdf's: http://www.springerlink.com/content/nq6404/?MUD=MP Slides Articles	
Assessment	Written exam (100% of the final grade)	
	Note: the exact format of assessment may change due to Coronavirus (COVID-19) regulations.	

CH3382	Molecular Engineering of Soft Materials in Health Care	4
Responsible Instructor	Dr. A.Y. Rwei	
Instructor	Dr. R. Eelkema	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course provides an overview of the design, analysis and engineering principles for soft materials in health applications on a molecular scale. Such soft materials cover a wide range of health applications, including drug delivery, imaging, tissue engineering, and implants. We will cover practical knowledge in the engineering and advancement of these biomaterials that enables their use in biological tissues, including the properties of materials (e.g., biocompatibility and biodegradability), the biological response associated to them (e.g., inflammation and clearance), and the practical concerns during development and commercialization (e.g., safety and quality control). Case studies of health applications (e.g., biomaterials for COVID vaccination, artificial organs, tumor imaging and diagnosis) will be covered in this course to further integrate your knowledge into real-world applications.	
Study Goals	After successful completion of this course, the student should be able to: -understand the general properties required for healthcare materials -understand the interaction between living systems and soft materials -explain the process of biodegradation -explain the lab-to-commercialization process of biomaterials -identify the applications of biomaterials (e.g., drug delivery, tissue engineering, imaging, implants) -evaluate the safety of biomaterials -design biomaterials based on their application in health	
Education Method	Standard 2-hour lectures are followed by self-study sessions	
Assessment	Written report on case studies	

CH3421	Computational Transport Phenomena	6
Responsible Instructor	Prof.dr.ir. H.E.A. van den Akker	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	MSc level Transport Phenomena: Applied Transport Phenomena for Chemical Engineering students, or Advanced Transport Phenomena for Applied Physics students	
Summary	This elective course Computational Transport Phenomena is complementary to Molecular Transport Phenomena (about the fundamental processes at molecular level) and Applied Transport Phenomena (how to scale and reduce the model equations and introduce techniques to get approximate solutions which still reflect the relevant mechanisms and processes), all of the MSc program Chemical Engineering. For Applied Physics MSc students of the Physics for Fluids Engineering track, this course builds on the modules Advanced Fluid Dynamics and Advanced Physical Transport Phenomena. Computational Transport Phenomena is about numerical techniques for solving the Navier-Stokes and other transport equations - all being partial differential equations which do not have an exact analytical solution. The course is intended to provide some guidance as to a number of numerical issues and on simulation strategies such as size and type of the computational grid (including 2D vs 3D). The course, however, is not just a tutorial as to how to run Fluent for a variety of applications. Most of the course is devoted to the wide variety of models available for simulating turbulent flows, multi-phase flows (including free-surface flows) and micro-fluidics. If time permits, simulating chemical reactions will also be discussed. While most of the course will deal with finite volume techniques, as used in the common commercial solvers and OpenFoam, attention will also be paid to Lattice Boltzmann techniques.	
Course Contents	An introductory course on Computational Fluid Dynamics (CFD) and Computational Transport Phenomena (CTP): elementary fluid mechanics; numerical and computational aspects; turbulence & turbulence modelling; RANS vs Large-Eddy Simulations; operations and mixing & transport processes in process equipment among which cyclones; fluid-particle interaction; two-phase flows (Euler-Lagrangian vs Euler-Euler); chemical reactions	
Study Goals	Building an understanding of CFD, its promises and its limitations; acquire some experience in numerical and computational exercises; understanding turbulence and multiphase flows and their modelling; getting a good idea about the effects of all types of models on the CFD outcomes; becoming capable of interpreting and assessing CFD results	
Education Method	The course comprises regular classes (where new material is presented by the Professor) as well as hands-on workshops for exploring ANSYS/Fluent CFD software in a Studio classroom type of environment, with the help of Fluent tutorials, a TA being present for assistance. In these workshops and also in the final assignment, students work in teams of two. The emphasis in the course is on the physics and the models underpinning a software code such as Fluent as well as on the impact of boundary conditions, input parameters and numerics. Make sure you will have sufficient time during the semester to be an active participant in the course and to stay on board till the very end. You are also obliged this towards your fellow team member. Attending CLASS AND WORKSHOP is therefore COMPULSORY.	
Assessment	The grading is based on two elements: (a) the active participation in the mandatory weekly tutorials (counts for maximum 10% of the final grade), and (b) the grading of the report on the final assignment (counts for minimum 90% of the final grade). The latter report should make clear that the student has understood the course material, at least as far as relevant to the final assignment. This report should be a critical assessment (using literature data) of the simulation results carried out for the final assignment. A more detailed scheme of aspects contributing to the eventual grading of the final report is to be found on Brightspace. The effort in writing such a report is not to be underestimated! Students are allowed to improve their grades on the basis of the review by the teacher but cannot improve their grade by more than 1.5 on the scale 1-10. The lecturer may invite students for an oral assessment of their final assignment report.	
Remarks	This course will only be given if at least 10 students will participate. The lectures will be given by Professor Van den Akker, on Monday afternoons.	
Elective	Yes	
Tags	Fluid Mechanics Modelling Numeric Methods	

CH3513	Electrochemistry for Renewable Energy	4
Responsible Instructor	D.A. Vermaas	
Instructor	Dr. A.J. Houtepen	
Contact Hours / Week x/x/x/x	0/0/6/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Thermodynamics (e.g. 4051CHTHEY) and standard chemistry background (e.g., BSc education MST or similar)	
Course Contents	Electrochemistry provides an important platform to convert (renewable) electricity into valuable products, chemicals, fuels and coatings. This course provides an overview of several electrochemical processes of societal importance, linking fundamental understanding with applied utilization. The scope of the course will cover processes from the nanoscale to the meter-scale, and methods to understand fundamental phenomena in practical systems. This course will include: - Electrochemical potentials and definitions (electromotive force, formal potential, Donnan potential) - Types of electrochemical reactions (reversibility, multi-step reactions) - Electrochemistry at liquid-solid, liquid-liquid, solid-solid and gas-solid interfaces - Electrode reaction kinetics in steady and unsteady state (Cottrell equation) - Microscopic theories of charge transfer and (Marcus model, energy states, tunneling) - Ion transport through diffusion and migration - Electrochemical characterization techniques (cyclic voltammetry, chronopotentiometry, impedance, etc.) - Electric double layers (Helmholtz model, Gouy-Chapman model) - Examples of water electrolysis, CO₂ reduction - Industrial electrochemical systems (large scale electrolyzers, batteries, etc.) - Electrochemical equipment (potentiostats, reference electrodes, electrodes, electrolytes, membranes) - Experimental setup (including practicum)	
Study Goals	After this course, the student is able to: - Explain electrochemical responses in complex electrochemical cells (steady and non-steady state, and mono- or multi-ionic systems) to peers; - Demonstrate how theories for electrochemistry thermodynamics, kinetics (Marcus model, Tafel model) and electrolyte processes are applied to basic redox systems - Break down limiting factors (overpotentials or limiting currents) in given practical electrochemical conversions - Set up and perform suitable electrochemical tests for characterizing redox couples in practical electrochemical conversions	
Education Method	This course consists of: Instruction videos, Lectures and exercise sessions (combined), Practicum, Guest lectures	
Literature and Study Materials	Bard & Faulkner, Electrochemical methods fundamentals and applications	
Assessment	Grading is based on: - Homework assignments (0%, mandatory) - Practicum report (30%) - Exam at the end of the course (70%)	
Permitted Materials during Tests	Calculator (no graphical) Formula sheet	
Elective	Yes	
Continuing Courses	Possible follow-up courses are: CH3222: Energy Storage in Batteries (Q3) ME45203: Electrochemical Energy Storage 2: Analytical Modelling (Q4) MSc thesis	

CH3531	Functional Ceramics	3
Responsible Instructor	Dr. E.M. Kelder	
Instructor	Dr. P. Braga Groszewicz	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	BSc: general and inorganic chemistry, basic quantum chemistry	
Course Contents	Ceramic materials show a wide variety in chemical, optical and electrical properties. They are used in many different applications, ranging from catalysts, fuel cells, batteries and sensors to solar cells, photo-catalysts and luminescent nanoparticles. This course will treat the basic properties of functional ceramic materials. The following topics will be included: - crystal structures - bonding in inorganic solids - basic electronic structure and properties of semiconductors - defect chemistry including point defects, non-stoichiometry and doping - ionic and electronic charge transport - semiconductor junctions and space charges.	
Study Goals	After this course, the student is able to: - Correlate crystal structure and band structure to optical and electrical properties of ceramic materials - Describe the operating principles of simple devices based on semiconductor junctions (solar cells, diodes, photo-electrodes) and ionic conduction (Nernst sensors, (Li-ion)-batteries, solid oxide fuel cells (SOFCs), and catalysts) - Identify and predict relationships between point defects, dopants, space charges, and ionic/electronic conduction in both semiconductors and ionic conductors, and perform simple calculations	
Education Method	Lectures, self study, instruction classes	
Books	- "Physical Ceramics" by Y-M Chiang, D. P. Birnie, and W.D. Kingery, John Wiley & Sons, ISBN: 0-471-59873-9	
Reader	- Several handouts and the lecture slides will be made available on Brightspace	
Assessment	It is foreseen that the exam will be a closed book exam. Examples have been posted on Brightspace. It is stressed that the situation with respect to COVID-19 may change, and an open book exam may thus be expected instead.	
Permitted Materials during Tests	- Calculator, closed book (overview of most important equations will be handed out during exam, the reference sheets)	

CH3542	Inorganic Materials	3
Responsible Instructor	Prof.dr. E.A. Pidko	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	BSc: general chemistry, physical chemistry, thermodynamics, basic concepts of chemistry and molecular structure, chemical kinetics	
Course Contents	Introduction - The technology of materials - Recap: periodic table and periodic trends in chemistry - Types of matter: structure and bonding Inorganic Molecules - Coordination complexes and Organometallic chemistry - Inorganic Polymers Solid State Chemistry - Crystal Chemistry - Amorphous Solids and Icosahedral Phases - Introduction to structural and functional ceramics - Intermetallics - Intrinsic properties of solid state materials Solid State Reactions - Types of reactions of solids and kinetics of solid state reactions - The chemistry of ceramics and sintering Surface Chemistry of Inorganic Materials - Chemistry and physics of inorganic solid surfaces - Heterogeneous catalysis - Inorganic colloids - Growth of crystalline solids from liquids - Coordination chemistry and organometallic chemistry of inorganic surfaces Synthesis of Inorganic Materials - Solid-state processes - Synthesis from solutions and melts - Sol-gel method - Gas-phase techniques - Synthesis and modification of inorganic polymers / MOFs Design of Functional Inorganic Materials - Introduction to materials design - Combination of properties of composites - Functional materials (active materials, sensors, etc) - Nanostructured materials - Heterogeneous catalysis	
Study Goals	After this course, the student: - Has a broad knowledge of the principles and concepts of contemporary inorganic chemistry of materials - Can discuss and define the chemical origin of specific functionality of the inorganic materials	
Education Method	Lectures, self-studies, assignment	
Assessment	Written exam (70%) + Assignment (30%)	
	Note: the exact format of assessment may change due to Coronavirus (COVID-19) regulations.	

CH3564	Particle Technology for Health and Energy	3
Responsible Instructor	Dr.ir. J.R. van Ommen	
Instructor	Dr.ir. G.M.H. Meesters	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	It is useful if you took the course Interfaces & Particles before; if not, you can get up to speed with some limited self-study.	
Course Contents	The course starts with the basics of general particle technology / solids processing. After the introduction (I), we split the group in two parts: one focusing more on health/pharma and food applications of particle technology (H), and one on energy and circularity applications of particle technology (E). You are, of course, free to choose what you prefer. Topics: (I) Basic properties & characterization of particles (I) Particles in a fluid, particle-particle interaction (I) Motivation of applying (nano)particles; difference in behavior between small particles and large particles (I) Particle production & functionalization (H) Pharmaceutical processing (H) Nanoparticle formulation & controlled release (H) Safety, Health & Environment aspects of particles (E) Battery applications of particles (E) Catalysis, critical materials & circularity (E) Large scale energy processes involving particles	
Study Goals	1. The student is able to describe the synthesis, characterization and application of particles. 2. The student is able to use the scientific literature to find relevant information on particles. 3. The student is able to develop and apply model descriptions for particle synthesis, characterization and applications (i.e., both qualitative and quantitative). 4. The student is able to apply the obtained knowledge and skills to analyse defined cases in the area of health and food (H), or energy and circularity (E). 5. The student knows and understands the main applications of particles in the area of health and food (H), or energy and circularity (E).	
Education Method	Lectures, assignments	
Literature and Study Materials		
Assessment	Essay + multiple choice test	
Permitted Materials during Tests		

CH3622	Process Intensification (PI)	3
Responsible Instructor	Dr. A.A. Kiss	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge		
Course Contents	1. Introduction to Process Intensification(PI): - sustainability-related issues in process industry; - definitions of Process Intensification; - fundamental principles and approaches of PI. 2. How to design a sustainable, inherently safer processing plant - presentation of PI case study assignments. 3. PI Approaches: - STRUCTURE - PI approach in spatial domain (incl. "FOCUS ON" guest lecture) - ENERGY - PI approach in thermodynamic domain (incl. "FOCUS ON" guest lecture) - SYNERGY - PI approach in functional domain (incl. "FOCUS ON" guest lecture) - TIME - PI approach in temporal domain (incl. "FOCUS ON" guest lecture)	
Study Goals	Basic knowledge in Process Intensification	
Education Method	Lectures	
Literature and Study Materials	1. Lecture notes via Brightspace. 2. Book "Fundamentals of Process Intensification" by A. Stankiewicz, T. van Gerven and G. Stefanidis, Wiley-VCH 2019	
Assessment	Written exam	
Permitted Materials during Tests	Note: the exact format of assessment may change due to Coronavirus (COVID-19) regulations.	
Continuing Courses	CH3622-P Process Intensification Project	

CH3632	Chemistry and Physics of Solar Cells	6
Responsible Instructor	Dr. A.J. Houtepen	
Instructor	Dr. T.J. Savenije	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge		
Course Contents	The course deals with the working of various solar cell technologies. Traditional (first and second generation) solar cells are based on semiconductors that are prepared by physical methods (usually involving high temperature and/or high vacuum methods). However, various more recent solar cell technologies (so-called third generation solar cells) are prepared via chemical approaches. Examples are dye sensitised solar cells, organic solar cells, colloidal quantum-dot solar cells and perovskite solar cells. These new technologies have received a lot of attention since they can be produced much more cheaply and in some cases offer the prospect of increased efficiency. In this course we will discuss the operation of solar cells in general and will pay particular attention to the chemistry of recent solar cell technologies and the related challenges and opportunities. Understanding the operation of solar cells requires knowledge about the physics of semiconductors. Therefore the first part of the course will focus on the properties of semiconductors, following the book "Semiconductor physics and devices". In the second part of the course this knowledge will be applied to the operation of solar cells. Finally it will be discussed how the same principles that govern the operation of solar cells also apply to applications such as Light Emitting Diodes, Lasers, Photoelectrochemical cells and sensors.	
Study Goals	After this course the student: - Has a basic understanding of the chemistry and physics of solid state materials. - Can explain the origin of electronic bands in semiconductors and the occupation of those band by charge carriers - Can discuss the factors that govern charge transport in semiconductors - Can discuss electronic doping and the formation of junctions - Can interpret the effect of specific junctions on the behavior of charge carriers - Can explain the operation of p-n junction solar cells - Knows various types of solar cell technologies, is able to discuss their respective promises and challenges - Is able to discuss the operation of LEDs, lasers and sensors in terms of the properties of semiconductor materials	
Education Method	Lectures and exercise classes	
Literature and Study Materials	The first part of the course will follow the book "Semiconductor Physics and Devices" by Donald Neamen (4th edition). The second part of course will use handouts and selected scientific papers.	
Assessment	Examination will take place via a written final exam.	
Permitted Materials during Tests	Calculator and formula sheet	
Elective	Yes	
Tags	Chemistry Energy Physics Semiconductors Sustainability	

CH3672	Computational Materials Science	3
Responsible Instructor	Dr. F.C. Grozema	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Basic knowledge of quantum mechanics and electronic properties of materials (For example "Structure property relationships of advanced chemical products" (CH3172) and/or "Chemistry of Solar Cells" (CH3632))	
Course Contents	- Basic knowledge of methods for electronic structure calculations: Hartree-Fock, Density Functional Theory, Electron correlation - Potential energy surfaces and geometry optimizations (vibrations, transition states and chemical reactions) - Electrostatic properties of molecules (dipole moment, charge distribution, polarisability) - Optical properties of molecules: Absorption, fluorescence. - Interactions between molecules: Vd Waals, electrostatic - Electronic (band) structure of solid materials	
Study Goals	- Student has an overview of the existing molecular simulation techniques. - Student has basic knowledge of quantum chemical methods. - Student is able to choose a suitable simulation technique for a specific problem - Student is able to evaluate and interpret the result of a typical simulation	
Education Method	Lectures and extensive hands-on computer exercises	
Literature and Study Materials	Lecturer will provide study material	
Assessment	Assignment (Hands-on computer simulation on practical problem + written report)	
	Note: the exact format of assessment may change due to Coronavirus (COVID-19) regulations.	

CH3763	Nuclear Medicine	3
Responsible Instructor	Dr.ir. A.G. Denkova	
Responsible Instructor	Ir. R.M. de Kruijff	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Nuclear Medicine is responsible for the diagnosis of more than 30 million patients yearly as well as for the treatment of cancer, especially when tumours have spread all over the body. In this course we will cover the basics of NM including production of radioisotopes, preparation of radiopharmaceuticals, imaging modalities and radionuclide therapy. As part of this course several guest lectures are incorporated given by researchers working in Nuclear Medicine departments who can provide insight into real practice in the clinic.	
Study Goals	- Be able to explain which properties of radionuclides are relevant for imaging and therapy - Be able to explain the working principle of nuclear imaging techniques - Be able to provide basic radiolabelling approaches for radiopharmaceuticals - Be able to name and explain different production routes for radionuclides - Be able explain the working mechanism of radionuclide therapy and distinguish between the different radionuclide therapies - Understand the damage of radiation induced to cells	
Education Method	Lectures	
Assessment	70 % case study and 30 % exam. At least a 5 needs to be obtained for the exam to pass the course. Exams will be written closed book on campus if possible. Otherwise open book on line with an oral check. The case study consists of a written part and a presentation. The oral part counts for 25 % (out of 70 %), and the written part counts for the rest, 45 % (out of the 70%).	
	Note: the exact format of assessment may change due to Coronavirus (COVID-19) regulations.	

CH3765	Advanced Materials Analysis by Radiation Instrumentation	3
Responsible Instructor	Dr.drs.ir. L.J. Bannenberg	
Instructor	Dr. S. Ganapathy	
Instructor	Dr. S.R. Parnell	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	This course focuses on the characterization of materials using X-ray and neutron scattering, nuclear magnetic resonance (NMR) and X-ray absorption techniques. During the course, a wide range of experimental techniques will be presented that are relevant for chemists and chemical engineers, such as X-ray/neutron diffraction, reflectometry and small-angle scattering to determine the structure of materials at length scales from the Angstrom (atoms) to the micrometer length scales (aggregates, colloids, etc.) as well as nuclear magnetic resonance and X-ray photoelectron spectroscopy to determine, amongst others, the presence and chemical environment of atoms in a material. The measurement techniques will be presented from an application point of view, focusing on the basic principle behind the measurement technique, the information that can be derived from the measurement technique, and how a typical experiment would look like and can be performed. All measurement techniques will be illustrated with a broad range of examples applicable to chemical engineers and chemists (e.g. applications in soft matter, battery materials, solar cell materials, perovskites, thin films, etc.). An experimental project in which students can apply one technique and a tour along the facilities of the Reactor Institute Delft will be part of the course.	
Study Goals	By the end of this course, the student should be able to: - Explain the principles of several X-ray and neutron scattering techniques (diffraction, reflectometry, small-angle scattering etc.) used to obtain structural information about materials at various length scales. - Explain the principles of Nuclear Magnetic Resonance spectroscopy. - Explain the principles of X-ray spectroscopy techniques such as X-ray Photoelectron Spectroscopy. - Argue based on the interaction of X-rays and neutrons with matter whether X-rays or neutron radiation is the preferred option to answer a specific science question. - Argue which material characterization technique is most suitable to answer a specific science question. - Perform a basic material characterization experiment using X-ray or neutron radiation. - Interpret data obtained from a material characterization experiment.	
Education Method	Plenary lectures (2h). Problem sets are provided by the instructors after each lecture and that can be solved by the student. Solutions to the problem sets will be provided and questions can be asked about the problem sets during the next lecture. A mandatory experimental project where students perform an experiment in groups is part of the course. Excursions to the facilities of the Reactor Institute Delft are also organized.	
Literature and Study Materials	All course material will be incorporated in the lecture notes/sheets and problem sets. In case the student prefers background reading with a more mathematical approach about neutron and X-ray scattering, we recommend the book D.S. Sivia, Elementary Scattering Theory for X-ray and Neutron Users, Oxford University Press, ISBN 978-0-19-922868-3 (not required at all). Other topic-specific recommendations will be made throughout the course.	
Assessment	The final grade of the course will be based on the group experimental project (small report + presentation) that will count for 30% and a written final exam with a weight of 70%. The grade for the group project will also be valid for the resit, but not for next years edition of the course. Depending on the number of students, the resit may be an oral exam.	

CH3771	Nuclear Chemistry	6
Responsible Instructor	Dr.ir. A.G. Denkova	
Responsible Instructor	Ir. R.M. de Kruijff	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge		
Course Contents	This course is designed for students having either chemistry or physics background who are interested in learning more about nuclear chemistry and applied radiochemistry. This course covers topics from the Bing Bang to the production of radionuclides for medical and industrial applications. This course also provides hands-on-experience through practicals where you get to work with radioactive materials. Students should complete this course with an in depth knowledge of nuclear and radiochemistry.	
Study Goals	1. Identify the factors that affect nuclear stability 2. Explain the different kinds of radioactive decay 3. Interpret a radioactive decay series 4. Understand the interaction of radiation with matter 5. Be able to calculate ranges of different ionising particles 6. Distinguish between different radionuclide production routes 7. Be able to calculate the specific activity of the produced radionuclides 8. Be able to use the principle of Nuclear analytical techniques 9. Be able to apply radiotracer principles 10. Identify the best separation techniques for a specific radionuclide 11. Be able to perform radiodating calculations	
Education Method	Oral lectures on campus if possible. Practical if allowed	
Literature and Study Materials		
Assessment	A midterm exam counts for 20 % and a final exam for 80 % of the grade. The midterm exam will be 'a closed book' written exam on campus. The duration will be 1 hour. The final exam will be a 3 hour written, closed book exam on campus. Attending the practicals is obligatory to pass the course.	
Permitted Materials during Tests		
Studyload/Week	This is a 6 ECTS course composed of 24 lecture hours, 134 self-study hours, a midterm examination of 1 hour, and a final examination of 3 hours. This class will meet twice per week for two hours each day.	

CH3783	Materials Chemistry for The Nuclear Fuel Cycle	3
Responsible Instructor	Prof.dr.ir. J. Sietsma	
Responsible Instructor	A.L. Smith	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	General chemistry, basics of chemical thermodynamics	
Course Contents	This is a 3 ECTS course composed of 28 total lecture hours, 56 self-study and a final examination. This course is designed for Chemical Engineering, Applied Physics, Materials Science & Engineering, and Sustainable Energy Technology students that are interested in developing a working knowledge of nuclear materials chemistry and the nuclear fuel cycle. The course will cover various aspects of solution and solid state chemistry as well as materials science that play a key role at each step of the nuclear fuel cycle, from the metallurgy of uranium to the disposal of spent reactor fuel or high level waste. While the physics and engineering of controlled fission are central to the generation of electricity by nuclear reactors, chemistry in general and especially materials chemistry dominate all other aspects of the nuclear fuel cycle. This course will give students a comprehensive knowledge of the traditional fuel cycle (the uranium once-through cycle) in Light Water Reactors (LWRs). It will also cover some of the proposed nuclear fuel cycles that may well carry nuclear power through the coming decades. As an outcome of the course, the students will be able to compare and contrast existing and innovative fuel cycles, and discuss the pros and cons of each. In addition, the course will cover some of the unique properties of the actinide 5f-elements. The first three of the actinide series - Th, Pa and U - occur naturally, while the elements following uranium in the Periodic Table (transuranium elements) are man-made. The actinides are essential to nuclear power generation, but also find applications in many other areas of industry, medicine and research. The course will address 5f-electronic configurations, oxidation states, redox potentials, inorganic and structural chemistry, as well thermodynamic properties of actinide compounds. The actinides behaviour in the environment and in the geosphere will be discussed, together with analytical tools for their identification. The biological and environmental hazards associated with these elements pose certain risks which must be controlled and minimized. Due to their significant role and because the final destination of transuranic elements originating from the nuclear fuel cycle is still an open issue, the actinides chemistry and physics continue to be one of the major areas of nuclear research. The materials science part of the course will cover the main features of materials microstructures and behaviour in conditions that are relevant for nuclear reactors. The various types of defects at the atomic or micrometre scale will be explained as well as their consequences in terms of energy. This forms the basis of understanding the changes that materials undergo when subjected to elevated temperature and high-energy radiation. Changes in the microstructure and the defect configurations lead to changes in properties and behaviour and can thus jeopardise the material performance, forming a threat to safety. The course is aimed at master level students with general background in chemistry, and/or materials science.	
Study Goals	1. Students can demonstrate how chemistry and materials science influence almost all aspects of the fuel cycle. 2. Students can describe the key steps involved in the traditional nuclear fuel cycle and explain the involved chemical processes. 3. Students can apply the concepts addressed in the course to discuss alternative fuel cycles, not covered in the class, and explain the fundamental differences with the presented traditional fuel cycle. 4. Students can justify, at each step of the fuel cycle, the behaviour of the different actinide elements in view of their chemical and physical properties (electronic structure, redox potentials, solubility, complexation, speciation, thermodynamics, etc) 5. Students can predict and explain the behaviour of fission products in the irradiated fuel, during the reprocessing and in geological environment. 6. Students are able to predict the environmental behaviour of various actinide species. 7. Students can discuss major analytical techniques for actinide studies. 8. Students can perform numerical evaluations to predict the behaviour of the actinide elements and fission products at each step of the fuel cycle. 9. Students are able to discuss the major features of materials microstructures and behaviour when subjected to the conditions in a nuclear reactor. 10. Students write an essay and make a presentation on a supplemental topic related to the course material, but not covered in class, in which they apply the key concepts presented during the lectures and perform critical analysis of the information.	
Education Method	Oral lectures / Supported self-study	
Literature and Study Materials	Main: Course reader Additional reading: N. Tsoulfanidis. The Nuclear Fuel Cycle, American Nuclear Society, 2013; The Nuclear Fuel Cycle: from Ore to Wastes, P.D. Wilson (ed.), Oxford University Press, 1996. The Chemistry of the Actinide and Transactinide Elements (L.R. Morss, N.M. Edelstein, J. Fuger (Ed.)), Springer Netherlands, 2011 S.A. Cotton, Lanthanide and Actinide Chemistry, John Wiley & Sons, 2006 W. Loveland, D.J. Morrissey, G.T. Seaborg. Modern Nuclear Chemistry, Wiley-Interscience, 2006; G.R. Choppin, J. Rydberg, J.-O. Liljenzin, C. Ekberg, Radiochemistry and Nuclear Chemistry, Academic Press, 2013. Nuclear Fuel Cycle Science and Engineering, I. Crossland (ed.), Woodhead Publishing, 2012 M. Halka, B. Nordstrom, Lanthanides and Actinides, Facts on File Inc., 2011 W. Loveland, D.J. Morrissey, G.T. Seaborg, Modern Nuclear Chemistry, Wiley-Interscience, 2006. G.R. Choppin, J. Rydberg, J.-O. Liljenzin, C. Ekberg, Radiochemistry and Nuclear Chemistry, Academic Press, 2013. William D. Callister Jr., David G. Rethwisch, Materials Science and Engineering: An Introduction, ISBN 978-0470419977, Ed. Wiley	
Assessment	The final grade is composed of the weighted average of the Final exam (50%), presentation (15%) and essay (35%). Note: the exact format of assessment may change due to Coronavirus (COVID-19) regulations.	
Elective	Yes	
Tags	Chemistry	
Location	Reactor Institute Delft, Radiation Science and Technology department (preferred)	

CH3921	Sustainable Polymer Materials	3
Responsible Instructor	Dr. W.F. Jager	
Contact Hours / Week x/x/x/x	x/x/4/x	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	BSc. Chemistry or Chemical Engineering	
Course Contents	This course will focus on the question how the current oil-based polymer industry that produces non-degradable polymers that accumulate in our environment in massive quantities can be transformed in an industry that is green, sustainable and fits in a more circular economy. In this course we will provide background regarding (the 12 principles of) Green Chemistry. Various metrics that are used to quantify the greenness and sustainability of a process or a product will be introduced and discussed. We will make an inventory regarding the current practice in polymer manufacturing and polymer waste treatment. Subsequently we will introduce various scenarios towards (more) sustainable polymers, and discuss the pros and cons of these approaches.	
Study Goals	After attending this course the student will: (1) have detailed knowledge of the current industrial production of polymers (2) know various scenarios to make polymers more green and sustainable (3) be able to use the 12 Principles of Green Chemistry to assess the sustainability of a given polymer (4) be able to critically assess the purport of green claims	
Education Method	Frontal lectures, homework and discussions in class. Lecture notes, and references for scientific articles and tutorials will be provided via Brightspace.	
Literature and Study Materials	selected articles will be available via Brightspace	
Books	suggested reading: 'Green Chemistry - An Introductory Text' (2nd ed., M. Lancaster) RSC Publishing (ISBN: 978-1-84755-873-2)	
Assessment	(1) Written exam, 75% of the overall grade (2) written assignment about a topic of Green Sustainable Polymers (groups of 3-4 students), 25% of the overall grade	

LM3311	Green Chemistry and Sustainable Technology	3
Responsible Instructor	Dr. F. Hollmann	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4 5	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Organic Chemistry (LB1045 or 4051OCSTRY) Chemical Biology (LB2621) OR Catalysis (4052KATALY)	
Course Contents	Depleting fossil fuels and environmental concerns force us to change the way we produce our goods. The transition of the chemical industry to becoming sustainable represents one of the major challenges to be faced in the 21st century. Catalytic reactions will increasingly substitute the existing energy- and resource-intensive production procedures; hence reducing or preventing toxic pollutants. Biotechnology will play a major role in this transition and complement advanced chemical production technologies. This course will highlight the emerging role of catalysis in the chemical industry. A major focus will lie on biotechnological production of chemicals (fermentation processes, isolated enzymes). These methods will be related to their chemocatalytic counterparts. Thus, ample examples including the underlying mechanistic principles - of bio- and chemocatalytic processes will be discussed. Finally, the emerging field of renewable feedstocks will be discussed not only from a chemical standpoint, but also on socioeconomical and ecological grounds. The 12 Principles of Green Chemistry will serve as guideline to assess the scope and limitations of the examples discussed.	
Study Goals	After attending this course the student will: (1) have detailed knowledge of sustainable catalysts (enzyme- and chemo-catalysis) (2) know industrial examples of sustainable manufacturing practice (3) be able to use the 12 Principles of Green Chemistry to assess the sustainability of a given process (4) be able to critically assess the purport of green claims	
Education Method	Lecture Sheets, specialized books, articles and patents	
Literature and Study Materials	selected articles will be available via Brightspace	
Books	suggested reading: 'Green Chemistry - An Introductory Text' (2nd ed., M. Lancaster) RSC Publishing (ISBN: 978-1-84755-873-2) and/or 'Green Chemistry and Catalysis' (R. A. Sheldon, Isabella Arends, Ulf Hanefeld), Wiley VCH (SBN: 978-3-527-30715-9)	
Assessment	(1) Written exam (digital off-campus (open book) or on-campus, depending on the COVID-19 measures), 75% of the overall grade (2a) only for SET students: written assignment about a topic of Green Chemistry (groups of 3-4 students), 25% of the overall grade (2b) for non-SET students: group presentation about a topic of Green Chemistry (groups of 4-5 students), 25% of the overall grade	

LM3731	Advanced Biocatalysis	6
Responsible Instructor	Prof.dr. U. Hanefeld	
Module Manager	Prof.dr. U. Hanefeld	
Module Manager	C.E. Paul	
Contact Hours / Week x/x/x/x	0/4/3/0	
Education Period	2 3	
Start Education	2	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	BSc degree LST or equivalent	
Course Contents	This course gives an introduction to the use of enzymes for the synthesis of chemicals and pharmaceuticals. Starting from a basic knowledge in biochemistry, enzymology and organic chemistry, we will study proteins for their capacity to act as catalysts for different classes of reactions (such as reductions, oxidations and C-C bond formations). The importance of enzymes for the synthesis of chiral compounds will be an important theme. In addition the advantage and technique of immobilization of enzymes will be treated.	
Study Goals	After this course the student should have acquired in depth knowledge about enzymes as biocatalysts and their application in Organic Chemistry. Based on this knowledge the student should be able to choose a suitable biocatalyst for an industrially important reaction and he/she should be able to explain why which type of approach is more suitable for the production of a given bulk or fine chemical. This includes an in depth analysis of the enzyme, the working conditions and the mechanism. Moreover, the students will acquire presentation skills via a structured preparation and presentation of a lecture on a self chosen topic.	
Education Method	Lectures, Self-study and a lecture by the student	
Literature and Study Materials	Handouts/Brightspace Books: K. Faber, Biotransformations in Organic Chemistry, 7th ed., Springer-Verlag, Berlin, 2018. available via the TU library - Soderberg, Timothy, "Organic Chemistry with a Biological Emphasis Volume I" (2019). Chemistry Publications. 1. https://digitalcommons.morris.umn.edu/chem_facpubs/1 - Soderberg, Timothy, "Organic Chemistry with a Biological Emphasis Volume II" (2019). Chemistry Publications. 2. https://digitalcommons.morris.umn.edu/chem_facpubs/2 Literature studies in the lecture theatre.	
Assessment	Written examination (80%, digital off campus (open book) or on campus (no open book), depending on COVID-19 campus regulations), Presentation skills will be examined as part of a review and presentation of a self-chosen scientific paper (20%). Resit period 4, 2022-2023: oral, on request.	
Enrolment / Application	not later than one week before the first lecture via Brightspace (courses > Master > LS&T).	
Schedule	1. Introduction 2. Why are enzymes such good catalysts/kinetics 3. Classes of enzymes 4. Hydrolysis/enantioselectivity 5. Reaction media: water or no water 6. C-C bond formation 7. Reductions 8. Oxidations 9. Immobilisation 10. Other relevant enzyme classes 11. Comparison of the enzymes and their application.	
Location	Delft	

ME45203	Electrochemical Energy Storage 2: Analytical Modelling	4
Responsible Instructor	J.W. Haverkort	
Instructor	Dr. R. Kortlever	
Contact Hours / Week x/x/x/x	0.0.0.4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Familiarity with transport phenomena, basic fluid dynamics, and elementary differential equations. A working knowledge of electrochemistry and in particular the Butler-Volmer equation is also assumed. Ideal preparation include the courses ME45201, CH3513, or CH3222. A previous course on fuel cells, batteries, or similar, or additional self-study may also be sufficient.	
Course Contents	Electrochemistry plays a pivotal role in the transition towards a sustainable energy future. Renewable electrical energy can be stored in batteries or used to synthesize chemicals or fuels in electrochemical reactors, like an electrolyzer producing hydrogen. This course will introduce to a student, the knowledge and skills required to understand, model and optimize such electrochemical devices. Content overview: Porous electrode theory, multiphase flow, and transport models will be used to derive analytical current-voltage relations of various electrochemical devices including fuel cells, batteries, redox flow batteries, and electrolyzers. Some of the topics that will be discussed in this course include: - The Nernst-Planck equations for ionic transport in binary electrolytes. - Porous electrode theory using the concept of an electrochemical Thiele modulus. - The flooded agglomerate and relative permeability models for gas diffusion electrodes. - The single particle and Newman's P2D battery models. - Tobias model for gas-liquid electrolyzers.	
Study Goals	After successful completion of this course, the student is able to independently: 1. explain and apply porous electrode theory. 2. derive 0D, 1D, and 2D mathematical models to describe the current-voltage relationship of various electrochemical cells including flow, multiple phases, and porous electrodes. 3. optimize an electrochemical cell design with respect to, for example, energy efficiency.	
Education Method	Two lecture hours + two exercise hours per week.	
Course Relations	This course is a natural sequel to ME45201, CH3513, and CH3222 for students interested in analytical modelling of electrochemical systems.	
Literature and Study Materials	Reader, lectures slides, exercises.	
Assessment	Written exam.	

SEN1121	Complex Systems Engineering	5
Module Manager	Prof.dr. F.M. Brazier	
Instructor	Dr. P.H.G. van Langen	
Instructor	Y. Huang	
Instructor	Prof.dr. F.M. Brazier	
Responsible for assignments	Prof.dr. F.M. Brazier	
Co-responsible for assignments	Y. Huang	
Contact Hours / Week x/x/x/x	4/0/0/0 and interactive lab sessions	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The course introduces design thinking in socio-technical systems as the core orientation of the entire SEPAM master. Students learn about designing complex technological systems in multi-actor environments. It starts from understanding socio-technical systems towards creating artefacts to realize change in socio-technical systems. Different perspectives on systems design are discussed to provide students with a background for working with designers from different disciplines. Thereby, the course lays the foundation for further design-oriented courses. Typical questions the module addresses are: - How to explore a problem situation? - How to oversee a design challenge? - How to formulate a design task? - What is a suitable design approach for specific design challenges? Students follow lectures and work together in groups on a specific challenge to put the knowledge into practice: learning by doing.	
Study Goals	After completion of the course, the student is able to - use and differentiate discuss concepts and terminology related to the design in socio-technical systems - analyze a problem situation in complex, technological, large-scale systems in multi-actor environments - use and combine methods and tools for requirements analysis and conceptual design - use and compare methods and tools that facilitate systems design and engineering - apply and compare methods and tools related to decision, project and risk management - determine a suitable design approach for specific socio-technical design challenges - use and combine methods and tools for systems analysis and modelling to address complexity and uncertainty in design processes	
Education Method	Learning by being prepared and participating: online interactive lectures (for which participation is essential), and labs on campus in which groups work together supported by TAs and teaching staff.	
Literature and Study Materials	The main content of this course is provided on the slides used during the lecture, together with papers, articles and book chapters as appropriate. These reading materials are made available on Brightspace.	
Assessment	The overall grade for this course consists of 2 sub grades: 1) Exam: The exam focuses both on the theoretical knowledge and the ability to apply and reflect on the theory taught. The digital exam is open book and contributes to 70% of the final grade. The exam will be held on campus unless due to unforeseen circumstances or measures resulting from COVID-19, this is not possible. In this case the exam will be held remotely. 2) Assignments: During the labs, theory from the lectures is applied to a practical case. The assignments contribute to 30% of the final grade. Both parts must be completed with a passing grade (at least 5.75).	
Tags	Analysis Challenging Design Group Dynamics/Project Organisation Group work Modelling Projects Sustainability	

SET3070	Thermochemistry of Biomass Conversion	4
Responsible Instructor	Prof.dr.ir. W. de Jong	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	To start this course you need to have a good knowledge of BSc. level thermodynamics, chemistry, kinetics, basic reactor modelling (batch, plug flow, continuously well stirred reactor), basic process engineering (mass, energy, momentum balances), and physical transport phenomena. In case you think you miss some of this pre-existing knowledge, please perform selfstudy of the first six chapters of the book 'Biomass as a Sustainable Energy Source for the Future', Wiley (2015), eds. De Jong and Van Ommen	
Summary	Biomass is available in abundance on a global level; it is naturally stored solar energy in a wide diversity of chemical structures. Among the renewable energy sources, biomass can be stored most easily, of course depending once again on the resources characteristics. Due to the heterogeneous nature of biomass also many different conversion technologies have been developed in the context of energy conversion chains with combustion being the oldest performed by mankind. This course deals with the background and design and implementation of different technologies to convert biomass thermochemically.	
Course Contents	In this course you will learn about the following aspects of biomass conversion based on thermo-chemical conversion technologies. - Biomass Characterisation (biomass types, properties, analyses) - Pretreatment (harvesting, size reduction, drying, torrefaction) - Thermochemical conversion (combustion, gasification and pyrolysis; reactor types) - Transportation fuels / biorefinery (syngas conversion, economics) - Process design (structured design approach)	
Study Goals	During this course, the students will obtain knowledge of and insight into energy conversion technologies based on biomass thermo-chemical conversion processes for a wide variety of applications, like supply of power, heat and fuels. They will be offered practical problems to solve and projects to perform in this area. These main learning objectives can be divided into several smaller learning objectives: 1) The student has knowledge on the typical biomass composition, related properties (like e.g. heating value, ash melting behaviour) and can critically identify and select fuel characterization techniques. 2) The student is able to identify and perform basic capacity calculations concerning thermo-chemical biomass conversion systems for small and larger scale (combined) heat and power generation as well as systems for transportation fuel production. 3) The student is able to set up reaction equations (stoichiometry) for a wide range of energy supply related fuel conversion processes. 4) The student can set up and perform heat and mass balance calculations regarding single phase and simple multi-phase biomass thermo-chemical conversion reactors and extended (downstream) systems for power, heat and fuels generation. 5) The student knows how to perform basic ideal gas phase chemical reaction equilibrium calculations to calculate product compositions and extents of reactions given initial reaction mixture composition, pressure and temperature. 6) The student can understand and describe reaction kinetic expressions and also apply them in order to solve (design) problems related to the conversion of fuels and the formation of emission components. 7) The student demonstrates good knowledge of mass and heat transfer mechanisms and phenomena applied to the chemical conversion of both liquid and solid biomass (derived) fuels. 8) The student can derive equations for idealised model reactors (well-stirred reactor and plug flow reactor concepts) and apply these simplified reactor concepts to solve engineering problems related to the thermo-chemical conversion of biomass fuels, in particular combustion, gasification and pyrolysis. 9) The student is able to perform calculations concerning the conversion of small particles; important in this respect is the identification of the regime of conversion. Related dimensionless numbers such as the Thiele Modulus and the Biot number are key indicators in this respect. 10) The student is able to understand the basics of thermo-chemical conversion processes and to translate this into designing and evaluating such (sometimes hybrid) systems, including their economic evaluation based on characteristic parameters like NPV, POT, and ROI. 11) The student demonstrates good communication skills, both oral and written, concerning presentation of (applied) scientific results.	
Education Method	This course consists of six 2-hour lectures, also using blended learning methods. Each lecture is followed by a 2-hour work session or excursion(s)/lab visit(s) in the same week -depending on the actual COVID situation- focused on working out minor assignments including literature surveys, problem solving and work on the major project. In the week of the exams you will present the major assignment project results.	
Course Relations	This course is a core course in the Sustainable Energy Biomass Energy profile.	
Literature and Study Materials	Book: "Biomass, a sustainable energy source for the future", de Jong and van Ommen, Wiley., 1st edition, Wiley (2015). Lecture slides	
Assessment	The final grade consists of three parts: * the solution to the (three) assignments (team work); 40% of the grade * a larger final assignment (team work); 50% of the grade * evaluation of the final assignment's presentation; 10% of the grade The assessment method may be adapted to actual student population. You can get a chance to resit the larger final assignment part of the course in case of having obtained an insufficient grade for this part by means of a repair additional work for the same assignment to obtain a sufficient grade. If you pass only part of the course, all parts need to be retaken. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

SET3085	Hydrogen Technology	4
Responsible Instructor	Prof.dr. F.M. Mulder	
Instructor	Prof.dr. J.J.C. Geerlings	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Knowledge on the content of the previous SET courses (the need for storage, batteries) is assumed, as well as the chemical and thermodynamics background that is necessary for the MSc SET.	
Course Contents	Hydrogen is a vital element in the energy transition. While sustainable energy sources such as solar and wind plants will power the planet, this energy needs to be converted in chemical bonds to provide storage, fuel and resources to our future energy system. Hydrogen is the most simple chemical bond that can be made by (electro)chemistry. Hence, we will discuss the various (electro-)chemical ways that hydrogen can be produced and discuss the main impediments for the implementation of this technology. Once the hydrogen is produced it will need to be stored and converted. Hence, we will discuss the basics of different storage methods: as a gas, liquid, adsorbed or absorbed inside appropriate solids. Furthermore, the use of hydrogen as a fuel in several types of fuel cells will be discussed.	
Study Goals	After this course students: Students are able to distinguish the various sustainable methods to produce hydrogen, assess their pro/cons and describe their operation in thermodynamic terms. Students have a basic understanding of the physical nature of hydrogen and are able to assess the various ways of storing it. Students have a basic understanding of various fuel cell technologies and are able to assess their efficiency and describe their operation in thermodynamic terms.	
Education Method	This course consists of 14 2-hour lectures. These lecture also include some problem solving.	
Course Relations	This course is a core course in the Sustainable Energy Energy Storage profile.	
Literature and Study Materials	The exam will be based on the lecture slides and on selected research papers as provided on Brightspace	
Assessment	Written exam	
	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

Year	2022/2023
Organization	Applied Sciences
Education	Master Chemical Engineering

Thesis Project 2022

Year	2022/2023
Organization	Applied Sciences
Education	Master Chemical Engineering

Design Module 2022

CH3803	Product & Process Design	6
Responsible Instructor	Prof.dr.ir. A.B. de Haan	
Instructor	Dr.ir. G.M.H. Meesters	
Contact Hours / Week x/x/x/x	0/0/8/4	
Education Period	3 4	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	This course is an obligatory preparatory course for the CDP project (CH3843) and therefore full participation - proven by presence, making assignments and tests and sitting for the exam - is required in order to be admitted to the CDP project (see also CH3843).	
Course Contents	- Product Design & Material Supply Chains - Relation between Product Performance and Composition - Process Design Methodology - Process Integration - Process Flow Sheet Modelling - Energy and Mass Integration - Process and Product Evaluation and Optimisation - Health, Safety and Environmental aspects of design	
Study Goals	The student should be able to: - understand the generic design cycle - decompose a process design in hierarchical levels - apply process synthesis, analysis and evaluation methods - identify opportunities for new products - select performance specifications - identify relevant aspect of safety, health, environment and sustainability - perform an economical evaluation of products and processes	
Education Method	Lectures and team assignments. PRESENCE IS COMPULSORY.	
Computer Use	Aspen	
Literature and Study Materials	Seider, W.D., Seader, J.D., Lewin, D.R., Widagdo, S., Gani R. and Ng K.M., Product and Process Design Principles. Synthesis, Analysis, and Evaluation., Ed. 4 EMEA, John Wiley & Sons, Inc., 2017	
Assessment	3 Parts: Individual Product test, Individual Aspen Test, Individual Equipment Design Test and Group Assignments A peer review is conducted as part of the final assessment and used to establish individual group assignment grades. Pass requirements: Individual Product Test ≥ 5.0 Individual Aspen Test ≥ 5.0 Individual Equipment Design Test ≥ 5.0 Group Assignments ≥ 5.0 Final PPD grade ≥ 5.75 Conditions retakes: No retakes for group assignments For individual test retakes last mark counts Final PPD grade comprises Individual Product test grade (20%), Individual Aspen Test (20%), Individual Equipment Design Test (20%) and peer review corrected individual Group Assignment grade (40%) Note: the exact format of assessment may change due to Coronavirus (COVID-19) regulations.	
Permitted Materials during Tests	Calculator	
Enrolment / Application	students not enrolled in the MSc Chemical Engineering must contact the responsible instructor before January 20	
Studyload/Week	Required to be present during lecture days for reason of group assignments.	
	Presence lectures "Market Assessment" & "Safety" and workshops "Sustainability" & "Creativity" mandatory	

CH3843	Design Project	12
Responsible Instructor	Prof.dr.ir. A.B. de Haan	
Instructor	Dr.ir. G.M.H. Meesters	
Contact Hours / Week x/x/x/x	0/0/0/40	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Required for	MSc Chemical Engineering	
Course Contents	The design project includes the generation of a conceptual design for an integrated manufacturing process and/or a chemical product/device. The assignments are of high practical relevance as they are mostly delivered and supervised by industry.	
Study Goals	To offer students a realistic experience in making a conceptual product/process design and achieving a high degree of integration of chemical engineering know-how. Teamwork and communication skills are being trained since design projects are carried out by groups of students.	
Education Method	Project Team Work (5-6 students, mixed education background and nationalities) with intermediate reporting and presentations.	
Literature and Study Materials	Seider, W.D., Seader, J.D., Lewin, D.R., Widagdo, S. Gani, R. and Ng K.M., Product and Process Design Principles. Synthesis, Analysis, and Evaluation., Edition 4 EMEA, John Wiley & Sons, Inc., 2017 Tassoul M., "Creative Facilitation", 3rd Edition (2009) VSSD Course materials Product and Process Design (CH3804) CDP Manual and other documents/references as provided on Brightspace	
Prerequisites	A proof of full participation in the preparatory PPD course (CH3803) is required and at least 12 ec of other chemical engineering master courses/electives must have been completed to be admitted to the Design Project (see also PPD). Presence, making assignments and sitting for the exam are considered as proof for participation in the PPD course. Students doing homologation (bridging) courses as part of their master's programme must have completed all (except at most one) of these courses.	
Assessment	Based on concept design report, final design report, presentations (kick-off, concept, final) and project defence during final presentation and for supervisors. Divided over: Theory and concepts (40%) = quality of design Implementation (30%) = group performance, use of creativity tools, innovativeness, concrete result Communication (30%) = progress meetings, reports, presentations A peer review is conducted as part of the final assessment and used to establish individual marks Final reports will be screened for plagiarism. Note: the exact format of assessment may change due to Coronavirus (COVID-19) regulations.	
Studyload/Week	8 weeks fulltime	

TPM330A	Engineering Ethics and Risk	4
Module Manager	Dr. F. Santoni De Sio	
Instructor	Prof.dr.ir. G.L.L.M.E. Reniers	
Co-responsible for assignments	Dr. J.B. van Grunsven	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 3	
Course Language	English	
Course Contents	This course is reserved to the students of the Master Chemical Engineering. Students from all other Masters are invited to join the Ethics and Engineering course under the codes WM0320TU (3 ECTS) or WM0329TU (6 ECTS). Students specifically interested in the Ethics of Technological Risk are invited to consider the thematic course: WM0375TU. The first part of this course is identical to the Ethics and Engineering course (WM032TU). In this part you will explore the ethical and social aspects and problems related to technology and to your future work as professional or manager in the design, development, management or control of technology. You will be introduced to and make exercises with a range of relevant aspects and concepts, including professional codes, philosophical ethics, individual and collective responsibilities, value pluralism and relativism, responsibility within organisations, responsible conduct of companies and the role of law. You will analyse legal, political and organisational backgrounds to existing and emerging ethical and social problems of technology, and you will explore possibilities for resolving, diminishing or preventing these problems. The second part of the course (Risk) will be a series of guest lectures within the broader field of risk and safety management. Several industrial sectors will be discussed and how risk and safety are seen as important topics with discussion of risks (can be calculation, mgt, treatment, etc.) within the specific sector.	
Study Goals	After having completed the Ethics part of the course course you: - can better recognise and analyse ethical and social aspects and problems inherent in technology and in the work of professionals and managers active in the design, development, management and control of technology. - have insight into how these ethical and social aspects and problems are related to legal, political and organisational backgrounds. - are able to explore and assess possibilities for solving or diminishing existing and emerging ethical and social problems that attach to technology and the work of professionals and managers. - are better prepared to perform your future work as a professional or manager in the design, development, production and control of technology in an ethical and socially responsible way. After having completed the Risk part of the course you can recognize and understand risk and different activities in relation to risk (e.g. risk management, risk treatment, risk analysis, risk calculation, etc.) in a broad context. You obtain insights into the risk concept and dealing with risks in an organisational context, and you will be better prepared to look for the right information to optimize your decision-making.	
Education Method	Ethics: A series of lectures and interactive work sessions with partial assessment, concluded with a written test. Risk: 5 or 6 (guest) lectures.	
Literature and Study Materials	Chapters from the Reader Ethics and Engineering, available at Nextprint and as PDF files on Brightspace; articles available as PDF on Brightspace; Powerpoint slides, materials used in the working groups, and lecture notes. The powerpoint presentations of the guest lectures on Risk will be made available on Brightspace.	
Assessment	For the Ethics part (3 EC points): Written exam (60%), presentation and active participation during the working group sessions (40%). In order to pass the course, a minimal grade of 5.5 should be obtained *both* in the exam *and* in the working group, and the average grade should be higher than 5.75. It is possible to re-take the exam without re-taking the working groups, in this case the working group grade will be kept. If you re-take the exam, it is also possible to re-attend the working groups to improve the final grade. For the Risk part (1 EC point): written exam, closed book, open reflection question	
Enrolment / Application	Enrolment via Brightspace is required for this course. This is needed in order to plan the workgroups. The Brightspace page will be activated a couple of weeks before the start of the course. Please enroll as soon as possible and check Brightspace for more information on the enrollment procedure and timeline.	
Remarks	This course is reserved to the students of the Master Chemical Engineering. Students from all other Masters are invited to join the Ethics and Engineering course under the codes WM0320TU (3 ECTS) or WM0329TU (6 ECTS). Students specifically interested in the Ethics of Technological Risk are invited to consider the thematic course: WM0375TU.	

Year	2022/2023
Organization	Applied Sciences
Education	Master Chemical Engineering

Professional & Social Orientation 2022

Introduction 1

Students can further personalise their programme by choosing the Professional & Societal Orientation (PSO) of their preference (30 EC). The Scientific and Social Orientation is intended to help students better prepare for their future career by choosing one of the following SSO variant:

- Research and Development: Industrial internship (18 EC) + elective courses (12 EC).
Chemical Engineering electives are listed under Scientific and Social Orientation.
- Management of Technology: Educate yourself in the field of Management and Entrepreneurship
- Education: great for people that pursue a teaching career at Dutch high schools.
- Study Abroad: A coherent package of courses at one of our partner universities abroad.
- Science & Engineering: Best for students who pursue an academic career.

Research and Development 2022

EC Program	30
Introduction 1	Research and Development This SSO variant is especially tailored for students who will work in industry after completing their master education. Research and Development consists of: - Industrial Internship (CH3702, 18 credits) - Electives (12 EC): at least 6 EC in Chemical Engineering Electives (see approved list on the left side). A maximum of 6 EC can be used for free electives (modules offered by other TU Delft MSc programmes and/or modules that focus on transferable/language skills). Please note that the total EC of the MEP and R&D should be 60EC. This programme is especially tailored for students who will work in industry after completing their MSc education. Three modules are available: RD1 (30EC) - a regular Internship (18EC) complemented with 12EC of Free Electives - only in combination with a 30EC MEP (CH3930) RD2 (30EC) - an extended Internship (24EC) complemented with 6EC of Free Electives - only in combination with a 30EC MEP (CH3930) RD3 (60EC) - a regular Internship (18EC) - - only in combination with a 42EC MEP (CH3930)

CH3702	Industrial Internship	18
Responsible Instructor	Prof.dr. J.H. van Esch	
Project Coordinator	L. van der Elst	
Contact Hours / Week x/x/x/x	-	
Education Period	1 2 3 4 Summer Holidays	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	You will be working in a Chemical Engineering related organization full-time for a period of 3 months. Your Master Chemical Engineering knowledge is necessary for the project. The industrial internship is guided by an internal (TU Delft) and an external (company) supervisor. The mark for the internship is awarded by these two supervisors. The mark is based on a series of criteria, including an individual report of the student, in which the experiences concerning the main goals, including the experience in the working environment are discussed. During the industrial internship you are expected to get to know the organization. To do so, you need to start gathering information about the organization of your choice prior to the start and make a list of questions you would like to have answered. During the internship you need to make time to get acquainted with other departments within the organization and interview, for example, a number of people in different parts of the organization. IMPORTANT: It is desirable for the internship to have a Chemical Engineering content. However, in some cases it may be acceptable if this component is lacking, provided that the work requires problem solving or design skills for a chemical engineer. The criteria formulated above do not exclude activities in a startup company (maybe initiated by the student) to be accepted as internships. The criterion is that there should be an internship supervisor with sufficiently strong involvement to be able to act as such. The internship is in principle intended as a way for students to broaden their awareness of working environments outside academia. This excludes certain organisations. For very good students who aspire a career in academia, a possibility for an academic internship is through a grant from the Justus and Louise van Effen fund, which is restricted to very good students and which provides funding for internships at top-20 engineering universities only. For more information, see https://www.tudelft.nl/en/student/study-and-career/study-internship-abroad/funding-scholarships/justus-louise-van-effen-research-grant/. The student is responsible for finding an internship position. The Internship Office TNW only has an assisting, coordinating and administrative role. We therefore advice students to start looking for an internship well in advance: International students are advised to start the orientation process during the first year of their Master. Dutch students are advised to start the orientation process at least 6 months in advance if planning to stay in the Netherlands, and at least 9 months if planning to go abroad. Allowed: Business, chemical engineering content; TNO (or foreign equivalents such as Fraunhofer Gesellschaft); ECN; NFI; patent office; consultancy (provided technical / quantitative); financial analysis (provided strong quantitative, or related to financing a chemical plant or business). Not allowed: university, Dutch or abroad (except with van Effen grant, see above); FOM institute (such as Amolf, NIKHEF, DIFFER) and foreign equivalents (such as Max Planck Institut, ICFO); Academic hospital (unless practical work is done similar to an internship that could be done in a 'normal' hospital, and not in combination with a thesis project in a (different) academic hospital); Nationale Denktank. Exceptions are only possible if there are special reasons, after approval of the Board of Examiners. If the Board of Examiners has given permission to do the thesis project outside the university, the internship can't be done at the same organization as the thesis project.	
Study Goals	The goal of the internship is: - To become familiar with a professional working environment for a chemical engineer. - To use academic knowledge and skills, acquired in the degree course, to solve problems or be active in design related to Chemical Engineering, in a professional working environment. Skills should be acquired on the following topics: 1. Performance: Productivity, Independence, Practical skills, Theoretical knowledge, Creative thinking, Time management; 2. Communication: Reporting, Presentation, Interpersonal; 3. Competences: Motivation & enthusiasm, Critical attitude, Taking initiative.	
Education Method	Practical training on the job, the company supervisor is responsible for day-to-day supervision. During the internship period the student will also report to the TU Delft instructor 2-3 times about progress made on the project.	
Literature and Study Materials	The Brightspace page http://brightspace.tudelft.nl > catalog > search for Internship Office Applied Sciences (enrol!) provides more information about the industrial internship. On this page you can find the step-by-step guide explaining all procedures, forms, internship offers, experiences from other students and additional information e.g. about grants and funding.	
Assessment	You can contact the Internship Office through: InternshipOffice-TNW@tudelft.nl The mark for the internship is awarded by the two supervisors. The mark is based on a series of criteria, including an individual report of the student, in which the experiences concerning the main goals, including the experience in the working environment are discussed. For the assessment you need to submit two reports: - A scientific report using the format and guidelines of the company for your TU Delft supervisor and company supervisor; - An evaluation report for the internship office. Two weeks before the end of your internship, the Internship Office TNW will send your company supervisor and TU Delft supervisor the internship review form. Your company supervisor needs to assess your internship at the end of your internship period and send his review form to the TU Delft supervisor. Your TU Delft supervisor needs to assess your internship within two weeks after the end of your internship and give a mark. The TU Delft supervisor will then send the assessment form (including the company supervisor review form) to the Internship Office TNW. In principle, the mark awarded by the company supervisor counts for 2/3 towards the final mark; that of the TU Delft supervisor	

for 1/3. The TU Delft supervisor can overrule this if he/she has good reasons for it, which should be communicated to the student and the internship office.

The Internship Grading Scheme will be used as a guideline to determine the mark. Please consult the Internship Office TNW/AS Brightspace page for the detailed grading scheme.

Your mark will be processed as soon as the Internship Office TNW has received:

The review form with an original signature from both supervisors. Your supervisors can request the review form by sending an email to InternshipOffice-TNW@tudelft.nl

A digital copy of both reports.

When the nature of the project requires so, it is possible to censor parts of the report. Details like numbers and names may be left out of the report (or striked-through), for example, as long as the report represents the achievements of the project and the TU Delft supervisor is able to assess the internship.

Enrolment / Application

Prior to the start, the internship needs to be approved by the TU Delft instructor*.

When the TU Delft instructor has approved the internship, fill out the Internship application form (you can download from the Internship Office TNW/AS Brightspace page) and make sure both the TU Delft instructor and company supervisor have read and signed the form. The form should be submitted prior to the start of your internship.

*A TU Delft supervisor is a member of faculty (tenured or appointed on a tenure track position, within the faculty of Applied Sciences) involved in the Chemical Engineering programme, and thus appointed as an examiner in the programme.

Students are only allowed to perform the Chemical Engineering Industrial Internship in case it is part of the (individual) study programme (e.g. Research and Development Orientation). Not intended for students with the Orientations (30 EC): MoT, Study Abroad, Education or Scientific Orientation.

Year	2022/2023
Organization	Applied Sciences
Education	Master Chemical Engineering

Study Abroad 2022

Contact for students	International office Applied Sciences, Physics building, Room A217, InternationalOffice-TNW@tudelft.nl
In association with the University of	See the Brightspace page of the international Office TNW for the most recent information.
EC Program	30 EC
Program Structure 1	This orientation is especially tailored for students who will do a PhD after completing their master education. It consists of a semester, project and/or courses, at a foreign university. The programme has to be approved in advance by the board of examiners. This program is only in combination with a 30EC MEP (CH3930) This programme consists of one semester of studies at a foreign university. A package of 30 EC of courses (optionally including a research project of a maximum of 20 EC) must be completed. The International Office and the Programme Management are responsible for admission taking into account the average grade, feasibility of study plan and prevention of study delay. After admission, the contents of the Study Abroad programme must be approved by the board of examiners.

Year	2022/2023
Organization	Applied Sciences
Education	Master Chemical Engineering

Education 2022

EC Program	30
Introduction 1	The educational programmes are aimed at Dutch-speaking students only, because they are oriented towards the Dutch school system and because it includes internships (Schoolpracticum) at Dutch schools. Consequently the educational programme modules are taught in Dutch. The programme consists of Basisdeel/Ed1 (30 EC) and Verdiepingsdeel/Ed2 (30 EC). The minor Education (Basisdeel/Ed1) can be done during the bachelor programme and leads to certification as a tweedegraads secondary school teacher with limited qualification (beperkte bevoegdheid). If a student has done the minor Education, only the Verdiepingsdeel/Ed2 of 30 EC remains for the master programme scientific and social orientation. The combination of the minor Education and Ed2 specialisation leads to certification as a fully-qualified eerstegraads (grade-one) secondary school teacher. The certificate will be attached to the master diploma. Students that did not take the minor Education can follow the Basisdeel/Ed1 specialisation as part of their master programme and then do the Verdiepingsdeel/Ed2 as a post-master course in order to become fully qualified. The programme has to be approved by coordinator of Science Education and Communication.

Year	2022/2023
Organization	Applied Sciences
Education	Master Chemical Engineering

Education Basis 2022

SL3462	Educational Sciences		6
Responsible Instructor	S. Schut		
Responsible Instructor	Drs. R.T.G. van Uffelen		
Contact Hours / Week x/x/x/x	Periode 1: 2 uur per week, periode 2: 3 uur per week.		
Education Period	1 2 3 4		
Start Education	1 3		
Exam Period	1 2 3 4		
Course Language	Dutch		
Course Contents	Onderwijskunde is een vakgebied dat gericht is op het begrijpen en verbeteren van onderwijs. Het heeft betrekking op leerprocessen en interactie binnen en buiten de klas, op de docent, de schoolorganisatie, maar ook op beleid en bestuur van het onderwijs. De functie van onderwijskunde in de opleiding van leraren is dat de docenten in opleiding (DIOs) zich eigen maken wat toepasbaar is uit deze wetenschappelijke discipline bij het lesgeven van alle dag en om hen te helpen een onderzoekende houding aan te nemen. In de colleges van het vak Onderwijskunde passeren daarom onderwerpen de revue die vanuit een onderwijskundig en pedagogisch (opvoedkundig) perspectief een licht werpen op de onderwijspraktijk. Dat betreft ook onderwerpen die niet direct verbonden zijn aan het ontwikkelen van je eigen lespraktijk, maar wel wezenlijk zijn voor je begrip van het onderwijs en de schoolorganisatie.		
Study Goals	Onderwijskunde is een theoretisch vak dat je de gelegenheid geeft te ontdekken en uit te diepen wat zich in het hoofd van leerlingen, in de klas en op school afspeelt en de rol jij daarbij als docent speelt. Onderwijskunde is een kaderstellend vak waarin basiskennis op het gebied van onderwijzen en leren aan bod komt. In Onderwijskunde staan de volgende leerdoelen centraal: 1. Je hebt kennis van belangrijke theorieën in relatie tot interpersoonlijk docentgedrag. 2. Je hebt kennis van belangrijke theorieën uit de adolescentiepsychologie die relevant zijn voor het onderwijs. 3. Je hebt kennis over de pedagogische opdracht van de school. 4. Je hebt kennis over de belangrijkste leer- en motivatietheorieën (en hun ontstaansgeschiedenis) en je bent in staat deze theorieën aan elkaar en aan actuele onderwijssituaties en -ontwikkelingen te verbinden. 5. Je hebt kennis over hoe leerlingen leren, in termen van motivatie, leeractiviteiten en leeroriëntaties en je kunt deze kennis koppelen aan de belangrijkste leertheorieën. 6. Je kunt non-verbale en verbale gedragingen van de docent waarnemen en je bevindingen schriftelijk presenteren. 7. Je kunt met behulp van het hardop-denkmethode de denkstappen en metacognitieve activiteiten van de leerlingen in kaart brengen. 8. Je kunt een ontwikkelingsgebied van de adolescentie in kaart brengen en dit verbinden aan het lesgeven en de begeleiding van de adolescent. 9. Je kunt een onderwijskundig onderwerp in kaart brengen en aan de hand van diverse bronnen je bevindingen schriftelijk presenteren		
Education Method	Colleges, interactieve sessies en groepswork.		
Prerequisites	Geen		
Assessment	Aanwezigheid bij de colleges is verplicht. Het cijfer is gebaseerd op drie groepsopdrachten en een individuele opdracht. Het cijfer van de drie groepsopdrachten wordt gemiddeld. Dit gemiddelde en het cijfer voor de individuele opdracht tellen 50/50 mee voor het eindcijfer.		

SL4201	Pedagogy of STEM education	4
Responsible Instructor	Dr. W.T.M. Caspers	
Responsible Instructor	Dr. L. Koopman	
Contact Hours / Week x/x/x/x	4 uur per week	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 3	
Course Language	Dutch	
Course Contents	Bij vakdidactiek gaat het om een analyse van de lesstof. Enerzijds vanuit het vakgebied en het daarvan afgeleide schoolvak en anderzijds vanuit leerlingdenkbeelden en voorkennis. Deze analyse wordt gebruikt voor het ontwikkelen van een aantal lessen waarin vaktaal, voorkennis en de inhoud van het schoolvak belangrijke aspecten zijn. Bij Inleiding Bètididactiek ga je aan de slag met het ontwikkelen van algemene didactische competenties. De volgende zaken komen aan bod: 1. het Delftse lesvoorbereidingsmodel met aandacht voor (a) leerdoelen & succescriteria; (b) voorkennis en preconcepten; (c) formatieve toetsing; (d) werkvormen; 2. enkele activerende werkvormen, waaronder het directe instructiemodel; 3. enkele vormen van formatieve toetsing; 4. opstellen en verhelderen van leerdoelen en succescriteria.	
Study Goals	De student kan: 1. Vanuit vakdidactische leerdoelen haar eigen onderwijs voorbereiden, uitvoeren, evalueren en bijstellen. Hierbij ligt de focus op het leren van de leerlingen. Er wordt gebruik gemaakt van theoretische handvatten en verzamelde data. 2. Een les voorbereiden, uitvoeren en evalueren waarbij: a. Een optimaal leerrendement van de leerlingen wordt nagestreefd. b. Gebruik wordt gemaakt van theoretische handvatten en verzamelde data. c. De aandacht van de toehoorders wordt verkregen en behouden met behulp van effectieve presentatietechnieken en activerende werkvormen.	
Education Method	Het vak bestaat uit twee onderdelen: een hoor- en een werkcollege. Het hoorcollege wordt door alle studenten, van alle schoolvakken gevolgd. Het hoorcollege richt zich vooral op de vakdidactische basiskennis die voor alle schoolvakken nodig is. Het hoorcollege reikt daarvoor de gereedschappen aan die alle studenten minimaal dienen te beheersen voor het geven van goed onderwijs. Het hoorcollege wordt gegeven in de vorm van instructievideo's en voorbereidende opdrachten op BrightSpace (ca. 2 uur per week). De student is zelf verantwoordelijk voor het bekijken en maken hiervoor voorafgaand aan het betreffende werkcollege. Het werkcollege (2 uur per week) wordt in een groep van ca. 12 studenten gevolgd. Het werkcollege wordt gegeven door een student-assistent. In het werkcollege komt vooral het oefen van vaardigheden aan bod en het leren hanteren van de gereedschappen die in het college zijn aangereikt. Om je de gereedschappen eigen te maken en je te leren ontwikkelen als docent werken we, net als in de stage, met een PDCA-cyclus (Plan, Do, Check, Act). Je kiest daarin zelf een vakdidactische leervraag. Daarnaast leer je de in het hoorcollege aangereikte gereedschappen toe te passen in de stage door het maken van lesplannen.	
Assessment	Leerdoel 1 wordt aangetoond door middel van twee keer een PDCA-opdracht op basis van een vakdidactische leervraag. De leervraag is naar keuze van de student. De PDCA-opdracht wordt schriftelijk vastgelegd (circa 3 à 4 A4, exclusief bijlagen). Zie voor uitleg over de PDCA-opdracht ook de BrightSpace-pagina van Schoolpracticum Basis. Leerdoel 2 wordt aangetoond door middel van twee lesplannen volgens het Delftse lesmodel en bijbehorende lesevaluaties. Er zijn twee inlevermomenten voor summatieve evaluatie (voldoende of onvoldoende): één halverwege kwartaal 1, één aan het eind van kwartaal 1. Bij elk inlevermoment wordt dus één PDCA-opdracht en één lesplan ingeleverd. Tussentijds is er de mogelijkheid om werk in te leveren voor formatieve evaluatie. Voor de afronding van het vak dienen beide PDCA-opdrachten en beide lesplannen met evaluaties met een voldoende afgerond te worden. Daarnaast is actieve deelname voor het werkcollege vereist. Er kan maximaal één werkcollege gemist worden, met kennisgeving vooraf. Let op: ook bij Schoolpracticum Basis wordt er beoordeeld op basis van PDCA-opdrachten. Deze zijn echter gericht op pedagogisch-didactische leervragen. Bij deze twee vakken kunnen derhalve niet dezelfde PDCA-opdrachten ingeleverd worden. Eindcijfer: Voldoende / onvoldoende Alle onderdelen moeten voldoende worden afgerond.	

SL4202	Professional Learning Community	1
Responsible Instructor	Dr. J.G. Spandaw	
Instructor	S. Schaap	
Instructor	Ir. E.M. Holla	
Instructor	S. de Haan	
Instructor	R. Ockhorst	
Instructor	Dr. W.T.M. Caspers	
Instructor	Dr. P.J.J.M. Dekkers	
Instructor	Drs.ir. M.A. Botke	
Instructor	Dr. L. Koopman	
Instructor	V.R. Landgraf	
Instructor	Drs. M.J. den Otter	
Contact Hours / Week x/x/x/x	Periode 1: 2 uur per week, periode 2: 1 uur per week	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	Dutch	
Course Contents	De PLG is ondersteunend aan de andere vakken die in het eerste semester gegeven worden. Bij de PLG worden de PDCA-opdrachten voor Inleiding Bètadidactiek en de beide schoolpractica uit het eerste semester besproken. Het gaat dan om verhelderen van de opdrachten en het bespreken van tussentijdse uitwerkingen van de studenten. In de bijeenkomsten leggen studenten vast wat ze de komende week gaan doen, proberen en waarom. Dit houdt de PDCA-cyclus op gang.	
Study Goals	De student kan werken met de PDCA-cyclus: leren eigen leerdoelen stellen, manieren identificeren om in de praktijk te werken aan deze leerdoelen, feedback te verzamelen, evalueren van de leerdoelen en reflecteren op het resultaat en formuleren van verbeterpunten op basis van theorie en verzamelde data.	
Education Method	Interactieve sessies	
Prerequisites	Geen	
Assessment	Aanwezigheid en actieve deelname.	
	PLG wordt beoordeeld als voldoende/onvoldoende.	

SL4230	Chemistry Teaching Methodology	4
Responsible Instructor	Drs. M.J. den Otter	
Responsible Instructor	Drs.ir. J. Hofman	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	Dutch	

SL4235	Chemistry Foundation Teaching Placement	15
Responsible Instructor	Drs.ir. J. Hofman	
Responsible Instructor	Drs. M.J. den Otter	
Practical Coordinator	Drs. R.T.G. van Uffelen	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	1 2 3 4	
Course Language	Dutch	

Year	2022/2023
Organization	Applied Sciences
Education	Master Chemical Engineering

Education Verdieping 2022

SL3012	Personal Professional Development	3
Responsible Instructor	Drs.ir. M.A. Botke	
Contact Hours / Week x/x/x/x	1 uur per week	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	1 2 3 4	
Course Language	Dutch	
Course Contents	Als docent ben je nooit uitgeleerd. In dit vak hebben we het over professionele ontwikkeling van jezelf als docent in een school. Je doet enkele opdrachten die inzicht geven in je professionele identiteit. Daarnaast werk je aan het realiseren van een innovatie op je (stage)school. Daarbij ligt de nadruk op het proces van innoveren en de samenwerking met je collegas. Dit vak wordt vormgegeven als een Professionele Leergemeenschap (PLG). Er is daardoor veel ruimte voor uitwisseling over onder meer professionele identiteit, het vormgeven van innovaties en richting geven aan je eigen ontwikkeling.	
Study Goals	De student kan aspecten van zijn/haar eigen professionele identiteit beschrijven. De student kan de professionele identiteit van collega-docenten beschrijven op basis van een interview. De student kan reflecteren op het proces van innoveren in de school en dit relateren aan de professionele identiteit van zichzelf en de collegas die betrokken zijn bij de innovatie. De student is in staat constructief bij te dragen aan een professionele leergemeenschap met peers.	
Education Method	Deelname aan de professionele leergemeenschap, opdrachten of professionele identiteit, werken aan innovatie in de (stage)school.	
Prerequisites	Om deel te kunnen nemen aan vakken in het verdiepingsdeel van het beroepsgerichte jaar, moet het basisdeel afgerond zijn.	
Assessment	Toetsing vindt plaats op grond van Aanwezigheid en actieve deelname in de PLG De opdrachten over professionele identiteit Verslaglegging over het realiseren van een innovatie in de (stage)school, met focus op het proces en de samenwerking. Het vak wordt beoordeeld met een voldoende als alle bovenstaande punten als voldoende zijn beoordeeld. Er geldt bij dit vak een aanwezigheidsplicht.	

SL4300	Design and Research in Education	10
Responsible Instructor	Drs. M.J. den Otter	
Responsible Instructor	A.J. Boer	
Responsible Instructor	Prof.dr. M.J. de Vries	
Responsible Instructor	S. de Haan	
Responsible Instructor	Drs. P.L.F. van der Hoeven	
Responsible Instructor	Dr. J.G. Spandaw	
Responsible Instructor	Dr. P.J.J.M. Dekkers	
Course Coordinator	A.J. Boer	
Contact Hours / Week x/x/x/x	4 uur per week	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	1 2 3 4	
Course Language	Dutch	
Course Contents	Onderwijs vraagt continue verbeteringen. In dit vak kom je in aanraking met twee manieren om op een concrete en systematische wijze je onderwijs te analyseren en te verbeteren op basis van je observaties gecombineerd met theorie. Deze twee manieren zijn action research en lesson study. In dit vak ontwerp je een bescheiden onderwijsinnovatie die aan je stage is gekoppeld en ga je door middel van onderzoek na wat het functioneren daarvan beïnvloedt en hoe deze innovatie verbeterd kan worden. Daarnaast ga je verkennen hoe je deze innovatie breder kunt implementeren. Daarbij werk je vanuit een systeemperspectief aan strategische communicatie in innovatieprocessen. De vraag daarbij is hoe je deze innovatie breder kunt implementeren.	
Study Goals	De student kan: - een kleine onderwijsinnovatie ontwerpen op basis van theorieën over ontwerpen en onderwijsinnovaties; - een onderzoek opzetten en uitvoeren naar de effecten van deze onderwijsinnovatie op het leren van leerlingen; - vanuit een systeemperspectief beschrijven hoe deze onderwijsinnovatie breder geïmplementeerd kan worden en hiervoor een strategische communicatie beschrijven.	
Education Method	Colleges, interactieve sessies, groepswork. Het ontwerp implementeer je en onderzoek je in de stage.	
Prerequisites	Om deel te kunnen nemen aan vakken in het verdiepingsdeel van het beroepsgerichte jaar, moet het basisdeel afgerond zijn.	
Assessment	Je levert twee producten in: (een beschrijving van) de onderwijsinnovatie en een onderzoeksverslag. Dat laatste bevat zowel het onderzoek naar je innovatie als de strategie om deze breder te implementeren. Er wordt aan de hand van een rubric beoordeeld op de kwaliteit van het ontwerp, de kwaliteit van het onderzoek, functioneren student en de kwaliteit van het verslag. De rubric vind je op BrightSpace. Alle onderdelen moeten als voldoende beoordeeld zijn. Het eindcijfer is het gemiddelde van deze vier onderdelen. Er geldt bij dit vak een aanwezigheidsplicht.	

SL4330	Advanced Chemistry Teaching Methodology	5
Responsible Instructor	Drs.ir. J. Hofman	
Responsible Instructor	Drs. M.J. den Otter	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	2 4	
Course Language	Dutch	

SL4335	Advanced Chemistry Teaching Placement	12
Responsible Instructor	Drs.ir. J. Hofman	
Responsible Instructor	Drs. M.J. den Otter	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	1 2 3 4	
Course Language	Dutch	

Year	2022/2023
Organization	Applied Sciences
Education	Master Chemical Engineering

Management of Technology 2022

**In association with the
Faculty of** TBM

Introduction 1 This programme is offered by the faculty of Technology, Policy and Management. The programme consists of the first or the second semester of the MSc Management of Technology.

The MoT is only in combination with a MEP of 30 EC (CH3930).

Year	2022/2023
Organization	Applied Sciences
Education	Master Chemical Engineering

1st Semester MoT Modules 2022

MOT1412	Technology Dynamics	5
Module Manager	Dr. C. Werker	
Instructor	Dr. C.C.R. Penna	
Co-responsible for assignments	Dr.ir. U. Pesch	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	This course has reached its maximum capacity and therefore we can only admit our regular (TPM and orientation) students. There is limited place available for students from EPA, CoSEM and IE students. MOT STUDENTS HAVE PRIORITY. Bachelor students are excluded from participation.	
Course Contents	In this course we discuss technical innovations and technological development as the result of human choices. Therefore, you learn how to identify the factors that determine these choices, i.e. institutions, behaviour and choices of innovative agents, their relationships, knowledge and collaboration, economic value as well as legitimacy and societal values. Based on that you will figure out how processes of socio-technological change take place as well as study how innovative agents can steer new innovations into a societally responsive direction. You study these issues by using the following theory-based topics: 1) The framework of innovation systems by discussing how innovative agents and institutions drive and shape them on the sectorial/industrial/technological level as well as on the regional, national or global level, in particular by figuring out which role management and policy can play in influencing innovation processes and technological change in different kinds of innovation systems. 2) Methods to integrate societal values into and systems, particularly to find out how to make innovation responsible.	
Study Goals	At the end of the course, students are expected to: 1. analyse innovations as the result of human choices, 2. identify the actions and interactions that are relevant to innovation processes, 3. analyse innovations in a specific technological and geographical context, 4. apply different concepts and methods for specific innovation processes, 5. reflect on the societal implications of innovation and technological change and the challenges that society poses to technology and industry, 6. give recommendations for steering an innovation process into a more societally responsive direction, 7. collect, identify and read relevant literature to do the above mentioned analysis. 8. write up the results of the analyses in a concise and clear way, 9. give feedback to other students.	
Education Method	The course is organized as a series of lectures mostly given in the form of blended learning. Moreover, students can apply the knowledge obtained in the lectures during interactive groups work.	
Assessment	The mark of students is determined by a written exam. Prior to the written exam there are two mock exams: one will cover the first part, the other the second part of the course. Participation in both mock exams as well as correcting the exam of a fellow student both times is mandatory for taking the written exam. Normally the mock exams and written exam take place on-campus. If there are unanticipated circumstances of measures, particularly because of COVID-19, the (mock) exam(s) will take place as a timed open book exam.	
Enrolment / Application	MOT and orientation students will be automatically enrolled in Brightspace. EPA, CoSEM and IE students who wish to apply for this course can do so by filling in the following form: https://tudelft.fra1.qualtrics.com/jfe/form/SV_1FcBgaGLXnjIP0G Please be aware that application does not guarantee that you will be admitted. Since we only have a limited of places available we recommend you to look for an alternative course. On this webpage you can find a list of alternative courses: https://www.tudelft.nl/en/student/faculties/tpm-student-portal/education/courses-and-examinations.	

MOT1421	Economic Foundations	5
Module Manager	Dr. S.T.H. Storm	
Responsible for assignments	Dr. S.T.H. Storm	
Co-responsible for assignments	Dr. C.W.M. Naastepad	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	There is limited participation for students from other Master programs. MOT STUDENTS HAVE PRIORITY. Bachelor students are excluded from participation	
Summary	Economics is about how we work, what we produce, how we interact, cooperate and compete, and how we distribute and ultimately use what we have produced. Because the economy is too important to be left to the economists, this course provides an introduction to economics that explains how a market economy works, why it sometimes does not work as well as we would like, how its performance can be improved by regulation and policy, and how over time it changes and develops. The course does not assume that the student has any prior knowledge of economics. The course has two parts: (1) microeconomics, which covers the standard topics of supply and demand, forms of market competition, imperfect competition, efficiency, labour markets and technological progress by firms; and (2) macroeconomics, which examines aggregate demand, growth, welfare, inflation, unemployment and globalization. In both parts, special attention is given to (economic factors explaining) the extraordinary technological dynamism and productivity growth of market economies and to governments role in improving micro- and macroeconomic performance. Microeconomics (2,5 ECTS) focuses on markets, and takes both the firm as the government as principal actors. How are firms' attempts to realize their objectives (in terms of profits, sales or market shares) influenced by demand, technology and market structure? How can public regulation improve market performance which falls below norms considered acceptable? We identify important market failures, due to which markets under-perform, and specifically investigate firm behaviour in (realistic) oligopolistic markets. Macroeconomics (2,5 ECTS) deals with the workings of the economy as a whole, why the market system often underperforms, and how macroeconomic performance can be improved by fiscal and/or monetary policy. Attention will be given to conflicts between achieving different goals of macroeconomic policy at the same time, and to the ways in which globalization affects the capacity of governments to regulate their economies. The answer to this question holds major lessons for public policy.	
Course Contents	Microeconomics: demand and supply; consumer behavior; market competition; perfect competition; monopoly; oligopoly; market failures; externalities; public goods; regulation; efficiency; welfare; production; profit maximization; choice of technique; technological progress; game theory; dominant strategy; labour markets; employment; worker-management relations. Macroeconomics: aggregate demand and aggregate supply; circular flow of incomes; neoclassical economics; Keynesian economics; unemployment; fiscal policy; monetary policy; money creation; crowding out; stabilization policies; labour market regulation; technology policy; foreign direct investment; financial markets.	
Study Goals	At the end of this course, students will be able to: 1. analyze firm behavior in markets of perfect competition, monopoly, and oligopoly as well as the impact of firm behavior on economic welfare. 2. identify and appraise the rationale, scope and limits of public market regulation (in situations of "imperfect markets" and/or "market failures"). 3. derive and document the impact(-s) of regulation on the static efficiency and dynamic efficiency of markets. 4. analyze and weigh up the short-term and long-term impacts of macroeconomic policy on income, employment, productivity and competitiveness. 5. critically examine and reflect on the existing alternative theories and models. 6. understand how money is created and how the financial sector works, and how this affects the real economy.	
Education Method	Lectures. Students are expected to prepare for each class by reading the indicated literature.	
Books	Lecture Notes, Question sets and papers/articles - all made available on Brightspace.	
Assessment	In normal circumstances, the exam is a closed-book (in-person, on campus) multiple-choice examination of 3 hours. In case of unforeseen circumstances or due to measures resulting from COVID-19, the exam will be given in the form of an on-line (remote) open-book & timed multiple-choice examination of 3 hours.	
Special Information	None.	

MOT1442	Social and Scientific Values	5
Module Manager	M.B.O.T. Klenk	
Instructor	M.B.O.T. Klenk	
Instructor	Dr. J.J. Casey	
Responsible for assignments	M.B.O.T. Klenk	
Co-responsible for assignments	Prof.dr. S. Roeser	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	This course has reached its maximum capacity and therefore we can only admit our regular (TPM and orientation) students. There is limited place available for students from EPA, CoSEM and IE students. MOT STUDENTS HAVE PRIORITY. Bachelor students are excluded from participation.	
Course Contents	Part I: Scientific Values (Philosophy of Science and Decision Theory) 1. The role of values in science and technology: changing the world vs. describing and understanding the world. The contrast between descriptive claims or arguments vs. normative claims or arguments. 2. The role of reasoning in science. The strength and quality of arguments. The notion of (scientific) proof. The contrast between deductive vs. inductive or ampliative reasoning. 3. The difference between natural and social science. How humans differ from natural things and what this means for the scientific investigation of human behaviour. The contrast between explanatory vs. normative models of action and decision making. 4. The notions of rationality and optimization in action. Different forms of uncertainty and related regimes of decision making. Elementary managerial decision theory. Part II: Social Values (Ethics) 1. Introduction and overview. Getting acquainted with the notions of responsibility and other ethical notions related to managing technology and (individuals working in) the organisations that develop, produce and control technology. Basic meta-ethical questions: is ethics relative and subjective or, alternatively, might there be moral truths or universal moral principles? What are the implications of answers to these questions for management of technology? 2. Introduction into the most important normative ethical theories: utilitarianism and deontology. Exploration how these theories (should) play a role in critical reflection about management of technology. 3. Methods and theories about the right ethical theories to guide action.	
Study Goals	This module explores scientific and moral values in the management of technology. The module presents and discusses science and technology as practices involving a wide variety of judgements that are based on or express values. Values internal to science and technology are discussed in relation to the rationality of belief (e.g. scientific evidence) and action (e.g. decision making). Moral values are discussed by focusing on the ethical and social aspects and problems of technology and of professionals and managers active in the development, production and control of technology, and it explores possibilities for resolving, diminishing, preventing these problems and decision making about them, via for example design for values, responsible innovation and emotional deliberation.	
Education Method	The module is taught in the form of a mixture of (online) lectures and tutorials. Attendance is obligatory. The methodology used is philosophical analysis, reflection and discussion. We will use mostly MS Teams to share information about the course, have online debates, and share educational material.	
Assessment	The assessment method for this course consists of two individual exams and a group-assignment consisting of an essay. Each exam will consist of a multiple choice and an open question part. The final grade for this course is determined according to the following scheme: students receive partial grades for the Scientific Values part of the course and for the Social Values part of the course. The final course grade is the plain average of these two partial grades, provided each is at least 5.76. The partial grade for the Scientific Values part is determined by the grade for a written exam. The partial grade for the Social Values part is determined as the plain average of a grade for a written exam and a grade for a group essay, where both grades need to be at least 5.76. Students do not receive a grade for the course in Osiris until all provisions indicated in the preceding have been met. As long as this is not the case, students can access an overview of their grades so far in Brightspace. Sufficient grades received for assessment components remain valid until a sufficient grade for the course has been reached. If unforeseen circumstances occur, such as COVID-19 related measures that prohibit exam-taking on campus, we will deviate from the above determined assessment method as follows: The individual exam(s) affected by the circumstances will be held online as a Möbius open book exam.	
Enrolment / Application	MOT and orientation students will be automatically enrolled in Brightspace. EPA, CoSEM and IE students who wish to apply for this course can do so by filling in the following form: https://tudelft.fra1.qualtrics.com/jfe/form/SV_3RkkO6GetqrStb8 Please be aware that application does not guarantee that you will be admitted. Since we only have a limited of places available we recommend you to look for an alternative course. On this webpage you can find a list of alternative courses: https://www.tudelft.nl/en/student/faculties/tpm-student-portal/education/courses-and-examinations	
Tags	Ethics Philosophy	

MOT1461	Financial Management	5
Module Manager	Dr.ir. Z. Roosenboom-Kwee	
Instructor	A.A. Ralcheva	
Responsible for assignments	Dr.ir. Z. Roosenboom-Kwee	
Responsible for assignments	Dr.ir. Z. Roosenboom-Kwee	
Co-responsible for assignments	Dr. S.T.H. Storm	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	This course has reached its maximum capacity and therefore we can only admit our regular (TPM and orientation) students. There is limited place available for students from EPA, CoSEM and IE students. MOT STUDENTS HAVE PRIORITY. Bachelor students are excluded from participation.	
Expected prior knowledge	None	
Summary	This course is about how corporations make financial decisions. It is designed to equip technical/engineering students with fundamental financial management skills.	
Course Contents	The course covers topics such as financial statement analysis, types of financing and investment decisions, and financial consequences of those decisions. While investment decisions involve spending money, financing decisions involve raising it. Important to highlight is that corporations finance their investments by borrowing, by retaining and reinvesting cash flow, and by selling additional shares of stock to the corporations shareholders. Hence, there will be among others three broad financial questions discussed in the lectures. First, what investments should the corporation make? Second, how should it pay for those investments? Third, what plans can the firm make to ensure that it stays financially strong and healthy?	
Study Goals	At the end of the course, students are expected to acquire the ability to: evaluate financial performance of companies through financial statement analysis evaluate investment opportunities using investment appraisal techniques such as internal rate of return (IRR), net present value (NPV), payback period, profitability index understand and discuss different types of long-term (debt and equity) financing such as venture capital, initial public offering, corporate debt) financial instruments and markets perform valuation of bonds and stocks analyze choices of capital structure and portfolio investment evaluate investment and capital structure decision risks (e.g. financial distress) understand and analyse dividend payout policy	
Education Method	The lectures and workshops are on campus. However, depending on the COVID-19 measures, the lectures and workshops may be given through a hybrid form or through online and remote learning.	
Literature and Study Materials	Lecture slides and textbook	
Books	Jonathan Berk and Peter DeMarzo (2020). Corporate Finance: The Core. Fifth Edition, Pearson. ISBN: 9781292304151. The fourth edition is also acceptable.	
Assessment	The course assessment is a written exam through Digital exam on campus. The exam is open questions and closed book. Only a calculator and a pen are allowed. Dictionary is not allowed. In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be modified as follows: Digital Exam (to be announced)	
Enrolment / Application	MOT and orientation students will be automatically enrolled in Brightspace. EPA, CoSEM and IE students who wish to apply for this course can do so by filling in the following form: https://tudelft.fra1.qualtrics.com/jfe/form/SV_bdQf6ZAuEILSEjI Please be aware that application does not guarantee that you will be admitted. Since we only have a limited of places available we recommend you to look for an alternative course. On this webpage you can find a list of alternative courses: https://www.tudelft.nl/en/student/faculties/tpm-student-portal/education/courses-and-examinations	

MOT1524	Leadership and Technology Management	5
Module Manager	Dr. R.M. Verburg	
Responsible for assignments	Dr. R.M. Verburg	
Co-responsible for assignments	Dr. L. Rook	
Contact Hours / Week x/x/x/x	3/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	This course has reached its maximum capacity and therefore we can only admit our regular (TPM and orientation) students. There is limited place available for students from EPA, CoSEM and IE students. MOT STUDENTS HAVE PRIORITY. Bachelor students are excluded from participation.	
Course Contents	Leadership and management in the context of high technology firms and digital business are the key theme of this course. The core of successful management in technology-based environments is the recognition of the value of employees for organizational success. Selection, development and performance management of employees are core activities of personnel management. Such practices are no longer seen as tasks that are the responsibility of a separate personnel department, but they become part of top managements personnel strategy. Line managers are increasingly expected to play an active role in implementing management policies and practices. Leadership appears to be a critical determinant of organizational effectiveness whether we are discussing a football team, an army, a street gang or a multinational corporation. Leadership is the subject of intense academic research and debate and journalistic commentary. In this course we will highlight the critical roles of leadership on different levels for the success of technology-based companies.	
Study Goals	At the end of this course, students are expected to competently: - analyze the nature of leadership management within advance technology organizations - provide arguments how leaders contributes to organizational performance - recognize knowledge processes in organizations - recognize management practices and analyze the success of these within different organizations - show the need for alignment between management practices - show the need for alignment between management practices and business strategy - recognize best practices with regard to the selection and performance management of employees - qualify scientific research in the area of Leadership and Organizational Behavior - recognize the current and future challenges of technology firms in leading and managing people	
Education Method	The course consists of weekly lectures. The lectures are dynamic and participative including: - Theory presentations and interactive lectures by the staff (in class and online) - Presentations by a guest speaker from practice - Presentations by students who work together in expertise groups - Case write-ups	
Assessment	Individual final examination The final exam will be a closed book exam on the content of this course. Hence, students who take the course seriously and structurally prepare the literature and assignments will be very likely to pass the exam. The exam will consist of Open Questions. A minimum grade of 5.75 in the exam is necessary for a passing this course. In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be modified as follows: Take-home assignment (open book), i.e. an Individual assignment based on open-ended question(s), typically to be delivered within period of a few days	
Enrolment / Application	MOT and orientation students will be automatically enrolled in Brightspace. EPA, CoSEM and IE students who wish to apply for this course can do so by filling in the following form: https://tudelft.fra1.qualtrics.com/jfe/form/SV_8jZbDsClxC1FHBV Please be aware that application does not guarantee that you will be admitted. Since we only have a limited of places available we recommend you to look for an alternative course. On this webpage you can find a list of alternative courses: https://www.tudelft.nl/en/student/faculties/tpm-student-portal/education/courses-and-examinations	

MOT2313	Research Methods	5
Module Manager	Dr.ir. G.A. de Reuver	
Instructor	Dr. L. Rook	
Co-responsible for assignments	Dr. L. Rook	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	This course has reached its maximum capacity and therefore we can only admit our regular (TPM and orientation) students. There is limited place available for students from EPA, CoSEM and IE students. MOT STUDENTS HAVE PRIORITY. Bachelor students are excluded from participation.	
Course Contents	Student learn how to design and execute research, as well as how to interpret and critically assess the outcomes of research. Both quantitative and qualitative research methods are taught. To do so, students learn the fundamentals of research design as well as quantitative and qualitative research methods. This course is designed to help students develop and execute research steps from the initial research problem until drawing conclusions. These steps include conceptualization of research ideas/problems, development of a relevant theoretical framework and related hypotheses, defining and operationalizing core concepts, and developing an appropriate research design. Students learn how to scrutinize the findings and draw implications drawn from statistical analysis and qualitative analysis of data. Topics that will be covered among others: - define research problem, research objectives and questions - conduct research process of scientific investigation - develop a conceptual model - formulation of hypotheses - operationalize and develop measurement of concepts - develop research design, as well as make motivated decisions on data collection and data analysis methods - evaluate different types of quantitative techniques (hypothesis testing, non-parametrics, regression analysis, multivariate analyses) - interpret and critically reflect upon findings from statistical analysis - qualitative data analysis, based on coding principles - interpretation and presentation of results - reflection on reliability and validity of research results - research ethics	
Study Goals	1. Create a research design for a research problem relevant to the context of MOT To do so, sub-goals are to: 1a. Understand core concepts of research design (i.e. theory, population, sample, hypotheses, research strategy, operationalization, data collection) 1b. Apply these core concepts to design a research project (i.e. from a broad topic towards research questions, conceptualization, research strategy, research design, data collection approach, sampling, operationalization, and selection of analysis methods) 2. Critically evaluate outcomes of qualitative and quantitative research in the MOT context To do so, sub-goals are to: 2a. Interpret statistical analysis findings 2b. Scrutinize / Reflect upon findings / conclusions from both qualitative and quantitative research (i.e. validity, reliability, traceability, ethics) 2c. Suggest alternative / future research directions	
Education Method	Course contains three main parts: - Research design - Qualitative research methods - Quantitative research methods Part 1 and 2.1 consist of: - Lectures, which are a mix of instruction and formative assignments. - Summative assignments (weekly, group of 2 students) in which a research design is created Part 2.2 consists of: - Lectures and practice	
Literature and Study Materials	To be announced via BrightSpace. At least one textbook on research methods for business.	
Assessment	Summative assignments (15% = 0.75 EC): Group of 2 students, weekly assignment, week 1-5 of the course. The summative assignments cover part 1 and 2.1 of the course. Digital exam (85% = 4.25 EC; at least 5.0 to pass the course). The exam covers part 1, 2.1 and 2.2 of the course. The exam contains open questions and is closed book. In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed exam will be modified as follows : Timed digital (Möbius) open book exam	
Enrolment / Application	MOT and orientation students will be automatically enrolled in Brightspace. EPA, CoSEM and IE students who wish to apply for this course can do so by filling in the following form: https://tudelft.fra1.qualtrics.com/jfe/form/SV_3ObBU6LgvAgQ0Hs Please be aware that application does not guarantee that you will be admitted. Since we only have a limited of places available we recommend you to look for an alternative course. On this webpage you can find a list of alternative courses: https://www.tudelft.nl/en/student/faculties/tpm-student-portal/education/courses-and-examinations	

Year	2022/2023
Organization	Applied Sciences
Education	Master Chemical Engineering

2nd Semester MoT Modules 2022

MOT1003	Integration Moment	5
Module Manager	Dr. R.M. Verburg	
Instructor	Prof.dr.ir. A. Verbraeck	
Instructor	Dr.ir. G.A. de Reuver	
Instructor	Dr. N. Pachos-Fokialis	
Instructor	H.A. Ofe	
Instructor	Dr. S. Hinrichs-Krapels	
Responsible for assignments	Dr. R.M. Verburg	
Co-responsible for assignments	Prof.dr.ir. A. Verbraeck	
Contact Hours / Week x/x/x/x	0/0/0/9	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Required for	Please, be aware that students, who are not enrolled in the MOT program, cannot take this course as an elective. This course constitutes an integration of the Management of Technology master, which requires knowledge of the core modules of the program. Therefore, the course is not an elective but for MOT students and for a selected group of students who are enrolled in the MOT orientation program only.	
Course Contents	The Master program in Management of Technology (MOT) educates students with a bachelor degree in engineering as technology managers, innovation managers, analysts of technological markets (either as scientists or consultants), and entrepreneurs in technology-based, internationally-oriented competitive business environments. The ultimate objective of the program is to improve the quality of technology and innovation management in the different engineering mono-disciplines in practice. Students contribute to scientific work in the area of MOT and have the opportunity to apply their knowledge in real-world settings, such as advanced technology organizations, laboratories, and high-tech business ventures. The program deliberately aims at an international and diverse group of students. Students of MOT are all rooted in different engineering mono-disciplines and work together in order to combine insights from the different engineering disciplines and to apply the different aspects of technology and innovation management. In this particular course, we aim to integrate knowledge and skills as put forward in the first year of the MOT program by applying this knowledge and skills through a consultancy assignment for MOT related business cases of organizations and companies. You will work as a technology management consultancy firm for a client with the aim to reflect on knowledge and skills needed to work in a typical MOT environment.	
Study Goals	At the end of this course, students are expected to: At the end of this course, students are expected to: 1. Apply MOT concepts from the first-year program on a case study. 2. Show a thorough understanding of the (inter-) relationships between the different parts of the MOT program. 3. Write a well-argued consultancy report and present your advice to a client.	
Education Method	Class members are divided into teams of three and assigned the task of doing a consultancy project. Each team will collaborate with three other teams on a business case for an actual client. Participants are supported by different MOT coaches during the progress of the assignment. The aim is to provide a solution to a current business problem in a coherent consultancy report in which the following four aspects as underlined in the MOT program are integrated A. Organizational angle (see e.g. MOT1524, MOT1531), B. Commercial angle (see e.g. MOT1434, MOT2421), C. Financial (investment) angle (see e.g. MOT1421, MOT1461), D. Safety, security, responsibility angle	
Assessment	Each of the partial reports will be graded on a scale from 1-10. The total report will be graded from a consistency standpoint as well, which can influence the mark of the partial report for a group. The grade for the final report will be determined by your case coach in collaboration with a specific content coach for technology, innovation, finance and security/safety on the basis of a detailed grading grid. The grid includes the following criteria: 1. The use of MOT concepts and theories(1-10) 2. Understanding of the core issue/topic(1-10) 3. Advice to the client (1-10) 4. Argumentation, and quality of the report(1-10) 5. The contribution of your particular group topic(1-10) The five criteria will weigh equally. You need a minimum of a 6,0 in total to pass the course. As the grading is project- based, the re-take opportunities are limited to re-writing (parts of) the report in the case of a grade lower than 6,0.	

MOT1435	Technology, Strategy and Entrepreneurship	5
Module Manager	Dr. G. van de Kaa	
Instructor	Dr.ing. V.E. Scholten	
Instructor	Prof.dr. C.P. van Beers	
Responsible for assignments	Dr. G. van de Kaa	
Co-responsible for assignments	Prof.dr. C.P. van Beers	
Contact Hours / Week x/x/x/x	0/0/5/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	This course has reached its maximum capacity and therefore we can only admit our regular (TPM and orientation) students. There is limited place available for students from EPA, CoSEM and IE students. MOT STUDENTS HAVE PRIORITY. Bachelor students are excluded from participation.	
Summary	The course Technology, Strategy and Entrepreneurship focuses on formulating and implementing technology strategy for large firms and entrepreneurs. However, before a manager can formulate a successful strategy, it is needed to understand the specifics of the external economic and societal environment in which the strategy will be implemented. Throughout the course, students will be acquainted with a variety of academic perspectives which are utilized by strategy and entrepreneurship researchers.	
Course Contents	Greater product and technology complexity and increased internationalization of innovation activities have increased costs and risks for innovators such that the boundaries of the firms have changed and are changing. Specifically, we explain the typical patterns of technological innovation in order to provide a foundation for the formulation of a technology strategy. Furthermore, we focus on identifying the reasons why most industries adopt a dominant design and why a particular firm's technology is adopted as the dominant design. We will examine the factors that affect the outcome of standards battles. These include e.g. timing of entry strategies, collaboration strategies, and appropriability strategies. We will provide a basic background of entrepreneurship. Specifically, we will pay attention to networking strategies for new firms (entrepreneurs). How do these firms attempt to achieve success with their new technology, in a new market? As the boundaries of the firms are getting more transparent due to greater complexity of technical projects, it is required to decide whether to collaborate with external partners, and with whom. This can be a good option for entrepreneurial firms as they lack the necessary complementary assets to successfully commercialize a new technology. We will explore collaboration strategies for entrepreneurial firms. Technological innovation is a much more uncertain activity for a firm than other activities. Strategic thinking and formulation require knowledge on the boundaries of the firm and the markets in which the firm operates, its value chain and its core competencies. Also, strategy formulation aims at controlling the external environment of the firm, which is characterized by uncertainties. A distinction can be made in changes in the external environment in time or by location. General Purpose Technologies affect strategy formulation in time and frugal or inclusive innovations by location. After formulating the sources and innovation strategies we will pay attention to the resources firms have for innovation processes and specific valuation techniques, like real options, conjoint analysis and data envelopment analysis.	
Study Goals	After the course students are able to - understand the theoretical background of technology and innovation strategy. - position a variety of strategy topics thereby taking into account sectoral and country context differences. - understand the theoretical background of entrepreneurship. - apply key technology strategy models in different sectoral and country contexts. - understand the relations between those models. - analyse the industry dynamics of technological innovation. - analyse companies technology strategy. - analyse and explain a company's technological innovation strategy vis-a-vis its strategic or entrepreneurial posture - analyse and explain the organization and structure of new product development teams based on the firm's technological innovation strategy	
Education Method	This is a blended course and the course includes regular sessions, games, and teaching cases. Each week will start with a spark which will be communicated through Brightspace. On the first day the teacher will discuss the spark with you. On the fourth day of the week, a regular interactive lecture is planned. On the fifth and final day of the week quizzes are available. Each regular session will consist of lectures. In the teaching case sessions students discuss questions posed in assigned teaching cases that relate to the topics studied in the regular sessions. For the regular sessions students study the literature and for the teaching case sessions students prepare the teaching cases. In the teaching games students apply the theory by implementing and evaluating various strategies in a real life situation.	
Literature and Study Materials	Book: Schilling, M. A. 2019. Strategic Management of Technological Innovation. New York, USA, 6th edition Journal articles: Will be posted in Brightspace.	
Assessment	Slides presented in class: Will be posted in Brightspace. The final grade for this course is determined as follows: 1. Individual exam on book, literature and lectures (70%) 2. quizzes (30%) o In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be modified as follows: the individual exam will be replaced with a timed take-home exam (open book). This will be a remote examination.	
Enrolment / Application	MOT and orientation students will be automatically enrolled in Brightspace. EPA, CoSEM and IE students who wish to apply for this course can do so by filling in the following form: https://tudelft.fra1.qualtrics.com/jfe/form/SV_3EpjZgyQcSYv7Fk Please be aware that application does not guarantee that you will be admitted. Since we only have a limited of places available we recommend you to look for an alternative course. On this webpage you can find a list of alternative courses: https://www.tudelft.nl/en/student/faculties/tpm-student-portal/education/courses-and-examinations	

MOT1452	Inter- and intra-organisational decision making	5
Module Manager	J. Lieu	
Instructor	Prof.mr.dr. J.A. de Bruijn	
Instructor	Dr. W.W. Veeneman	
Instructor	J. Lieu	
Instructor	Dr. L.D. Virla Alvarado	
Responsible for assignments	Dr. W.W. Veeneman	
Co-responsible for assignments	J. Lieu	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Required for	This course has reached its maximum capacity and therefore we can only admit our regular (TPM and orientation) students. There is limited place available for students from EPA, CoSEM and IE students. MOT STUDENTS HAVE PRIORITY. Bachelor students are excluded from participation.	
Course Contents	All organizations whether large or small, public or private, old or new make decisions on a regular basis. One might assume that these decisions taken inside and between organizations are well-planned and based on sufficient information. But in reality, information is often incomplete and contested among the actors making a decision. These actors may also have competing interests, for instance governments that favor different policies, or companies that pursue their individual profits. As a result, complex decisions often appear chaotic and ill-structured. The primary purpose of this course is to help understand why it is thus impractical to solve multi-actor decision problems using a purely analytical approach; Why carefully, pre-conceived plans can backfire when hierarchically imposed, and force even ministers and CEOs to the negotiation table; And why decision-making in networks can be characterized as a permanent struggle and exercise of power among interdependent actors. The course introduces students to theories and models which help explain how decisions are made in practice, i.e. what decision-makers really do, and why they do it that way. It acquaints students with networks and strategies that can be used to manage actors and interests. During lectures, students apply these theories to real-worlds examples through group discussions. In parallel, they analyze a case study in depth, reconstructing and evaluating the decision-making process, and writing a paper about it. By the end of the course, students will learn to design processes and strategies that enable change in multi-actor settings.	
Study Goals	At the end of this course, students will be able to achieve the following learning objectives: 1: Explain key concepts, perspectives, as well as negotiation and framing strategies on decision-making within multiple-actor networks; 2: Explain individual and collective decision biases in different types of organisations 3: Evaluate the decision process surrounding a complex real-world problem 4: Analyse a decision concerning a recent real-world wicked problem 5: Apply negotiation and framing strategies used in multi-actor decisions	
Education Method	This course consists of lectures, tutorials, self-study, and assignments. During lectures, we will synthesize the required readings. News events and large projects are often used as examples. (Students are expected to have read the relevant literature beforehand). During tutorials, students can discuss their papers as they progress with their case analysis. Interim assignments help students along the way.	
Literature and Study Materials	The required reading for the course includes (chapters of) the following books: 1. "Thinking, Fast and Slow", by Daniel Kahneman, 2013. 2. Hans de Bruijn et al., Management in Networks, Chapter 4: A process, not a project. Routledge, London. DOI: 10.4324/9781315453019 In addition, a number of book chapters and papers are uploaded/linked to on Brightspace. Some of them are mandatory reading; others give additional context or examples.	
Assessment	The final grade for this course is determined as follows: 1. Individual Assignment (35%): critical reflection of theoretical knowledge and reflection of personal learning process 2. Participation (15%): assess theoretical knowledge and motivate active participation and discussions during lectures. 3. Group Paper (50%): assess the ability to apply theory to a real case and write a well-argued paper in a multi-actor team setting. To pass, students need a minimum of 5.8 on both the individual assessment and group paper. Group papers are only graded if the mandatory assignments have been submitted on-time.	
Enrolment / Application	MOT and orientation students will be automatically enrolled in Brightspace. EPA, CoSEM and IE students who wish to apply for this course can do so by filling in the following form: https://tudelft.fra1.qualtrics.com/jfe/form/SV_2rY1TinXlrEIOFY Please be aware that application does not guarantee that you will be admitted. Since we only have a limited of places available we recommend you to look for an alternative course. On this webpage you can find a list of alternative courses: https://www.tudelft.nl/en/student/faculties/tpm-student-portal/education/courses-and-examinations	

MOT1531	Digital Business Process Management	5
Module Manager	Prof.dr.ir. M.F.W.H.A. Janssen	
Instructor	Dr. J. Ubacht	
Instructor	Dr. Y. Ding	
Responsible for assignments	Prof.dr.ir. M.F.W.H.A. Janssen	
Co-responsible for assignments	Dr. J. Ubacht	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	This course has a maximum capacity of 200 students. There is limited participation for students from other Master programs. MOT STUDENTS HAVE PRIORITY. Bachelor students are excluded from participation.	
Course Contents	Business processes are at the core of high-tech organizations and consist of a large number of activities aimed at creating value for customers. These processes can be executed by humans or by digital technology. In the digital age business processes are supported by, enabled and changed by technologies like the Internet of Things, Artificial Intelligence, Robotics and so on. Business processes are situated at the point where the business side meets the IT side and where technology should fulfill customer needs. If business processes fail or do not adapt to the needs of customers the whole business might fail. Business Process Management (BPM) is necessary for ensuring that processes are operating in concert and that these processes are adaptable to changes in the overall environment of an organization. In this course the relationship between strategy, technology and business processes will be illustrated, explained and discussed. Methods and tools for business process analysis and improvement will be presented, including a number of statistical and ICT-architecture methods. New solutions will be architected for high-tech organizations. The extent to which the nature of business processes is affected by contemporary technology constitutes the core of this course. In particular to analyze technologies and their impact on business process management and to architect new solutions in the organizational context of the firm. Topics - Aligning strategy and processes, value perspective, supply chains - Workflow, resources, BPM elements and control - Business process modeling (concepts, BPMN) + practical work - Process measurement, Balanced Score Card, Activity-based Costing (ABC) + practical work - Business process improvement strategies and approaches, path dependencies, capabilities, resource-based view, business process maturity - Improvement methodologies (LSS, lean, six sigma, theory of constraints, technology ..) - Statistical analyses and improvement + practical work - Compliance by design, build-in-controls in processes and technology - Business process automation and technology (BPM, ERP, workflow, SOA,...) - Business processes automation (case-based, process -based, use-based) - Agility and adaptability of BPM systems and the users, business process and agile enterprises - Business and knowledge rules - Internet of Things, data-driven organizations	
Study Goals	The overall aim of the course is to learn how to improve the relationship between business processes, strategy and technology for realizing organizational strategies. The focus is on realizing business processes enabled by digital technology for high tech organizations. In particular the objectives are - To analyze the relationship between business processes, strategy and technology - to understand new technologies and their impact on business processes - to be able to select and use methods and tools for improving business processes - to analyse and provide improvement suggestions for business processes and technology	
Education Method	-lectures -practical work -assignment	
Computer Use	Planning tool. There are suitable planning tools like Trello and Asana BPMN modelling tool. There are several modelling tools which are appropriated like Eunomia process builder (http://www.eunomia-process.com/), Bizagi (http://www.bizagi.com/) or LucidChart (https://www.lucidchart.com/) Activity Based Costing (ABS) calculations and simulations using a spreadsheets, like MS Excel.	
Literature and Study Materials	Parts of the book N. Bharosa, R. Van Wijk, N. De Winne & M. Janssen (2015). Challenging the Chain. Governing the Automated Exchange and Processing of Business Information. IOS Press http://www.iospress.nl/book/challenging-the-chain/ - open access ebook List of articles (containing classical papers and showing the state-of-the-art) Reader (containing an overview of main elements and explaining some methods) Exercises + answers (for practicing)	
Assessment	Students will collaborate in groups to make an assignment which should contain 1) the analysis 2) improvement and digitalization of a business process and 3) implementation aspects. Students are free to select a process that is critical or a company and can be improved by ICT. The assignment consists of three parts 1. Analysis: Students select a company, analyse their strategy and select a critical business process. The existing process should be modeled (in BPMN) and the resulting product, stakeholders role,s customer needs and technology used should be analysed. 2. Improvement: Based on the improvement methods and instruments a proposal for business process improvements should be made. 3. Realization: how the proposed business process should be realized by means of digital technology. The assignment should be handed in as a report. The report should adhere to the APA reference standards. The students are expected to prepare two presentations and some of the groups will be asked to present these during the lectures. Based on these presentations feedback will be given, which are relevant for all groups. Presentations will be assessed using a pass or fail. The report will be graded; this grade will reflect the final grade.	
Enrolment / Application	MOT and orientation students will be automatically enrolled in Brightspace. Other students who wish to apply for this course can do so by filling in the following form: https://tudelft.fra1.qualtrics.com/jfe/form/SV_787nY10vO4JW7zg Please be aware that application does not guarantee that you will be admitted. Since we only have a limited of places available we recommend you to look for an alternative course. On this webpage you can find a list of alternative courses: https://www.tudelft.nl/en/student/faculties/tpm-student-portal/education/courses-and-examinations You can apply until January 22.	

Elective	Yes
Tags	Economics Information & Communication Modelling Technology

MOT1534	High-tech Marketing	5
Module Manager	Dr. L. Rook	
Instructor	Dr. L. Rook	
Co-responsible for assignments	Dr. J.R. Ort	
Contact Hours / Week x/x/x/x	0/0/2x2/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	This course has reached its maximum capacity and therefore we can only admit our regular (TPM and orientation) students. There is limited place available for students from EPA, CoSEM and IE students. MOT STUDENTS HAVE PRIORITY. Bachelor students are excluded from participation.	
Course Contents	Companies in high-tech markets face constant uncertainties caused by market dynamics, technological progress and volatile customer preferences. In order to survive under such highly dynamic circumstances, companies must keep on innovating, and carefully select -- or modify -- appropriate high-tech marketing strategies. This course aims at familiarizing students with several theoretical and methodological approaches to high-tech marketing, and will also capture recent developments in marketing research, such as neuromarketing, and Web-based (2.0) and Mobile-based (3.0) technologies that render high-tech marketing prone to (online and offline) word-of-mouth dynamics. These issues will be discussed under reference to best practices (from various high-tech companies) that serve to illustrate ways in which to deal with high-tech marketing strategy problems.	
Study Goals	At the end of the course, students are expected to have gained knowledge on principles of high-tech marketing and will be able to recognize, analyze, reflect upon, and creatively respond to, high-tech marketing problems and challenges with regard to: - Marketing strategies and plans to deal with uncertainty in high-tech environments; - Customer segmentation, targeting, and positioning for business-to-business (B2B) and business-to-consumer (B2C) markets; - Product and service bundles development, (offline and online) word-of-mouth, advertising and promotion, incl. modern online tools; - Some marketing research techniques (various methodologies) aimed at analyzing customer preferences for regular as well as persuasive (Web 2.0/3.0) marketing strategic communications and technologies (recommender systems).	
Education Method	The course will consist of a series of lectures based on: - Theory discussions and interactive lectures based on the course materials; - Guest lectures from experts from academia and practice; - Analysis and evaluation in small groups of existing corporate marketing strategy problems in a self-chosen set of case companies. Analysis will be based on strategic assessment of the business case, and should be grounded in a customized marketing plan, and result in a creative proposal for a business-to-business (B2B) or business-to-consumer (B2C) marketing campaign. All these components in the mix should lead to specific recommendations on how the case company needs to cope with (some of) the most pressing high-tech marketing challenges. Individual academic reflections on the groupwork are required.	
Literature and Study Materials	The materials for this course are: - A course book (see below). - Academic articles from leading journals in Marketing Research (a.o., Marketing Science, J. of Marketing, J. of Marketing Research, J. of Consumer Research) and in the Automation, Computing Machinery (ACM) research community to be accessed via the TU Library online repositories.	
Books	t.b.a.	
Assessment	Students will be graded on the quality and rigor of a marketing research project to be executed in group setting, complemented with individual theoretical reflections upon the group assignment. Specifically: - A substantial group paper will (a) critically review the relevant literature on high-tech marketing to a relevant business case. At the heart of the group paper lies (b) empirical (qualitative and/or quantitative) analysis of a specific marketing problem, and (c) the adequate discussion thereof; analyses will inform (d) the development of a creative marketing campaign intended to solve the business challenge identified. - Each group paper must be supplemented with contributions (theoretical reflections) from each of the individual group members. - The research project may include peer review so as to determine the weight of individual contributions to the group work. - The assessment of the marketing research project will be based on a detailed grading rubric shared in advance. - A minimum final grade of 5.75 is a requirement for passing the course.	
Enrolment / Application	MOT and orientation students will be automatically enrolled in Brightspace. EPA, CoSEM and IE students who wish to apply for this course can do so by filling in the following form: https://tudelft.fra1.qualtrics.com/jfe/form/SV_24ybEwIWOWgnFs You can sign up until Sunday, January 22, 2023. Please be aware that application does not guarantee that you will be admitted. Since we only have a limited of places available we recommend you to look for an alternative course. On this webpage you can find a list of alternative courses: https://www.tudelft.nl/en/student/faculties/tpm-student-portal/education/courses-and-examinations	

MOT2421	Emerging and Breakthrough Technologies	5
Module Manager	Dr. J.R. Ortt	
Responsible for assignments	Dr. J.R. Ortt	
Co-responsible for assignments	Dr. L. Rook	
Contact Hours / Week x/x/x/x	0/0/0/8	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Required for	This course has reached its maximum capacity and therefore we can only admit our regular (TPM and orientation) students. There is limited place available for students from EPA, CoSEM and IE students. MOT STUDENTS HAVE PRIORITY. Bachelor students are excluded from participation.	
Course Contents	The course focuses on innovation processes in a high-tech environment. Firstly, innovation processes are studied on the micro-level, distinguishing stage-gate, agile and hybrid project approaches to innovation. Secondly, the course will discuss the development and diffusion of breakthrough technologies on the macro-level. The pattern, of the development and diffusion of such technologies, the actors and factors that explain the evolution of the pattern, and strategies that companies can adopt during the pattern are discussed. Thirdly, the course will shortly discuss how disciplines (defined as combination of related technologies) emerge and evolve. The three levels of analysis (project, pattern, discipline) together form a comprehensive perspective on innovation processes.	
Study Goals	Describe innovation processes on three levels: (1) project level, (2) level of the pattern of development and diffusion of technologies and (3) discipline level. Distinguish alternative theoretical perspectives on each level and the assumptions of these perspective. Be able to explore and analyse an innovation opportunity or problem, choose the level of analysis and apply the perspective on that level that fits the context of the opportunity or problem.	
Education Method	lectures and assignment. Normally lectures are face-to-face. If needed, due to circumstances, lectures will be on-line.	
Assessment	Exam and assignment: In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be modified as follows: 1. The exam will administered on-line using maple-ta or a similar program. 2. The assessment will remain the same.	
Enrolment / Application	MOT and orientation students will be automatically enrolled in Brightspace. EPA, CoSEM and IE students who wish to apply for this course can do so by filling in the following form: https://tudelft.fra1.qualtrics.com/jfe/form/SV_6mrjadvsL8ud8Eu Please be aware that application does not guarantee that you will be admitted. Since we only have a limited of places available we recommend you to look for an alternative course. On this webpage you can find a list of alternative courses: https://www.tudelft.nl/en/student/faculties/tpm-student-portal/education/courses-and-examinations	

Year	2022/2023
Organization	Applied Sciences
Education	Master Chemical Engineering

Science & Engineering 2022

Introduction 1

Science and Engineering (S&E) is suitable for students who pursue an academic career.
It consists of an external research project of 15EC, then the students must choose a MEP of 30 EC or 42 EC.
Consequently a choice of 3-15EC electives.

When choosing a MEP of 30 EC, then 15 EC of Electives must be choosen.
When choosing a MEP of 42 EC, then 3 EC of Electives must be choosen.

TUD4040	Joint Interdisciplinary Project	15
Responsible Instructor	Prof.dr.ir. J. Hellendoorn	
Project Coordinator	Ir. B.J.E. de Bruin	
Contact Hours / Week x/x/x/x	x/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Summary	JIP consists of three sets of activities: 1. The project which takes place at a company. Students are responsible for the project work (collaboration, integration of different perspectives, the content quality of their work). They plan their activities via a scrum method, are challenged to realise field trips, consult with experts and realise the necessary research work. The team keeps a Scrum log, run a blog for interaction with the outside world and every team member keeps a personal development log. 2. Meeting Professionals lectures and workshops about specialised topics, by academic staff or senior professionals from the companies involved. 3. Plenary meetings and design reviews for all the teams, company coaches and academic staff. At the meetings the students present their intermediate resp. final outcomes and plans, and are provided with feedback.	
Course Contents	The aim of the Joint Interdisciplinary Project is to prepare students to contribute to solving impactful technological challenges. The projects not only demand good engineering working knowledge but also experience with interdisciplinary and systems theory, and both knowledge and mindsets of innovation and entrepreneurial behavior. The project brief is provided by renowned companies like Airbus, Arcadis, etc. Teams of interdisciplinary student teams guided by a company coach and offered academic and industry expertise, are invited to realise an innovative problem solution to a complex problem and contributing to the sustainable development goals.	
Study Goals	1. Cognitive abilities attributable to interdisciplinary learning Demonstrate the ability to engage in perspective-taking; Develop structural knowledge pertaining to the problem; Integrate knowledge and modes of thinking drawn from two or more disciplines; Produce an interdisciplinary understanding of complex problem or intellectual question. 2. Scientific and intellectual development Capable to analyze scientific and societal consequences (economic, social, cultural, environmental) of the innovation; 3. Research and design capabilities Demonstrate engineering skills: technical skills, interpreting results, creativity, usability for company/institute; Demonstrate that they are capable to independently apply relevant theory and/or knowledge to research and/or design; 4. Collaboration and communication in an interdisciplinary team Demonstrate behavioral competences and skills: taking initiative, responsibility, showing communication skills, independency, collaboration and the ability to respect different disciplines and adapt to different cultures); Show ability to write a technical report: structured/consistent, language proficient, with correct use of literature/references, use of figures/tables/equations, and has a concise format (30 pages); Present work performed in a structured way through an oral presentation to their peers and customer. 5. Self-adjustment and reflection capabilities Plan and control the project efficiently considering resources and methodology; Being able to reflect on personal functioning in an evaluation report: reflect on personal objectives, indicate personal strengths/weaknesses. Indicate future personal improvement, drawing conclusions for future career. Cognitive abilities attributable to interdisciplinary learning: The ability to integrate (scientific and practical technological) knowledge from different disciplines to solve complex problems Scientific and intellectual development The capacity to evaluate the ethical, scientific and societal consequences of the proposed innovation Research and design capabilities The ability to create reasonable and relevant research or design , according to the academic standards of the involved disciplines Collaboration and communication in an interdisciplinary team Demonstrate behavioural competences and skills relevant for teamwork and effective communication with different stakeholders Self-adjustment and reflection capabilities To carry out regular reflections on professional and personal development and being able to improve upon those reflections Understand contemporary and societal issues in their work.	
Education Method	Full-time project work in an interdisciplinary team of about five students. The project work is interspersed with some just in time seminars and workshops about specialized subjects, methods or practical situations that play an essential role in interdisciplinary project work.	
Assessment	The assessment criteria pertain to the process, product and presentation and are assessed at an individual and team level, during 3 review sessions at three levels of accomplishment (each review assesses at a different level): Interdisciplinary work Scientific reasoning and ethical mindset Innovation process Presentation and communication Academic staff, supported by the input from company coaches, grade the students during the Problem Statement, the Midterm and the Final Review. The final group mark for JIP is differentiated per person and is based on: The team presentations - 30% The Problem definition, progress report and final report 30% The individual contribution to the team 20% The final individual reflection report 5% The Blog 5%	
Remarks	Project Outcomes (product, model, system) are presented at: 1. Problem Statement Review 2. Midterm meeting 3. Final Review 4. Symposium presentation	

Year	2022/2023
Organization	Applied Sciences
Education	Master Chemical Engineering

Free electives

AS3111	ATHENS	2
Course Coordinator	Drs. E. Jacobs	
Contact Hours / Week x/x/x/x	-	
Education Period	Different, to be announced	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	Twice a year you have the opportunity to follow a one-week intensive course, organised by several European Universities. This programme is called the ATHENS programme (Advanced Technology Higher Education Network, Socrates). More information: https://www.tudelft.nl/en/student/study-and-career/study-internship-abroad/university-exchange-partners-networks/athens/athens-outgoing-students/ and http://athensnetwork.eu/	
Study Goals	Recommended for 2nd year MSc students (only)! The courses allow you either to deepen your knowledge within your area of studies or to discover a new discipline. The courses are held in English and/or French. Each course includes about 30 hours of scientific course work. Another 10-15 hours go towards European Dimension activities; you will not only discover the university and the city, you will also meet other international students and have a great time.	
Education Method	7 day intensive course - study abroad (BME Budapest, KU Leuven, IST Lisbon, UPM Madrid, PM Millano, TUM Munich, CTU Prague, AUT Thessaloniki, NTNU Trondheim, TU Vienna, WUT Warschau, ITU Istanbul, UPB Bucharest, ParisTech)	
Practical Guide	https://www.tudelft.nl/en/student/study-and-career/study-internship-abroad/university-exchange-partners-networks/athens/athens-outgoing-students/	
Assessment	No grading. The result is indicated by a V ("Voldaan" = Pass), if a certificate is obtained at the foreign university and handed-in at the international-office of the faculty of Applied Sciences. It's not possible to earn credits more than once. No credits are awarded if you return to your home country.	
Enrolment / Application	Application Deadline: 1 October 2022 / 20 January 2023	

AS3121	Scientific Writing and Argumentation	3
Responsible Instructor	Dr. F.C. Grozema	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Expected prior knowledge	Excellent command of English language, level C1/C2	
Summary	Writing a well-argued text, on scientific results obtained in end project but also in engineering practice, is an important skill for every engineer. The aim of this course is to improve the argumentation and writing skills of students, making them aware of the importance of clear technical writing in engineering and science and offer tangible approaches to achieve this.	
Course Contents	1. Writing in science and engineering: the scientist as a professional writer; 2. Making a story sticky; 3. Story structures: a. Opening, Challenge, Action, Resolution (OCAR) b. Action, Background, Development, Climax, Ending (ABDC) c. Lead, Development (LD) d. Lead, Development, Resolution (LDR); 4. Targeting different audiences; 5. Internal structure: arcs of tension; 6. Good/bad paragraphs, sentences and words; 7. Flow: connecting the parts; 8. Different levels of reasoning: descriptive, relational, linear, multicomponent; 9. Data presentation, deceptive conclusions and overselling.	
Study Goals	1. Students are familiar with basic concepts of the structure of a story for different types of writing; 2. Student are familiar with the internal structure of paragraphs and sections; 3. Students are familiar with different types of reasoning; 4. Student can recognize and comment on the story structure (or absence of good structure) in a written text; 5. Students can design a story structure for a text targeted at a specific audience; 6. Student can recognize levels of reasoning and improve the level of reasoning in their own writing; 7. Students are familiar with the optimal way of presenting data and avoiding deceptive conclusions and overselling.	
Education Method	The course consists of 7 weeks, 4 hours a week consisting of instruction (~2 hours) and a practical part (~2 hours) writing shorts text, reviewing published scientific papers, reviewing/editing each others writing.	
Literature and Study Materials	Book: Writing Science by Joshua Schimel (Oxford University Press, 2012). Reader/handouts with additional material and examples of good and bad scientific writing.	
Assessment	Homework assignments (20%) and written exam (80%)	
	Note: the exact format of assessment may change due to Coronavirus (COVID-19) regulations.	
Elective	Yes	

AS3131	Art, Empathy & Ethics	4
Responsible Instructor	Dr. E. Mendes	
Instructor	Dr. F. Santoni De Sio	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 4	
Course Language	English	
Course Contents	Art is a sophisticated activity binding the rational and emotional counterparts of our species. It makes a unique contribution to society by offering various means of expression and exchange and it is a deep investigation tool for complex human issues. Art plays an important societal role in expanding the public imagination by engaging them in a challenging sensorial - intellectual dialogue, increasing the perception of themselves as individuals and society. This course invites you to experience and produce art rather than simply broaden your knowledge in contemporary art or history of art. Under the supervision of an artist and using the tools and media suggested by the artist, you will develop an art project around an ethical or a societal problem related to the design and use of technology. You will be led to perceive and portrait some relevant aspects of a chosen societal problem, questioning in a completely open and creative way, the consequences, limitations, advantages or any other aspect associated with a chosen technology, as perceived by those who experience the problem or the (mis)use of a that (imagined) technology. You will examine the situation through the eyes of your inner artist, connecting with society through an empathic pathway. As examples of tools and media we can cite, not exclusively, the case of visual arts: painting, printing, photography, installations, graffiti, films, etc. As examples of global societal problems, we can cite boat people, misuse of big data, consequences of genetically modified organisms, facial recognition, etc. The theme and art media will vary from educational period to educational period. Theme of this year : "Connectivity" This empathic and aesthetic exercise is to be complemented by a systematic conceptual reflection on the chosen ethical or societal problem, both carried out in parallel. That is achieved via self-study of proposed material on Ethics where you will explore the ethical and societal aspects and problems related to technologies. You will then write an Ethics Assignment on a topic of your choice that is coupled to your art trajectory and Final Art Assignment. In practice, this course will offer one compulsory weekly evening session for the development of the art project. All sessions are guided and supervised by an artist. Art sessions take place on campus at X Delft (as long as Covid-19 regulations and Campus resources allow for).	
Study Goals	After completion of this course, the students can: - recognise their own way to develop a concept or reasoning that has an emotional or empathic origin - reflect on the role of art in expressing and questioning risks associated to new technologies - understand how people immersed in societal problems perceive that problem in their daily lives - communicate via own artistic production, emotions, questions, ideas related to the risk associated to new technologies - reflect on and anticipate the emotional reaction of society or specific social groups to new technologies - empathically participate in the emotional-moral debate associated to state-of-the-art science and technologies - make use of divergent thinking strategies that incorporate creative (artistic) approaches - interpret various aspects of avant-garde art in order to challenge their personal responsibility (position) as engineers - improve their spoken and visual communication through creative expression	
Education Method	Compulsory week interactive evening session in the ateliers of X - Delft" (as long as Covid-19 regulations and Campus resources allow for). Self-study of Ethics material	
Assessment	Art part (60% of grade) - Active participation during the working art group sessions, via art assignments, artistic experimentation, brainstorming, etc. (30%) - Your artistic output in the form of a Final Art Assignment (including how it reflects the emotional-moral process that you went through and the critical thinking emanating from the emotional art experience) (30%) Ethics part (40% of grade) - A written short ethical assignment coupled/inspired by the art trajectory Minimum Grade - A minimum of 5.0 has to be obtained in each of the individual parts above (Art and Ethics)	
Enrolment / Application	Just email directly to E. Mendes (e.mendes@tudelft.nl) with cc: to Marlijn Helder (m.helder@tudelft.nl) with a motivation letter (max 1/2 A4), preferably two weeks before the course starts because places are limited. PLEASE, use email subject: enrolment AS3131 : Art, Empathy and Ethics	
Location	The weekly compulsory interactive Art sessions take in the Amphitheater of X - TU Delft" (former TU Delft Culture Center)	

TPM301B	Spoken English for Academic Purposes Intermediate	2
Module Manager	Drs. M.A. Swennen	
Instructor	S.F. Johnson	
Instructor	Drs. D.W. Laponder	
Instructor	Drs. K.I. van der Linden	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	3/3/3/3	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	This speaking course focuses on the language and skills students need to participate successfully at the university. Students will engage in a number of activities both inside and out of the classroom - to practise and improve their English. The vocabulary focus will be on general academic language, so that after the course students can interact with more confidence in an academic setting.	
Study Goals	The aim of the course is to develop language skills needed for the university environment. You will learn phrases and words for a range of skills needed here, for example participating in discussions; rephrasing and explaining; giving an opinion, (dis)agreeing, making suggestions; describing and presenting a concept.	
Education Method	Group and pair work, combined with assignments to do at home. Feedback on course components is given by the course lecturer and other students	
Literature and Study Materials	To be announced in Brightspace.	
Practical Guide	This is a course for both BSc and MSc students who would like to feel (more) confident when communicating with others at the university.	
Assessment	When you have taken our Placement Test you will be recommended a language level, either Intermediate or Advanced. With this recommendation you can sign up for both of the courses offered at that level. Class participation and weekly assignments. An 80% attendance rate is obligatory, which means you may only miss one session. To keep your place in the group you must attend the first session. The assignments must be submitted each week before the deadline set by the instructor.	
Enrolment / Application	You can drop the course with no penalty before the second session. Be aware of your "commitment" from the start, as you will still receive a grade [NV: Niet Verschenen--Not Attending], in Osiris even if you stop coming to class after session 2. Before enrolling in a group you must first take our Placement Test. This is not an actual test but a tool to ensure that you are taking the course that is the most suitable for your level of English. For more information on this 'test' please visit our website: www.itav.tudelft.nl Enrolment for the course (and the Placement Test) is via Brightspace. You do not need to register for this course in Osiris.	
Remarks	Please note that you must also choose a group to reserve a place in the class. Type 'TPM301B' in Course search (Catalog). Please make sure you select the correct academic year and quarter. Register in the course AND in a group that suits your schedule. You can drop the course with no penalty before the second session. Be aware of your "commitment" from the start, as you will still receive a grade [NV: Niet Verschenen--Not Attending], in Osiris even if you stop coming to class after session 2.	
Elective	Yes	
Tags	English proficiency Group work Small groups	
Category	BSc and MSc students	

TPM302B	Spoken English for Academic Purposes Advanced	2
Module Manager	Drs. M.A. Swennen	
Instructor	S.F. Johnson	
Instructor	Drs. D.W. Laponder	
Instructor	Drs. K.I. van der Linden	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	3/3/3/3	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	This speaking course focuses on expanding and improving the language students need to participate at the university. A particular focus point of the course is the vocabulary needed at a higher level of English, both in an academic and a more general context. A range of sources for language input is offered, such as podcasts, videos, websites, or (academic) articles. The sessions revolve around student interaction.	
Study Goals	The first aim of the course is to improve and expand the language skills needed at a higher level of English. These include paraphrasing and sharing information with others, using polite language, giving feedback, and presenting information. Another study course goal is to expand and improve the students' general vocabulary.	
Education Method	Group and pair work, combined with (individual) assignments to do at home. Feedback on course components is given by the course lecturer and other students.	
Literature and Study Materials	To be announced in Brightspace.	
Practical Guide	This is a course for both BSc and MSc students who would like to expand and improve their English language skills. When you have taken our Placement Test you will be recommended a language level. With this recommendation you can sign up for either of the courses offered at that level.	
Assessment	For this course you need a recommendation to take courses at the Advanced level. Class participation and weekly assignments. An 80% attendance rate is obligatory, which means you may only miss one session. To keep your place in the group you must attend the first session. The assignments must be submitted each week before the deadline set by the instructor.	
Enrolment / Application	You can drop the course with no penalty before the second session. Be aware of your "commitment" from the start, as you will still receive a grade [NV: Niet Verschenen--Not Attending], in Osiris even if you stop coming to class after session 2. Before enrolling in a group you must first take our Placement Test. This is not an actual test but a tool to ensure that you are taking the course that is the most suitable for your level of English. For more information on this 'test' please visit our website: www.itav.tudelft.nl Enrolment for the course (and the Placement Test) is via Brightspace. You do not need to register for this course in Osiris. Please note that you must also choose a group to reserve a place in the class. Type 'TPM302B' in Course search (Catalog). Please make sure you select the correct academic year and quarter. Register in the course AND in a group that suits your schedule.	
Remarks	The course will only be offered if there is a sufficient number of participants. Please note that it is not a presentation course.	
Elective	Yes	
Tags	English proficiency Group work Small groups	
Category	BSc and MSc level	

TPM303A	Intermediate Writing in English for the University	2
Module Manager	Drs. K.I. van der Linden	
Instructor	S.F. Johnson	
Instructor	Drs. D.W. Laponder	
Instructor	Drs. K.I. van der Linden	
Instructor	Drs. M.A. Swennen	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	2/2/2/2	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	This writing course is a 7 week course with 1 lesson a week (2 hours) plus approx. 6 hours of homework assignments per week. The course focuses on the acquisition of necessary language skills in the areas of grammar and usage, as well as reading and writing at a level appropriate to university study. Students will develop skills that develop an idea with relevant support for that idea by structuring the text according to current writing conventions, using various transitions to link ideas and demonstrating a knowledge of grammar and punctuation. Students will edit their own texts and work with other students in peer-editing groups.	
Study Goals	The aim of the course is to develop language skills and writing skills needed for producing academic texts. This course will give the student an understanding of the conventions of academic writing in English and some of the grammatical structures common in academic texts.	
Education Method	Individual, pair and group work, combined with assignments to do at home. Feedback on course components is given by the course lecturer and other students.	
Practical Guide	This is a course for both BSc and MSc students who would like to expand and improve their English writing skills. When you have taken our Placement Test you will be recommended a language level. With this recommendation you can sign up for any of the courses offered at that level. For this course you need a recommendation to take courses at the Intermediate level.	
Books	Advance in Academic Writing 1 - Steve Marshall This textbook is available in e-format (approx. 34) or a combined e-book and paper book (approx. 55). Details of how to obtain a copy will be given on Brightspace after enrolment.	
Assessment	Students will be given a final writing assignment. In borderline cases teachers will take the final decision, based on the student's class participation and homework assignments.	
Enrolment / Application	Before enrolling in a group you must first take our Placement Test. For more information on this test please visit our website: https://www.tudelft.nl/tbm/over-de-faculteit/afdelingen/stafafdelingen/itav/onderwijs/english-unit/placement-test For this course you need a recommendation to take courses at intermediate level. After receiving your Placement Test results you can enrol via Brightspace. Please note that you must also choose a group to reserve a place in the class. Once you have enrolled in the course, go to collaboration and select the group that suits your schedule. You do not need to register in Osiris for this course.	
Special Information	Note that you are required to spend an additional 6 hours (approx.) per week doing homework assignments. Please make sure that your schedule allows you this time.	
Remarks	An 80% attendance rate is obligatory, which means you may miss one session. To keep your place in the group you must attend the first session. You can drop the course with no penalty before the second session. Be aware of your "commitment" from the start, as you will still receive a grade [NV: Niet Verschenen--Not Attending], in Osiris even if you stop coming to class. The course is offered in Q2 and Q4. However, the course will only be offered if there are sufficient participants.	
Elective	Yes	
Tags	English proficiency	
Targetgroup	This course is available to BSc and MSc students	

TPM304A	Advanced Writing in English for the University	2
Module Manager	Drs. K.I. van der Linden	
Instructor	S.F. Johnson	
Instructor	Drs. D.W. Laponder	
Instructor	Drs. K.I. van der Linden	
Instructor	Drs. M.A. Swennen	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	2/2/2/2	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	This writing course is a 7 week course with 1 lesson a week (2 hours) plus approx. 6 hours of homework assignments per week. The course focuses on writing effective, well-structured and coherent academic texts in English. It looks at the writing process, including planning and structuring the text. Students will then develop paragraphs into longer academic texts and work on text coherence and cohesion. The course also pays attention to advanced academic vocabulary, sentence structure, punctuation and grammar.	
Study Goals	The aim of the course is to develop language skills and writing skills needed for producing academic texts. This course will give the student an understanding of the conventions of academic writing in English and some of the grammatical structures common in academic texts.	
Education Method	Individual, pair and group work, combined with assignments to do at home. Feedback on course components is given by the course lecturer and other students.	
Practical Guide	This is a course for both BSc and MSc students who already have a good understanding of academic writing, but who wish to expand and improve their English writing skills. When you have taken our Placement Test you will be recommended a language level. With this recommendation you can sign up for any of the courses offered at that level. For this course you need a recommendation to take courses at Advanced level.	
Books	Advance in Academic Writing 2 - Steve Marshall This textbook is available in e-format (approx. 34) or a combined e-book and paper book (approx. 55). Details of how to obtain a copy are given on Brightspace.	
Assessment	Students will be given a final writing assignment. In borderline cases teachers will take the final decision, based on the student's class participation and homework assignments.	
Enrolment / Application	Before enrolling in a group you must first take our Placement Test. For more information on this test please visit our website: https://www.tudelft.nl/tbm/over-de-faculteit/afdelingen/stafafdelingen/itav/onderwijs/english-unit/placement-test For this course you need a recommendation to take courses at Advanced level. After receiving your Placement Test results you can enroll via Brightspace. Please note that you must also choose a group to reserve a place in the class. Once you have enrolled in the course, go to collaboration and select the group that suits your schedule. You do not need to register in Osiris for this course.	
Special Information	Note that you are required to spend an additional 6 hours (approx.) per week doing homework assignments. Please make sure that your schedule allows you this time.	
Remarks	An 80% attendance rate is obligatory, which means you may miss one session. To keep your place in the group you must attend the first session. You can drop the course with no penalty before the second session. Be aware of your "commitment" from the start, as you will still receive a grade [NV: Niet Verschenen--Not Attending], in Osiris even if you stop coming to class. The course is offered in Q1 and Q3. However, the course will only be offered if there is a sufficient number of participants.	
Elective	Yes	
Tags	English proficiency	
Targetgroup	This course is available to BSc and MSc students	

TPM305A	Writing a Masters Thesis in English	2
Module Manager	Drs. D.W. Laponder	
Instructor	S.F. Johnson	
Instructor	Drs. D.W. Laponder	
Instructor	Drs. K.I. van der Linden	
Instructor	Drs. M.A. Swennen	
Responsible for assignments	Drs. D.W. Laponder	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	This course is designed for students who have started their MSc thesis or are about to start on their graduation project. The main focus is on the writing process, which means that matters such as grammar, academic style and precision of formulation are considered at length. Particular attention is also paid to the importance of structure and organising ideas. Aspects of the writing process will be backed up by weekly reading and writing assignments.	
Study Goals	The course is primarily designed for students working on their Masters thesis. As the name of the course indicates, the focus throughout is on writing. Samples of your academic writing will be corrected in considerable detail and you will be taught how to structure an essay, report or thesis. Plenty of tips will be given on ways of perfecting your English and invigorating your writing. Your systematic mistakes will be pointed out to you so that you can become critical about your own writing and go on to do your own editing.	
Education Method	Learning by doing and learning from lectures, corrections, peer review and tutor feedback.	
Books	Academic Writing for Graduate Students, Essential Tasks and Skills ,3rd edition by John M. Swales & Christine B. Feak. ISBN: 978-0-472-03475-8	
Assessment	Continuous assessment. Participants are expected to submit weekly writing assignments on a subject that is closely related to their field of study. An 85% attendance record is required. This means you may miss not more than one session (preferably none). If you fall behind with your coursework or cannot attend a minimum of six classes, you may be asked to leave the course. You will receive the course credits when all conditions are met.	
Elective	Yes	
Tags	English proficiency	
Category	MSc level	

WM0203TU	Oral Presentations	2
Module Manager	L. Meester	
Module Manager	I.M. Schouten	
Co-responsible for assignments	Drs. A.E. Kam	
Contact Hours / Week x/x/x/x	2/2/2/2	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	Dutch	
Course Contents	In job advertisements for engineers applicants are always required to possess excellent communication skills. This means among other things that they should be able to present their designs and research results fluently, for example during meetings with clients or with managers in their own company. A persuasive presentation does not only require thorough preparation of content, but also good style. It takes quite a bit of skill to come across understandable for any particular audience and to stay in control of the situation. For this purpose the course Oral Presentations offers students the possibility to gain experience in giving presentations under professional guidance in a small group.	
Study Goals	At the end of the course you should be able to: Present complex material clearly to both expert and non-expert audiences; Evaluate your own and other people's presentations Use presentation techniques such as: - Using various hand gestures - Using vocal variety - Keeping a good posture; Create attractive slides that support your presentation; Interact with your audience and respond to their reactions and questions professionally.	
Education Method	The first meeting will be an introduction to several presentation principles. The second, third, fourth, fifth and sixth meeting will be dedicated to practicing presentation skills. Students will be asked to prepare short presentations for this purpose, focusing on individual skills. For the seventh meeting students will be asked to give a final presentation about a subject that is related to their study. These presentations will be evaluated by a panel of students and the teacher. All presentations will be recorded on video. This will offer participants the possibility to evaluate their own presentations. Attendance is compulsory; no more than one lesson may be missed.	
Literature and Study Materials	Book: Van der Laaken, M. & van der Laaken, B. (2013). Presentation Techniques (Rev. ed.). Coutinho. Digital: online module in Brightspace.	
Assessment	A final mark will be based on the final presentations. Students who fail, will be given the opportunity to try again during an extra session, or to compensate for this by means of an assignment respectively.	
Enrolment / Application	Please enroll for a group in Brightspace: "wm0203tu Mondeling Presenteren". After enrollment you can subscribe in a group. Note: before subscribing check if you get ECTS for this course or if Oral Presentations (eng) is needed.	
Elective	Yes	
Category	BSc niveau	

WM1115TU	Dutch Elementary 1	3
Module Manager	Drs. G. Hoezen	
Responsible for assignments	Drs. G. Hoezen	
Co-responsible for assignments	Dr. S.J. van Boxtel	
Contact Hours / Week x/x/x/x	3/3/3/3	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	1 2 3 4	
Course Language	Dutch English	
Required for	After this course, participants can continue learning in Dutch Elementary 2 (course code WM1116TU)	
Expected prior knowledge	No prior knowledge on the Dutch language needed. Basic knowledge of English is required. If you already know about 1000 words in Dutch or more, we advise you to sit a level test. Contact delftsemethode@tudelft.nl .	
Parts	Preparing lessons at home by reading, listening and repeating Dutch texts, speaking during conversation class. Continual testing through listening tests, writing assignments and gap filling tests.	
Course Contents	Dutch Elementary 1 is a basic Dutch language course aimed at every day communication. You will learn the language through reading, analysing and listening to texts on various topics as preparation for the lessons. The lessons themselves will be mainly devoted to speaking. Regular testing is also part of this course.	
Study Goals	The goal of this course is to acquire basic Dutch language skills on CEF level A1.	
Education Method	Interactive group sessions, individual work through online course materials.	
Literature and Study Materials	Bondi Sciarone e.a., "Nederlands voor buitenlanders. De Delftse Methode". Text book. 5th revised edition.	
Assessment	The final grade is compiled of grades for speaking skills during class and several tests and assignments during the course period: writing assignments, listening tests and written gap filling tests. Attendance also contributes to the final grade. If this final grade is not sufficient, students sit a speaking test, a written test or a combination of both. These tests take place after the regular course period.	
Permitted Materials during Tests	No materials admitted during tests.	
Enrolment / Application	--- for MSc/BSc students --- There is a two-step-enrolment procedure. 1: Enrolment in Preliminary program for Dutch Elementary 1 course This programme takes place one week before the actual course. It is meant for receiving information about the learning method. On top of that, you will study the first two texts and sit the introductory test. If you pass, you can enrol for the actual course Dutch Elementary 1. 2: Enrolment in the actual course Dutch Elementary 1 (WM1115TU) You can enrol for one of the groups. --- for others (PhDs, employees) --- Go to the website www.dm.tudelft.nl for registration.	
Special Information	Brightspace, www.dm.tudelft.nl or delftsemethode@tudelft.nl	
Remarks	Regular courses take place at the campus in groups of max. 15 participants. We will also offer some groups in an online format. These groups have max. 9 participants. Check the website or Brightspace to see the current situation. There is an option to take this course as a self-study programme with a final test. Contact the secretarys office for registration. delftsemethode@tudelft.nl More information: www.dm.tudelft.nl	
Elective	Yes	
Tags	Broad Diverse Small groups	
Targetgroup	Students and others who want to learn basic Dutch and who have little or no prior knowledge of Dutch. People who already know some Dutch (approx. 1000 words or more) can contact delftsemethode@tudelft.nl for a level test.	
Self Test	If you want to do self-study, you can register for a written test + speaking test by sending an email to delftsemethode@tudelft.nl	
Category	MSc level	

WM1116TU	Dutch Elementary 2	3
Module Manager	Drs. G. Hoezen	
Responsible for assignments	Drs. G. Hoezen	
Co-responsible for assignments	Dr. S.J. van Boxtel	
Contact Hours / Week x/x/x/x	3/3/3/3	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	1 2 3 4	
Course Language	Dutch	
Required for	After this course, participants can continue learning in Dutch Intermediate 1 (course code WM1117TU)	
Expected prior knowledge	A completed course Elementary 1 (WM1115TU). It is also possible to enrol if you have a similar level: CEF A2. In that case you have to sit a level test. Contact delftsemethode@tudelft.nl .	
Parts	Preparing lessons at home by reading, listening and repeating Dutch texts, speaking during conversation class. Continual testing through listening tests, writing assignments and gap filling tests.	
Course Contents	Dutch Elementary 2 is the second part of a basic Dutch language course aimed at every day communication. You will learn the language through reading, analysing and listening to texts on various topics as preparation for the lessons. The lessons themselves will be mainly devoted to speaking. Regular testing is also part of this course.	
Study Goals	The goal of this course is to acquire basic Dutch language skills on CEF level A2.	
Education Method	Interactive group sessions, individual work through online course materials.	
Literature and Study Materials	Bondi Sciarone e.a., "Nederlands voor buitenlanders. De Delftse Methode". Text book. 5th revised edition.	
Assessment	The final grade is compiled of grades for speaking skills during class and several tests and assignments during the course period: writing assignments, listening tests and written gap filling tests. Attendance also contributes to the final grade. If this final grade is not sufficient, students sit a speaking test, a written test or a combination of both. These tests take place after the regular course period.	
Permitted Materials during Tests	No materials admitted during tests.	
Enrolment / Application	--- for MSc/BSc students --- Enrolment through Brightspace in the course Dutch Elementary 2 (WM1116TU) --- for others (PhDs, employees) --- Go to the website www.dm.tudelft.nl for registration.	
Special Information	Brightspace, www.dm.tudelft.nl or delftsemethode@tudelft.nl	
Remarks	Regular courses take place at the campus in groups of max. 15 participants. We will also offer some groups in an online format. These groups have max. 9 participants. Check the website or Brightspace to see the current situation. There is an option to take this course as a self-study programme with a final test. Contact the secretarys office for registration. delftsemethode@tudelft.nl More information: www.dm.tudelft.nl	
Elective	Yes	
Tags	Broad Diverse	
Targetgroup	Students and others who want to improve their Dutch knowledge. If you are not sure whether is course is on the right level for you, please contact delftsemethode@tudelft.nl for a level test.	
Category	MSc level	

WM1117TU	Dutch Intermediate 1	3
Module Manager	Drs. G. Hoezen	
Responsible for assignments	Drs. G. Hoezen	
Co-responsible for assignments	Dr. S.J. van Boxtel	
Contact Hours / Week x/x/x/x	3/3/3/3	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	1 2 3 4	
Course Language	Dutch	
Required for	After this course, participants can continue learning in Dutch Intermediate 2. This is a paid course, also for students and will not give you ECTS credits.	
Expected prior knowledge	A completed course Elementary 2 (WM1116TU). It is also possible to enrol if you have a similar level: CEF A2+. In that case you have to sit a level test. Contact delftsemethode@tudelft.nl .	
Parts	Preparing lessons at home by reading, listening and repeating Dutch texts, speaking during conversation class. Continual testing through listening tests, writing assignments and gap filling tests.	
Course Contents	Dutch Intermediate 1 is the second part of a basic Dutch language course aimed at every day communication. You will learn the language through reading, analysing and listening to texts on various topics as preparation for the lessons. The lessons themselves will be mainly devoted to speaking. Regular testing is also part of this course.	
Study Goals	The goal of this course is to acquire basic Dutch language skills on CEF level A2.	
Education Method	Interactive group sessions, individual work through online course materials.	
Literature and Study Materials	Wesdijk en Blom Sciarone e.a., "Tweede Ronde, herziene editie. Nederlands voor buitenlanders. De Delftse Methode". Text book.	
Assessment	The final grade is compiled of grades for speaking skills during class and several tests and assignments during the course period: writing assignments, listening tests and written gap filling tests. Attendance also contributes to the final grade. If this final grade is not sufficient, students sit a speaking test, a written test or a combination of both. These tests take place after the regular course period.	
Permitted Materials during Tests	No materials admitted during tests.	
Enrolment / Application	--- for MSc/BSc students --- Enrolment through Brightspace in the course Dutch Elementary 2 (WM1116TU) --- for others (PhDs, employees) --- Go to the website www.dm.tudelft.nl for registration	
Special Information	Brightspace, www.dm.tudelft.nl or delftsemethode@tudelft.nl	
Remarks	Regular courses take place at the campus in groups of max. 15 participants. We will also offer some groups in an online format. These groups have max. 9 participants. Check the website or Brightspace to see the current situation. There is an option to take this course as a self-study programme with a final test. Contact the secretaries office for registration. delftsemethode@tudelft.nl More information: www.dm.tudelft.nl	
Elective	Yes	
Tags	Broad Diverse	
Targetgroup	Students and others who want to improve their Dutch knowledge. If you are not sure whether is course is on the right level for you, please contact delftsemethode@tudelft.nl for a level test.	
Category	MSc level	

WM1135TU	Advanced English for the University	3
Module Manager	Drs. K.I. van der Linden	
Instructor	S.F. Johnson	
Instructor	Drs. K.I. van der Linden	
Instructor	Drs. M.A. Swennen	
Responsible for assignments	Drs. K.I. van der Linden	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	2/2/2/2	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	This course is given at C1 CEFR (Common European Framework of Reference) level. Advanced English for the University is the highest level integrated skills course offered by the English Unit within the ITAV institute at TPM. It is designed for those who already have a thorough working knowledge of English grammar and vocabulary but who still wish to further perfect their proficiency in order to become more adept in both writing and speaking. Time will be devoted to: Academic texts and writing conventions Academic vocabulary and collocations Grammar and punctuation Presentations and discussions Language learning strategies Advanced English for the University is a 14-week course with 1 lesson a week (2 hours) plus approx. 4 hours of homework assignments per week. The groups tend to be small (max. 18 participants) and the sessions are highly interactive. Attendance is mandatory. Students can miss a maximum of two lessons. Courses begin in the first week of September and in the first or second week of February . Please see the website for Placement Test dates and for any further course details: https://www.tudelft.nl/tbm/over-de-faculteit/afdelingen/stafafdelingen/itav/onderwijs/english-unit	
Study Goals	By the end of the course students should be familiar with different types of academic texts academic writing conventions academic vocabulary with a specific focus on collocations common grammatical errors punctuation conventions structural devices to improve coherence various language resources such as online dictionaries	
Education Method	Students will do in-class exercises completed either in small groups or in pairs. They will also do a group PowerPoint presentation and a discussion led in pairs. The feedback on course components will be given both by the lecturer and by fellow course participants.	
Literature and Study Materials	Course book: Cambridge Academic English - An integrated skills course for EAP (Student's Book; C1 Advanced). Authors: Martin Hewings and Craig Thaine Publisher: Cambridge University Press, 2012 ISBN: 978-0-521-16521-1	
Prerequisites	To qualify for the course students need to have received an Advanced English for the University recommendation following the completion of the Placement Test. Please see in this connection: https://www.tudelft.nl/tbm/over-de-faculteit/afdelingen/stafafdelingen/itav/onderwijs/english-unit/placement-test	
Assessment	There is no final exam. Assessment is based on the written assignments and on the in-class discussions and presentations. The six writing assignments each account for 10% of your final grade while your grade for the discussion and your grade for the presentation each accounts for 20% of your final grade.	
Enrolment / Application	After receiving their Placement Test results students can enrol via Brightspace by first registering for the course and then selecting the group of their choice.	
Remarks	More information can be found on Brightspace WM1135TU	
Elective	Yes	
Tags	English proficiency	
Targetgroup	BSc, MSc and PhD students are welcome	
Category	BSc and MSc level	

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